

BOARD OF EDUCATION, SOUTH KENSINGTON,
VICTORIA AND ALBERT MUSEUM.

MUSICAL INSTRUMENTS

BY

CARL ENGEL

WITH SEVENTY-EIGHT ILLUSTRATIONS



REVISED EDITION.

LONDON:

PRINTED FOR HIS MAJESTY'S STATIONERY OFFICE,

By WYMAN AND SONS, LIMITED, 109, FETTER LANE, E.C.

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*** START OF THE PROJECT GUTENBERG EBOOK MUSICAL
INSTRUMENTS [1908] ***



FIG. 1.—MUSIC. After an oil painting attributed to Melozzo da Forlì
(1438-1494).
National Gallery.

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NOTE.

IN the preparation of the revised edition of the late Dr. Engel's handbook, first published in 1875, care has been taken to make as few alterations as possible and to express no views from which he might have dissented.

The greatly enlarged chapter relating to post-mediæval instruments has been chiefly compiled from Dr. Engel's Descriptive Catalogue of the musical instruments in the Museum, published in 1874.

The pages relating to the Ancient Egyptians have been revised by Dr. W. M. Flinders Petrie, those dealing with the Greeks, Etruscans and Romans by Dr. Cecil H. Smith, and the description of Chinese and Japanese instruments by Dr. Stephen W. Bushell. The thanks of the Board are due to these gentlemen for their valuable co-operation.

CONTENTS.

	PAGE
NOTE	iii
LIST OF CONTENTS	v
” ” ILLUSTRATIONS	vii
CHAPTER I.—Introduction	1
” II.—Pre-Historic Relics and Ancient Egyptian	9
” III.—Assyrian and Hebrew	16
” IV.—Greek, Etruscan and Roman	27
” V.—Oriental	37
” VI.—American Indian	58
” VII.—European Instruments of the Middle Ages	83
” VIII.—European Instruments of the Middle Ages	92
” IX.—European Instruments of the Middle Ages	99
” X.—Post-Mediæval Instruments	104
APPENDIX	135
INDEX	139

LIST OF ILLUSTRATIONS.

FIG.		PAGE.
1.—MUSIC, after an oil painting attributed to Melozzo da Forlì (1438-1494)	<i>Frontispiece</i>	
2.—PAINTED WOODEN HARP. Ancient Egyptian. XVIIIth dynasty (B.C. 1450)	<i>Facing</i>	10
3.—BRONZE AND REED FLUTES. Ancient Egyptian. B.C. 600, or later	<i>Facing</i>	12
4.—BRONZE SISTRA. Ancient Egyptian. XXIIInd-XXVIth dynasty (B.C. 1000-600)	<i>Facing</i>	14
5.—SERIES OF BELLS. Ancient Egyptian. Late Period		15
6.—A MUSE WITH A HARP, AND TWO OTHERS WITH LYRES. From a Greek vase		29
7.—PAIR OF BRONZE FLUTES, with mouthpiece in the form of a bust of a Mænad holding a bunch of grapes. Greek	<i>Facing</i>	30
8.—A MUSE PLAYING THE DIAULOS. Greek		31
9.—WALL PAINTING of a youth wearing a myrtle wreath and playing on the DOUBLE PIPES. Said to have been found in a columbarium in the Vigna Ammendola on the Appian Way near Rome, about 1823. British Museum	<i>Facing</i>	34
10.TUBA, CORNU AND LITUUS. Roman		35

11.HSÜAN. Chinese —		42
12.(a) CH'IN (a species of Lute). Modern Chinese —		
(b) SHÊNG (Mouth Organ). Chinese. 19th century		
(c) YUEH-CH'IN (Moon Guitar). Chinese. 19th century	<i>Facing</i>	42
13.(a) KOTO (a species of Lute). Japanese. 19th century —		
(b) BIWA (a species of Guitar). Modern Japanese		
(c) SÂMISEN. Japanese	<i>Facing</i>	44
14.(a) SÂRINDA AND BOW. Indian (Bengal). 19th century —		
(b) RUDRA VINA. Southern Indian (Madras). 19th century		
(c) SÂRANGI AND BOW. Southern Indian. 19th century	<i>Facing</i>	48
15.(a) KEMÂNGEH OR SITÂRA OR FIDDLE. Persian. — About 1800		
(b) NUY (Flute). Persian. 19th century		
(c) SANTIR (Dulcimer) CASE. Persian	<i>Facing</i>	54
16.POTTERY WHISTLES, with finger-holes. Ancient — Mexican		59
17.POTTERY FLAGEOLETS, with finger-holes. (a) and — (c) Ancient Mexican; (b) from the Island of Sacrificios	<i>Facing</i>	60
18.BONE FLUTES. Ancient Peruvian, (a) and (b) — Truxillo; (c) Lima	<i>Facing</i>	60
19.HUAYRA-PUHURA, discovered in a Peruvian — tomb		64

20.	WOODEN TRUMPET. Used by Indians near the — Orinoco	65
21.	JURUPARIS, with and without cover. South — American	66
22.	BOTUTO. Used by Indians near the Orinoco —	68
23.	CITHARA. From a 9th century MS. formerly in — the monastery of St. Blasius in the Black Forest	84
24.	PSALTERIUM. From a 9th century MS. formerly — in the monastery of St. Blasius in the Black Forest	85
25.	CITHARA. From a 9th century MS. formerly in — the monastery of St. Blasius in the Black Forest	85
26.	KING PLAYING PSALTERY. After an engraving in — N. X. Willemin's <i>Monuments Français</i> <i>Inédits</i> , Vol. I., pl. 19, taken from <i>Hortus Deliciarum</i> , a MS. of the 12th century	86
27.	NABLUM. From a 9th century MS. at Angers —	86
28.	FEMALE PLAYING A SPECIES OF CITOLE. From a — 9th century MS. formerly in the monastery of St. Blasius in the Black Forest	86
29.	HARP. From a 9th century MS. formerly in the — monastery of St. Blasius in the Black Forest	87
30.	CRWTH. Welsh. 18th century —	90

Facing

31.ORGANISTRUM		93
—		
32.SACKBUT		94
—		
33.ORGAN. From a 12th century psalter in the		
— library of Trinity College, Cambridge		95
34.ORGAN (Grand Orgue). After an engraving in		
— N. X. Willemin's <i>Monuments Français Inédits</i>		96
35.BAS-RELIEF, representing a group of musicians,		
— formerly at the abbey of St. Georges de Boscherville. Late 11th century (?). After an engraving in N. X. Willemin's <i>Monuments Français Inédits</i>	<i>Facing</i>	98
36.HURDY-GURDY (Vielle). With arms of France		
— and crowned monogram of Henry II. on back and front. About 1550	<i>Facing</i>	100
37.TYMPANUM of the Glory Gate of the Cathedral		
— of Santiago de Compostella. Dated 1188. From a plaster cast in the Victoria and Albert Museum	<i>Facing</i>	100
38.MINSTREL GALLERY, Exeter Cathedral. 14th		
— century. From a plaster cast in the Victoria and Albert Museum	<i>Facing</i>	102
39.LUTE. Italian (Venetian). Beginning of the 17th		
— century	<i>Facing</i>	104
40.ANGEL PLAYING A LUTE. After an oil painting by		
— Ambrogio da Predis. Late 15th century	<i>Facing</i>	104
41.ARCHLUTE. Inscribed "Rauche in Chandos		
— Street, London, 1762"	<i>Facing</i>	104
42.CHITARRONE. Italian. Made by Buchenberg in		
— Rome, anno 1614	<i>Facing</i>	106

43.PANDURINA. French. Second half of 16th — century	<i>Facing</i>	108
44.GUITAR. French (?). 17th century —	<i>Facing</i>	108
45.QUINTERNA, OR CHITERNA. German. Dated 1539 —	<i>Facing</i>	108
46.CITHER. German. End of 17th century —	<i>Facing</i>	108
47.HARP THEORBO. Made by Harley. English. — About 1800	<i>Facing</i>	110
48.HARP VENTURA. English. Early 19th century —	<i>Facing</i>	110
49.BANDURIA. English. Early 19th century —	<i>Facing</i>	110
50.HARP. Old Irish —	<i>Facing</i>	110
51.HARP. French. About 1770 —	<i>Facing</i>	112
52.VIOLIN. Said to have belonged to James I. — English. Early 17th century	<i>Facing</i>	112
53.ANGEL PLAYING A VIOL. After an oil painting by — Ambrogio da Predis. Late 15th century	<i>Facing</i>	104
54.VIOLA DA GAMBA. Italian. About 1600 —	<i>Facing</i>	114
55.VIOLA DA GAMBA. Italian. 17th century —	<i>Facing</i>	114
56.VIOLA DI BARDONE, OR BARITON, WITH BOW. — German. 17th century	<i>Facing</i>	114
57.VIOLA D'AMORE. Probably English. Late 17th — century	<i>Facing</i>	116

58.DOUBLE-BASS, WITH BOW. Known as “The — Giant.” Italian. 17th century	<i>Facing</i>	116
59.SORDINO, OR POCHETTE. Probably German. Late — 17th or early 18th century	<i>Facing</i>	118
60.BÛCHE, OR SCHEITHOLZ. Made by Fleurot, of the — Val d’Ajol in the Vosges Mountains. Early 19th century	<i>Facing</i>	118
61.VIRGINAL. Formerly belonging to Queen — Elizabeth. Italian. Second half of 16th century	<i>Facing</i>	118
62.VIRGINAL. Flemish. Second half of 16th century —	<i>Facing</i>	118
63.SPINET. Made by Annibale dei Rossi of Milan. — Italian. Dated 1577	<i>Facing</i>	120
64.SPINET. Signed “Johannes Player fecit” — English. About 1700	<i>Facing</i>	120
65.CLAVICHORD. Inscribed “Barthold Fritz fecit, — Braunschweig, anno 1751.” German. 18th century	<i>Facing</i>	120
66.CLAVICEMBALO. Signed “Joanes Antonius Baffo, — Venetus.” Italian. Dated 1574	<i>Facing</i>	122
67.CLAVECIN. Made by Pascal Taskin of Paris. — French. Dated 1786	<i>Facing</i>	124
68.ORGAN-HARPSICHORD, OR CLAVIORGANUM. — Formerly in the chapel of Ightham Mote, near Sevenoaks, Kent. Probably English	<i>Facing</i>	124
69.TRIPLE FLAGEOLET. Italian. About 1820 —	<i>Facing</i>	124
70.FLAUTO DOLCE, OR FLUTE. Ivory. Inscribed — “Anciuti a Milan, 1740”	<i>Facing</i>	124

71.FLAGEOLET. Italian. Middle of 18th century —	<i>Facing</i>	126
72.OBOE. Made by Anciuti of Milan. Formerly in — the possession of the composer Rossini. Latter half of 18th century	<i>Facing</i>	126
73.BASSOON, species of. English. Late 18th, or — early 19th century	<i>Facing</i>	128
74.THE SERPENT. Made by Gerock Wolf, in — London. English. Early 19th century	<i>Facing</i>	128
75.SERINETTE OR BIRD ORGAN. French. Period of — Louis XIV.	<i>Facing</i>	128
76.ORGAN (Positive). German. Dated 1627 —	<i>Facing</i>	128
77.BAGPIPES. English. 18th century —	<i>Facing</i>	130
78.HANDEL'S HARPSICHORD. Made by Andreas — Ruckers, of Antwerp, 1651	<i>Facing</i>	134

MUSICAL INSTRUMENTS.

I.

INTRODUCTION.

MUSIC, in however primitive a stage of development it may be with some nations, is universally appreciated as one of the Fine Arts. The origin of vocal music may have been coeval with that of language; and the construction of musical instruments evidently dates with the earliest inventions which suggested themselves to human ingenuity. There exist even at the present day some savage tribes in Australia and South America who, although they have no more than the five first numerals in their language and are thereby unable to count the fingers of both hands together, nevertheless possess musical instruments of their own contrivance, with which they accompany their songs and dances.

Wood, metal, and the hide of animals are the most common substances used in the construction of musical instruments. In tropical countries bamboo or some similar kind of cane and gourds are especially made use of for this purpose. The ingenuity of man has contrived to employ in producing music, horn, bone, glass, pottery, slabs of sonorous stone—in fact, almost all vibrating matter. The strings of instruments have been made of the hair of animals, of silk, the runners of creeping plants, the fibrous roots of certain trees, of cane, catgut (which, absurdly referred to the cat, is from the sheep, goat, lamb, camel, and some other animals), metal, etc.

The mode in which individual nations or tribes are in the habit of embellishing their musical instruments is sometimes as characteristic as it is singular. The negroes in several districts of Western Africa affix to their drums human skulls. A war-trumpet of the king of Ashantee which was brought to England is surrounded by human jawbones. The Maoris in New Zealand carve around the mouth-hole of their trumpets a figure intended, it is said, to represent female lips. The materials for ornamentation chiefly employed by savages are bright colours, beads, shells, grasses, the bark of trees, feathers, stones, gilding, pieces of looking-glass inlaid like mosaic, etc. Uncivilised nations are sure to consider anything which is bright and glittering ornamental, especially if it is also scarce. Captain Tuckey saw in Congo a negro instrument which was ornamented with part of the broken frame of a looking-glass, to which were affixed in a semicircle a number of brass buttons with the head of Louis XVI.. on them,—perhaps a relic of some French sailor drowned near the coast years ago.

Again, musical instruments are not infrequently formed in the shape of certain animals. Thus, a kind of harmonicon of the Chinese represents the figure of a crouching tiger. The Burmese possess a stringed instrument in the shape of an alligator. Even more grotesque are the imitations of various beasts adopted by the Javanese. The natives of New Guinea have a singularly shaped drum, terminating in the head of a reptile. A wooden rattle like a bird is a favourite instrument of the Indians of Nootka Sound. In short, not only the inner construction of the instruments and their peculiar quality of sound exhibit in most nations certain distinctive characteristics, but it is also in great measure true as to their outward appearance.

An arrangement of the various kinds of musical instruments in a regular order, beginning with that kind which is the most universally known, and progressing gradually to the least usual, gives the following results. Instruments of percussion of indefinite sonorousness or, in other words, pulsatile instruments which have not a sound of a fixed pitch, as the drum, rattle, castanets, etc., are most universal. Wind instruments of the flute kind—including pipes, whistles, flutes, Pandean pipes, etc.—are also to be found almost everywhere.

Much the same is the case with wind instruments of the trumpet kind. These are often made of the horns, bones, and tusks of animals; frequently

of vegetable substances and of metal. Instruments of percussion of definite sonorousness are chiefly met with in China, Japan, Burmah, Siam, and Java. They not infrequently contain a series of tones produced by slabs of wood or metal, which are beaten with a sort of hammer, as our harmonicon is played.

Stringed instruments without a finger board, or any similar contrivance which enables the performer to produce a number of different tones on one string, are generally found among nations whose musical accomplishments have emerged from the earliest state of infancy. The strings are twanged with the fingers or with a piece of wood, horn, metal, or any other suitable substance serving as a *plectrum*; or are made to vibrate by being beaten with a hammer, as our dulcimer. Stringed instruments provided with a finger-board on which different tones are producible on one string by the performer shortening it more or less—as on the guitar and violin—are met with almost exclusively among nations in a somewhat advanced stage of musical progress. Such as are played with a bow are the least common; they are, however, known to the Chinese, Japanese, Hindus, Persians, Arabs, and a few other nations, besides those of Europe and their descendants in other countries.

Wind instruments of the organ kind—*i.e.*, such as are constructed of a number of tubes which can be sounded together by means of a common mouthpiece or some similar contrivance, and upon which therefore chords and combinations of chords, or harmony, can be produced—are comparatively of rare occurrence. Some interesting specimens of them exist in China, Japan, Laos, and Siam.

Besides these various kinds of sound-producing means employed in musical performances, a few others less widely diffused could be pointed out, which are of a construction not represented in any of our well-known European specimens. For instance, some nations have peculiar instruments of friction, which can hardly be classed with our instruments of percussion. Again, there are contrivances in which a number of strings are caused to vibrate by a current of air much as is the case with the Æolian harp; which might with equal propriety be considered either as stringed instruments or as wind instruments. In short, our usual classification of all the various species into three distinct divisions, *viz.*, *Stringed Instruments*, *Wind*

Instruments, and *Instruments of Percussion*, is not tenable if we extend our researches over the whole globe.

The collection at South Kensington contains several foreign instruments which cannot fail to prove interesting to the musician. Recent investigations have more and more elicited the fact that the music of every nation exhibits some distinctive characteristics which may afford valuable hints to a composer or performer. A familiarity with the popular songs of different countries is advisable on account of the remarkable originality of the airs; these mostly spring from the heart. Hence the natural and true expression, the delightful health and vigour by which they are generally distinguished. Our more artificial compositions are, on the other hand, not infrequently deficient in these charms, because they often emanate from the lingers or the pen rather than from the heart. Howbeit, the predominance of expressive melody and effective rhythm over harmonious combinations, so usual in the popular compositions of various nations, would alone suffice to recommend them to the careful attention of our modern musicians. The same may be said with regard to the surprising variety in construction and in manner of expression prevailing in the popular songs and dance-tunes of different countries. Indeed, every nation's musical effusions exhibit a character peculiarly their own, with which the musician would find it advantageous to familiarise himself.

Now, it will easily be understood that an acquaintance with the musical instruments of a nation conveys a more correct idea than could otherwise be obtained of the characteristic features of the nation's musical compositions. Furthermore, in many instances the construction of the instruments reveals to us the nature of the musical intervals, scales, modulations, and suchlike noteworthy facts. True, inquiries like these have hitherto not received from musicians the attention which they deserve. The adepts in most other arts are in this respect in advance. They are convinced that useful information may be gathered by investigating the productions even of uncivilised nations, and by thus tracing the gradual progress of an art from its primitive infancy to its highest degree of development.

Again, from an examination of the musical instruments of foreign nations we may derive valuable hints for the improvement of our own; or even for the invention of new. Several principles of construction have thus

been adopted by us from eastern nations. For instance, the *free reed* used in the harmonium is an importation from China. The organ builder Kratzenstein, who lived in St. Petersburg during the reign of Catherine II., happened to see the Chinese instrument *cheng*, which is of this construction, and it suggested to him, about the end of the 18th century, to apply the *free reed* to certain organ stops. At the present day instruments of the harmonium class have become such universal favourites in western Europe as almost to compete with the pianoforte.

Several other well-authenticated instances could be cited in which one instrument has suggested the construction of another of a superior kind. The prototype of our pianoforte was evidently the dulcimer, known at an early time to the Arabs and Persians, who call it *santir*. One of the old names given to the dulcimer by European nations is *cimbal*. The Poles at the present day call it *cymbaly*, and the Magyars in Hungary *cimbalom*. The *clavicembalo*, the predecessor of the pianoforte, was in fact nothing but a *cembalo* with a key board attached to it; and some of the old *clavicembali* still preserved, exhibit the trapezium shape, the round hole in the middle of the sound-board, and other peculiarities of the first dulcimer. Again, the gradual development of the dulcimer from a rude contrivance, consisting merely of a wooden board across which a few strings are stretched, is distinctly traceable by a reference to the musical instruments of nations in different stages of civilisation. The same is the case with our highly perfected harp, of which curious specimens, representing the instrument in its most primitive condition, are still to be found among several barbarous tribes. We might perhaps infer from its shape that it originally consisted of nothing more than an elastic stick bent by a string. The Damaras, a native tribe of South-western Africa, actually use their bow occasionally as a musical instrument when they are not engaged in war or in the chase. They tighten the string nearly in the middle by means of a leathern thong, whereby they obtain two distinct sounds, which, for want of a sound board, are of course very weak and scarcely audible to anyone but the performer. Some neighbouring tribes, however, possess a musical instrument very similar in appearance to the bow, to which they attach a gourd, hollowed and open at the top, which serves as a sound-board. Again, other African tribes have a similar instrument, superior in construction only inasmuch as it contains more than one string, and is provided with a sound-board

consisting of a suitable piece of sonorous wood. In short, the more improved we find these contrivances the closer they approach our harp. And it could be shown, if this were requisite for our present purpose, that much the same gradual progress towards perfection, which we observe in the African harp, is traceable in the harps of several nations in different parts of the world.

Moreover, a collection of musical instruments deserves the attention of the ethnologist as much as of the musician. Indeed, this may be asserted of national music in general; for it gives us an insight into the heart of man, reveals to us the feelings and predilections of different races on the globe, and affords us a clue to the natural affinity which exists between different families of men. Again, a collection must prove interesting in a historical point of view. Scholars will find among old instruments specimens which were in common use in England at the time of Queen Elizabeth, and which are not unfrequently mentioned in the literature of that period. In many instances the passages in which allusion is made to them can hardly be understood, if we are unacquainted with the shape and construction of the instruments. Furthermore, these relics of bygone times bring before our eyes the manners and customs of our forefathers, and assist us in understanding them correctly.

It will be seen that the modification which our orchestra has undergone, in the course of scarcely more than a century, is great indeed. Most of the instruments which were highly popular about a hundred years ago have either fallen into disuse or are now so much altered that they may almost be considered as new inventions. Among Asiatic nations, on the other hand, we meet with several instruments which have retained unchanged through many centuries their old construction and outward appearance. At South Kensington may be seen instruments still in use in Egypt and western Asia, precisely like specimens represented on monuments dating from a period of three thousand years ago. By a reference to the Eastern instruments of the present time we obtain therefore a key for investigating the earlier Egyptian and Assyrian representations of musical performances; and likewise, for appreciating more exactly the biblical records respecting the music of the Hebrews. Perhaps these evidences will convey to some inquirers a less high opinion than they have hitherto entertained, regarding the musical accomplishments of the Hebrew bands in the solemn processions of King

David or in Solomon's temple; but the opinion will be all the nearer to the truth.

There is another point of interest about such collections, and especially that at South Kensington, which must not be left unnoticed. Several instruments are remarkable on account of their elegant shape and tasteful ornamentation. This is particularly the case with some specimens from Asiatic countries. The beautiful designs with which they are embellished may afford valuable patterns for study and for adoption in works of art.

II.

PRE-HISTORIC RELICS AND ANCIENT EGYPTIAN.

A REALLY complete account of all the musical instruments from the earliest time known to us would require much more space than can here be afforded. We can attempt only a concise historical survey. We venture to hope that the illustrations interspersed throughout the text will to the intelligent reader elucidate many facts which, for the reason stated, are touched upon but cursorily.

PRE-HISTORIC RELICS.

A musical relic has been exhumed in the department of Dordogne in France, which was constructed in an age when the fauna of France included the reindeer, the rhinoceros and the mammoth, the hyæna, the bear, and the cave-lion. It is a small bone somewhat less than two inches in length, in which is a hole, evidently bored by means of one of the little flint knives which men used before acquaintance with the employment of metal for tools and weapons.^[1] Many of these flints were found in the same place with the bones. Only about half a dozen of the bones, of which a considerable number have been exhumed, possess the artificial hole.

M. Lartet surmises the perforated bone to have been used as a whistle in hunting animals. It is the first digital phalanx of a ruminant, drilled to a certain depth by a smooth cylindrical bore on its lower surface near the expanded upper articulation. On applying it to the lower lip and blowing into it a shrill sound is yielded. Three of these phalanges are of reindeer, one is of chamois. Again, among the relics which have been brought to light from the cave of Lombrive, in the department of Ariège, occur several eye-

teeth of the dog, which have a hole drilled into them near the root. Probably they also yield sounds, like those reindeer bones, or like the tube of a key. Another whistle—or rather a pipe, for it has three finger-holes by means of which different tones could be produced—was found in a burying-place, dating from the stone period, in the vicinity of Poitiers in France; it is rudely constructed from a fragment of stag's horn. It is blown at the end, like a *flûte à bec*, and the three-finger holes are placed equidistantly. Four distinct tones must have been easily obtainable on it: the lowest, when all the finger-holes were covered; the other three, by opening the finger-holes successively. From the character of the stone utensils and weapons discovered with this pipe it is conjectured that the burying-place from which it was exhumed dates from the latest time of the stone age. Therefore, however old it may be, it is a more recent contrivance than the reindeer-bone whistle from the cavern of the Dordogne.

THE ANCIENT EGYPTIANS.

The most ancient nations historically known possessed musical instruments which, though in acoustic construction greatly inferior to our own, exhibit a degree of perfection which could have been attained only after a long period of cultivation. Many tribes of the present day have not yet reached this stage of musical progress.



**FIG. 2.—PAINTED WOODEN HARP. Ancient Egyptian, XVIIIth dynasty
(B.C. 1450).
British Museum.**

As regards the instruments of the ancient Egyptians we now possess perhaps more detailed information than of those appertaining to any other nation of antiquity. This information we owe especially to the exactness with which the instruments are depicted in sculptures and paintings^[2]. Whoever has examined these interesting monuments with even ordinary care cannot but be convinced that the representations which they exhibit are faithful transcripts from life. Moreover, if there remained any doubt respecting the accuracy of the representations of the musical instruments it

might be dispelled by existing evidence. Several specimens have been discovered in tombs, preserved in a more or less perfect condition.

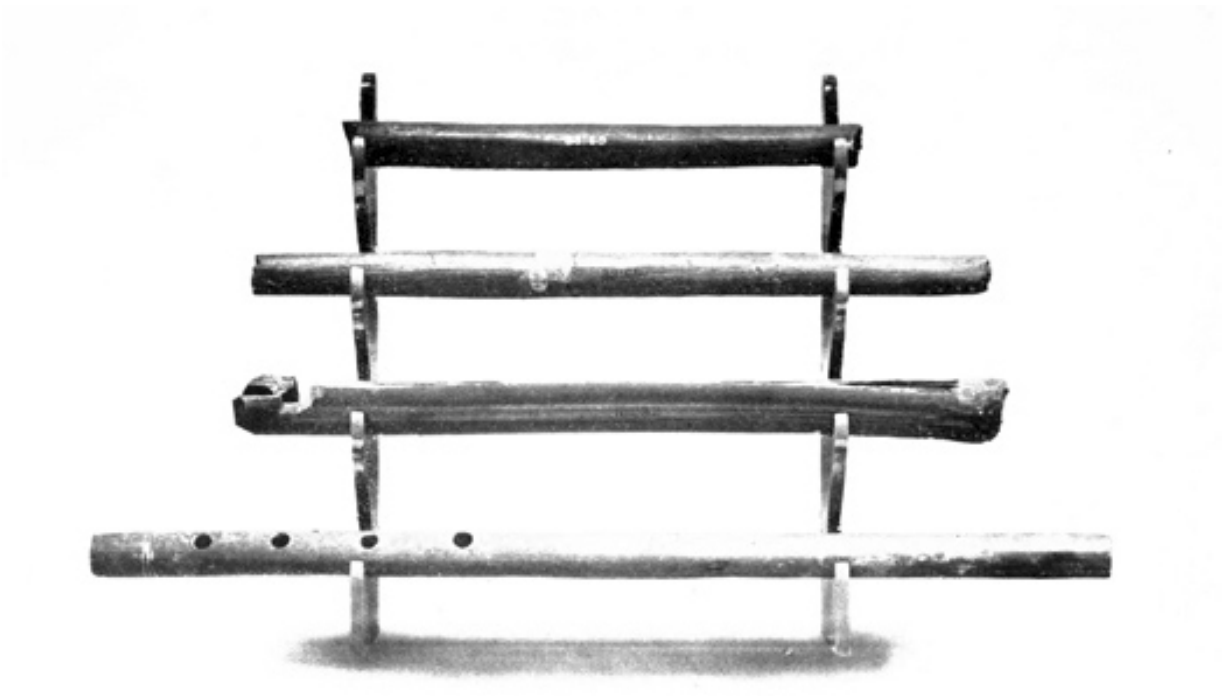
The Egyptians possessed various kinds of harps, some of which were elegantly shaped and tastefully ornamented. The largest were about 6½ feet high; and the small ones frequently had some sort of stand which enabled the performer to play upon the instrument while standing. The name of the harp was *bene*. Its frame had no front pillar; the tension of the strings therefore cannot have been anything like so strong as on our present harp. (Fig. 2.)

The Egyptian harps most remarkable for elegance of form and elaborate decoration are the two which were first noticed by Bruce who found them painted in fresco on the walls of a sepulchre at Thebes, supposed to be the tomb of Rameses III. who reigned about 1170 B.C. Bruce's discovery created a sensation among musicians. The fact that at so remote an age the Egyptians should have possessed harps which vie with our own in elegance and beauty of form appeared to some so incredible that the correctness of Bruce's representations, as engraved in his "Travels," was greatly doubted. Sketches of the same harps, taken subsequently and at different times from the frescoes, have since been published, but they differ more or less from each other in appearance and in the number of strings. A kind of triangular harp of the Egyptians was discovered in a well-preserved condition and is now deposited in the Louvre. It has twenty-one strings; a greater number than is generally represented on the monuments. All these instruments, however much they differed from each other in form, had one peculiarity in common, namely the absence of the fore pillar.

The *nefer*, a kind of guitar, was almost identical in construction with the Tamboura at the present day in use among several eastern nations. It was evidently a great favourite with the ancient Egyptians, and occurs in representations of concerts dating earlier than from B.C. 1500. The *nefer* affords the best proof that the Egyptians had made considerable progress in music at a very early age; since it shows that they understood how to produce on a few strings, by means of the finger-board, a greater number of notes than were obtainable even on their harps. The instrument had two or four strings, was played with a plectrum and appears to have been sometimes, if not always, provided with frets. In the British Museum is a

fragment of a fresco obtained from a tomb at Thebes, on which two female performers on the *nefer* are represented. The painter has distinctly indicated the frets.

Small pipes or flutes of the Egyptians have been discovered, made of reed, with three, four, five, or more finger-holes. There are some interesting examples in the British Museum; one of which has seven holes burnt in at the side (Fig. 3). Two straws were found with it of nearly the same length as the pipe, which is about one foot long. In some other pipes pieces of a kind of thick straw have also been found inserted into the tube, obviously serving for a similar purpose as the *reed* in our oboe or clarionet.



**FIG. 3.—BRONZE AND REED FLUTES. Ancient Egyptian.
B.C. 600 or later.
British Museum.**

The *sebā*, a single flute, was of considerable length, and the performer appears to have been obliged to extend his arms almost at full length in order to reach the furthest finger-hole. As *sebā* is also the name of the leg-bone (like the Latin *tibia*) it may be supposed that the Egyptian flute was

originally made of bone. Those, however, which have been found are of wood or reed.

A flute-concert is painted on one of the tombs in the pyramids of Gizeh and dates, according to Lepsius, from an age earlier than B.C. 2000. Eight musicians are performing on flutes. Three of them, one behind the other, are kneeling and holding their flutes in exactly the same manner. Facing these are three others, in a precisely similar position. A seventh is sitting on the ground to the left of the six, with his back turned towards them, but also in the act of blowing his flute, like the others. An eighth is standing at the right side of the group with his face turned towards them, holding his flute before him with both hands, as if he were going to put it to his mouth, or had just left off playing. He is clothed, while the others have only a narrow girdle round their loins. Perhaps he is the director of this singular band, or the *solo* performer who is waiting for the termination of the *tutti* before renewing his part of the performance. The division of the players into two sets, facing each other, suggests the possibility that the instruments were classed somewhat like the first and second violins, or the *flauto primo* and *flauto secondo* of our orchestras. The occasional employment of the interval of the third, or the fifth, as accompaniment to the melody, is not unusual even with nations less advanced in music than were the ancient Egyptians.

The Double-Pipe, called *mam*, appears to have been a very popular instrument, if we judge from the frequency of its occurrence in the representations of musical performances. Furthermore, the Egyptians had, as far as is known to us, two kinds of trumpets; three kinds of tambourines, or little hand drums; three kinds of drums, chiefly barrel-shaped; and various kinds of gongs, bells, cymbals, and castanets. The trumpet appears to have been usually of brass. A peculiar wind-instrument, somewhat the shape of a champagne bottle and perhaps made of pottery or wood, also occurs in the representations transmitted to us.

The Egyptian drum was from two to three feet in length, covered with parchment at both ends and braced by cords. The performer carried it before him, generally by means of a band over his shoulder, while he was beating it with his hands on both ends. Of another kind of drum an actual specimen has been found in the excavations made in the year 1823 at Thebes. It was 1½ feet high and 2 feet broad, and had cords for bracing it. A piece of catgut

encircled each end of the drum, being wound round each cord, by means of which the cords could be tightened or slackened at pleasure by pushing the two hands of catgut towards or from each other. It was beaten with two drumsticks slightly bent. The Egyptians had also straight drumsticks with a handle, and a knob at the end. The Berlin museum possesses some of these. The third kind of drum was almost identical with the *darabuka* of the modern Egyptians. The Tambourine was either round, like that which is at the present time in use in Europe as well as in the east; or it was of an oblong square shape, slightly incurved on the four sides.

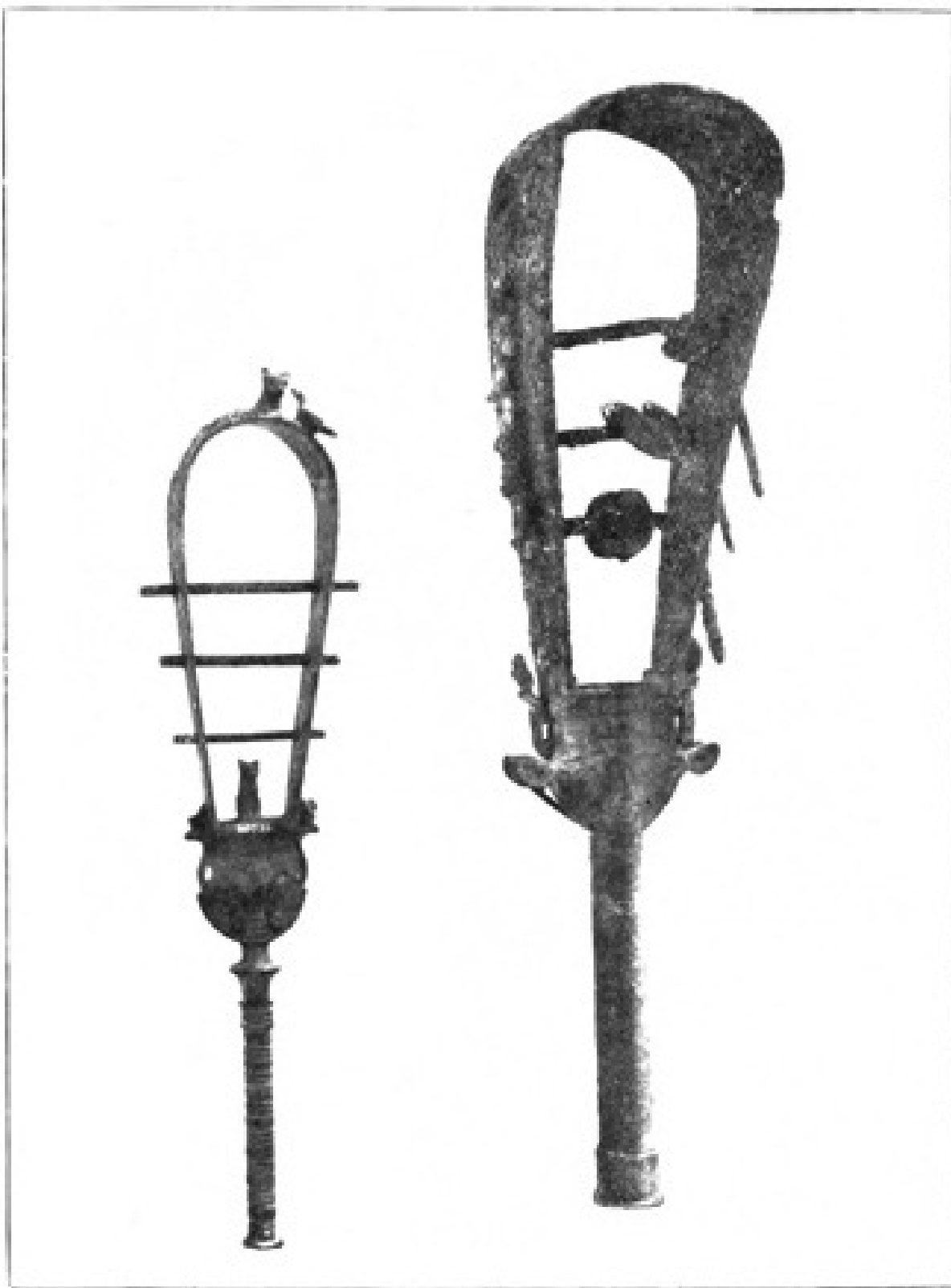


FIG. 4.—BRONZE SISTRA. Ancient Egyptian.

**XXIInd-XXVIth dynasty (B.C. 1000-600).
British Museum.**

The Sistrum consisted of a frame of bronze into which three or four metal bars were loosely inserted, so as to produce a jingling noise when the instrument was shaken. (Fig. 4.) The bars were often made in the form of snakes, or they terminated in the head of a goose. Not unfrequently a few metal rings were strung on the bars, to increase the noise. The frame was sometimes ornamented with the figure of a cat. The largest sistra which have been found are about eighteen inches in length, and the smallest about nine inches. The sistrum was principally used by females in religious performances. Its Egyptian name was *seshesh*.

The Egyptian cymbals closely resembled our own in shape. There are several pairs of them in the British museum. One pair was found in a coffin enclosing the mummy of a sacred musician, and is deposited in the same case with the mummy and coffin. Among the Egyptian antiquities in the British museum are also several small bells of bronze (Fig. 5). The largest is 2¼ inches in height, and the smallest three-quarters of an inch. Some of them have a hole at the side near the top wherein the clapper was fastened.



**FIG. 5.—SERIES OF BELLS. Ancient Egyptian. Late Period.
The smaller examples were sewn on wearing apparel.
British Museum.**

III.

ASSYRIAN AND HEBREW.

THE ASSYRIANS.

OUR acquaintance with the Assyrian instruments has been derived almost entirely from the famous bas-reliefs which have been excavated from the mounds of Nimroud, Khorsabad, and Kouyunjik (the site of the ancient Nineveh), situated near the river Tigris in the vicinity of the town of Mosul in Asiatic Turkey.

The Assyrian harp was about four feet high, and appears of larger size than it actually was on account of the ornamental appendages which were affixed to the lower part of its frame. It must have been but light in weight, since we find it not unfrequently represented in the hands of persons who are playing upon it while they are dancing. Like all the Oriental harps, modern as well as ancient, it was not provided with a front pillar. The upper portion of the frame contained the sound-holes, somewhat in the shape of an hourglass. Below them were the screws, or tuning-pegs, arranged in regular order. The strings were perhaps made of silk, like those which the Burmese use at the present time on their harps; or they may have been of catgut, which was used by the ancient Egyptians.

The largest assemblage of Assyrian musicians which has been discovered on any monument consists of eleven performers upon instruments, besides a chorus of singers. The first musician—probably the leader of the band, as he marches alone at the head of the procession—is playing upon a harp. Behind him are two men; one with a dulcimer and the other with a double-pipe; then follow two men with harps. Next come six female musicians, four of whom are playing upon harps, while one is

blowing a double-pipe and another is beating a small hand-drum covered only at the top. Close behind the instrumental performers are the singers, consisting of a chorus of females and children. They are clapping their hands in time with the music, and some of the musicians are dancing to the measure. One of the female singers is holding her hand to her throat in the same manner as the women in Syria, Arabia, and Persia are in the habit of doing at the present day when producing, on festive occasions, those peculiarly shrill sounds of rejoicing which have been repeatedly noticed by travellers.

The dulcimer is in too imperfect a state on the bas-relief to familiarize us with its construction. The slab representing the procession in which it occurs has been injured; the defect which extended over a portion of the dulcimer has been repaired, and it cannot be said that in repairing it much musical knowledge has been evinced.

The instrument of the Trigonon species was held horizontally, and was twanged with a rather long plectrum slightly bent at the end at which it was held by the performer. It is of frequent occurrence on the bas-reliefs. A number of them appear to have been generally played together. At any rate, we find almost invariably on the monuments two together, evidently implying "more than one," "a number." The left hand of the performer seems to have been occupied in checking the vibration of the strings when its discontinuance was required. From the position of the strings the performer could not have struck them as those of the dulcimer are struck. If he did not twang them, he may have drawn the plectrum across them. Indeed, for twanging, a short plectrum would have been more practical, considering that the strings are placed horizontally one above the other at regular distances. It is therefore by no means improbable that we have here a rude prototype of the violin bow.

The lyre occurs in three different forms, and is held horizontally in playing, or at least nearly so. Its front bar was generally either oblique or slightly curved. The strings were tied round the bar so as to allow of their being pushed upwards or downwards. In the former case the tension of the strings increases, and the notes become therefore higher; on the other hand, if the strings are pushed lower down the pitch of the notes must become

deeper. The lyre was played with a small plectrum as well as with the fingers.

The Assyrian trumpet was very similar to the Egyptian. Furthermore, we meet with three kinds of drums, of which one is especially noteworthy on account of its odd shape, somewhat resembling a sugar loaf; with the tambourine; with two kinds of cymbals; and with bells, of which a considerable number have been found in the mound of Nimroud. These bells, which have greatly withstood the devastation of time, are but small in size, the largest of them being only $3\frac{1}{4}$ inches in height and $2\frac{1}{2}$ inches in diameter. Most of them have a hole at the top, in which probably the clapper was fastened. They are made of copper mixed with 14 per cent. of tin.

Instrumental music was used by the Assyrians and Babylonians in their religious observances. This is obvious from the sculptures, and is to some extent confirmed by the mode of worship paid by command of king Nebuchadnezzar to the golden image; "Then an herald cried aloud, To you it is commanded, O people, nations, and languages, that at what time ye hear the sound of the cornet, flute, harp, sackbut, psaltery, dulcimer, and all kinds of musick, ye fall down and worship the golden image that Nebuchadnezzar the king has set up." The kings appear to have maintained at their courts musical bands, whose office it was to perform secular music at certain times of the day or on fixed occasions. Of king Darius we are told that, when he had cast Daniel into the den of lions, he "went to his palace, and passed the night fasting, neither were instruments of musick brought before him;" from which we may conclude that his band was in the habit of playing before him in the evening. A similar custom prevailed also at the court of Jerusalem, at least in the time of David and Solomon; both of whom appear to have had their royal private bands, besides a large number of singers and instrumental performers of sacred music who were engaged in the Temple.

THE HEBREWS.

As regards the musical instruments of the Hebrews, we are from biblical records acquainted with the names of many of them; but representations to

be trusted are still wanting, and it is chiefly from an examination of the ancient Egyptian and Assyrian instruments that we can conjecture almost to a certainty their construction and capabilities. From various indications, which it would be too circumstantial here to point out, we believe the Hebrews to have possessed the following instruments:

THE HARP.—There can be no doubt that the Hebrews possessed the harp, seeing that it was a common instrument among the Egyptians and Assyrians. But it is uncertain which of the Hebrew names of the stringed instruments occurring in the Bible really designates the harp.

THE DULCIMER.—Some writers on Hebrew music consider the *nevel* to have been a kind of dulcimer; others conjecture the same of the *psanterin* mentioned in the book of Daniel,—a name which appears to be synonymous with the *psalterion* of the Greeks, and from which also the present oriental dulcimer, *santir*, may have been derived. Some of the instruments mentioned in the book of Daniel may have been synonymous with some which occur in other parts of the Bible under Hebrew names; the names given in Daniel being Chaldæan. The *asor* was a ten-stringed instrument played with a plectrum, and is supposed to have borne some resemblance to the *nevel*.

THE LYRE.—This instrument is represented on some Hebrew coins generally ascribed to Judas Maccabæus, who lived in the second century before the Christian era. There are several of them in the British Museum; some are of silver, and the others of copper. On three of them are lyres with three strings, another has one with five, and another one with six strings. The two sides of the frame appear to have been made of the horns of animals, or they may have been of wood formed in imitation of two horns which originally were used. Lyres thus constructed are still found in Abyssinia. The Hebrew square-shaped lyre of the time of Simon Maccabæus is probably identical with the *psalterion*. The *kinnor*, the favourite instrument of king David, was most likely a lyre if not a small triangular harp. The lyre was evidently an universally known and favoured instrument among ancient eastern nations. Being more simple in construction than most other stringed instruments it undoubtedly preceded them in antiquity. The *kinnor* is mentioned in the Bible as the oldest stringed instrument, and as the invention of Jubal. Even if the name of one

particular stringed instrument is here used for stringed instruments in general, which may possibly be the case, it is only reasonable to suppose that the oldest and most universally known stringed instrument would be mentioned as a representative of the whole class rather than any other. Besides, the *kinnor* was a light and easily portable instrument; king David, according to the Rabbinic records, used to suspend it during the night over his pillow. All its uses mentioned in the Bible are especially applicable to the lyre. And the resemblance of the word *kinnor* to *kithara*, *kissar*, and similar names known to denote the lyre, also tends to confirm the supposition that it refers to this instrument. It is, however, not likely that the instruments of the Hebrews—indeed their music altogether—should have remained entirely unchanged during a period of many centuries. Some modifications were likely to occur even from accidental causes; such, for instance, as the influence of neighbouring nations when the Hebrews came into closer contact with them. Thus may be explained why the accounts of the Hebrew instruments given by Josephus, who lived in the first century of the Christian era, are not in exact accordance with those in the Bible. The lyres at the time of Simon Maccabæus may probably be different from those which were in use about a thousand years earlier, or at the time of David and Solomon, when the art of music with the Hebrews was at its zenith.

There appears to be a probability that a Hebrew lyre of the time of Joseph (about 1700 B.C.) is represented on an ancient Egyptian painting^[3] discovered in a tomb at Beni Hassan—which is the name of certain grottoes on the eastern bank of the Nile. Sir Gardner Wilkinson, in his “Manners and Customs of the Ancient Egyptians,” observes: “If, when we become better acquainted with the interpretation of hieroglyphics, the ‘strangers’ at Beni Hassan should prove to be the arrival of Jacob’s family in Egypt, we may examine the Jewish lyre drawn by an Egyptian artist. That this event took place about the period when the inmate of the tomb lived is highly probable—at least, if I am correct in considering Usertsen I. to be the Pharaoh who was the patron of Joseph; and it remains for us to decide whether the disagreement in the number of persons here introduced, thirty-seven being written over them in hieroglyphics, is a sufficient objection to their identity. It will not be foreign to the present subject to introduce those figures, which are curious, if only considered as illustrative of ancient customs at that early period, and which will be looked upon with unbounded interest should they

ever be found to refer to the Jews. The first figure is an Egyptian scribe, who presents an account of their arrival to a person seated, the owner of the tomb, and one of the principal officers of the reigning Pharaoh. The next, also an Egyptian, ushers them into his presence; and two advance bringing presents, the wild goat or ibex and the gazelle, the productions of their country. Four men, carrying bows and clubs, follow, leading an ass on which two children are placed in panniers, accompanied by a boy and four women; and, last of all, another ass laden, and two men—one holding a bow and club, the other a lyre, which he plays with the plectrum. All the men have beards, contrary to the custom of the Egyptians, but very general in the East at that period, and noticed as a peculiarity of foreign uncivilized nations throughout their sculptures. The men have sandals, the women a sort of boot reaching to the ankle, both which were worn by many Asiatic people. The lyre is rude, and differs in form from those generally used in Egypt.” In the engraving the lyre-player, another man, and some strange animals from this group, are represented.

THE TAMBOURA.—*Minnim*, *machalath*, and *nevel* are usually supposed to be the names of instruments of the lute or guitar kind. *Minnim*, however, appears more likely to imply stringed instruments in general than any particular instrument.

THE SINGLE PIPE.—*Chalil* and *nekeb* were the names of the Hebrew pipes or flutes.

THE DOUBLE PIPE.—Probably the *mishrokitha* mentioned in Daniel. The *mishrokitha* is represented in the drawings of our histories of music as a small organ, consisting of seven pipes placed in a box with a mouthpiece for blowing. But the shape of the pipes and of the box as well as the row of keys for the fingers exhibited in the representation of the *mishrokitha* have too much of the European type not to suggest that they are probably a product of the imagination. Respecting the illustrations of Hebrew instruments which usually accompany historical treatises on music and commentaries on the Bible, it ought to be borne in mind that most of them are merely the offspring of conjectures founded on some obscure hints in the Bible, or vague accounts by the Rabbins.

THE SYRINX OR PANDEAN PIPE.—Probably the *ugab*, which in the English authorised version of the Bible is rendered “organ.”

THE BAGPIPE.—The word *sumphonia*, which occurs in the book of Daniel, is, by Forkel and others, supposed to denote a bagpipe. It is remarkable that at the present day the bagpipe is called by the Italian peasantry *Zampogna*. Another Hebrew instrument, the *magrepha*, generally described as an organ, was more likely only a kind of bagpipe. The *magrepha* is not mentioned in the Bible but is described in the Talmud. In tract Erachin it is recorded to have been a powerful organ which stood in the temple at Jerusalem, and consisted of a case or wind-chest, with ten holes, containing ten pipes. Each pipe was capable of emitting ten different sounds, by means of finger-holes or some similar contrivance: thus one hundred different sounds could be produced on this instrument. Further, the *magrepha* is said to have been provided with two pairs of bellows and with ten keys, by means of which it was played with the fingers. Its tone was, according to the Rabbinic accounts, so loud that it could be heard at an incredibly long distance from the temple. Authorities so widely differ that we must leave it uncertain whether the much-lauded *magrepha* was a bagpipe, an organ, or a kettle-drum.

THE TRUMPET.—Three kinds are mentioned in the Bible, viz., the *keren*, the *shophar*, and the *chatzozerah*. The first two were more or less curved and might properly be considered as horns. Most commentators are of opinion that the *keren*—made of ram’s horn—was almost identical with the *shophar*, the only difference being that the latter was more curved than the former. The *shophar* is especially remarkable as being the only Hebrew musical instrument which has been preserved to the present day in the religious services of the Jews. It is still blown in the synagogue, as in time of old, at the Jewish new-year’s festival, according to the command of Moses (Numb. xxix. 1). The *chatzozerah* was a straight trumpet, about two feet in length, and was sometimes made of silver. Two of these straight trumpets are shown in the famous triumphal procession after the fall of Jerusalem on the arch of Titus.

THE DRUM.—There can be no doubt that the Hebrews had several kinds of drums. We know, however, only of the *toph*, which appears to have been a tambourine or a small hand-drum like the Egyptian *darabuka*. In the

English version of the Bible the word is rendered *timbrel* or *tabret*. This instrument was especially used in processions on occasions of rejoicing, and also frequently by females. We find it in the hands of Miriam, when she was celebrating with the Israelitish women in songs of joy the destruction of Pharaoh's host; and in the hands of Jephtha's daughter, when she went out to welcome her father. There exists at the present day in the East a small hand-drum called *doff*, *diff*, or *adufe*—a name which appears to be synonymous with the Hebrew *toph*.

THE SISTRUM.—Winer, Saalschütz, and several other commentators are of opinion that the *menaaneim*, mentioned in 2 Sam. vi. 5, denotes the sistrum. In the English Bible the original is translated *cymbals*.

CYMBALS.—The *tzeltzelim*, *metzilloth*, and *metzilthaim*, appear to have been cymbals or similar metallic instruments of percussion, differing in shape and sound.

BELLS.—The little bells on the vestments of the high-priest were called *phaamon*. Small golden bells were attached to the lower part of the robes of the high-priest in his sacred ministrations. The Jews have, at the present day, in their synagogues small bells fastened to the rolls of the Law containing the Pentateuch: a kind of ornamentation which is supposed to have been in use from time immemorial.

Besides the names of Hebrew instruments already given there occur several others in the Old Testament, upon the real meaning of which much diversity of opinion prevails. *Jobel* is by some commentators classed with the trumpets, but it is by others believed to designate a loud and cheerful blast of the trumpet, used on particular occasions. If *Jobel* (from which *jubilare* is supposed to be derived) is identical with the name *Jubal*, the inventor of musical instruments, it would appear that the Hebrews appreciated pre-eminently the exhilarating power of music. *Shalisbim* is supposed to denote a triangle. *Nechiloth*, *gittith*, and *machalath*, which occur in the headings of some psalms, are also by commentators supposed to be musical instruments. *Nechiloth* is said to have been a flute, and *gittith* and *machalath* to have been stringed instruments, and *machol* a kind of flute. Again, others maintain that the words denote peculiar modes of performance or certain favourite melodies to which the psalms were

directed to be sung, or chanted. According to the records of the Rabbins, the Hebrews in the time of David and Solomon possessed thirty-six different musical instruments. In the Bible only about half that number are mentioned.

Most nations of antiquity ascribed the invention of their musical instruments to their gods, or to certain superhuman beings. The Hebrews attributed it to man; Jubal is mentioned in Genesis as “the father of all such as handle the harp and organ” (*i.e.*, performers on stringed instruments and wind instruments). As instruments of percussion are almost invariably in use long before people are led to construct stringed and wind instruments it might perhaps be surmised that Jubal was not regarded as the inventor of all the Hebrew instruments, but rather as the first professional cultivator of instrumental music.

IV.

GREEK, ETRUSCAN AND ROMAN.

THE GREEKS.

MANY musical instruments of the ancient Greeks are known to us by name; but respecting their exact construction and capabilities there still prevails almost as much diversity of opinion as is the case with those of the Hebrews.

It is generally believed that the Greeks derived their musical system from the Egyptians. Pythagoras and other philosophers are said to have studied music in Egypt. It would, however, appear that the Egyptian influence upon Greece, as far as regards this art, has been overrated. Not only have the more perfect Egyptian instruments—such as the larger harps, the tamboura—never been much in favour with the Greeks, but almost all the stringed instruments which the Greeks possessed are stated to have been originally derived from Asia. Strabo says: “Those who regard the whole of Asia, as far as India, as consecrated to Bacchus, point to that country as the origin of a great portion of the present music. One author speaks of ‘striking forcibly the Asiatic kithara,’ another calls the pipes Berecynthian and Phrygian. Some of the instruments also have foreign names, as Nabras, Sambyke, Barbitos, Magadis, and many others.”

We know at present little more of these instruments than that they were in use in Greece. The Magadis is described as having twenty strings. The other three are known to have been stringed instruments. But they cannot have been anything like such universal favourites as the lyre, because this instrument and perhaps the *trigonon* are almost the only stringed instruments represented in the Greek paintings on pottery and other

monumental records. If, as might perhaps be suggested, their taste for beauty of form induced the Greeks to represent the elegant lyre in preference to other stringed instruments, we might at least expect to meet with the harp; an instrument which equals if it does not surpass the lyre in elegance of form.

The representation of a Muse with a harp, depicted on a splendid Greek vase now in the Munich Museum (*Mun. Vase Cat. No. 805*), may be noted as an exceptional instance. This valuable relic dates from the end of the fifth century B.C. The instrument resembles in construction as well as in shape the Assyrian harp, and has fifteen strings. The Muse is touching them with both hands, using the right hand for the treble and the left for the bass. She is seated, holding the instrument in her lap. The little tuning-pegs, which in number are not in accordance with the strings, are placed on the sound-board at the upper part of the frame, exactly as on the Assyrian harp. If we have here the Greek harp, it was more likely an importation from Asia than from Egypt. In short, as far as can be ascertained, the most complete of the Greek instruments appear to be of Asiatic origin. Especially from the nations who inhabited Asia Minor the Greeks are stated to have adopted several of the most popular. Thus we may read of the short and shrill-sounding pipes of the Carians; of the Phrygian pastoral flute; of the three-stringed *kithara* of the Lydians; and so on.

The Greeks had lyres of various kinds, more or less differing in construction, form, and size, and distinguished by different names; such as *lyra*, *kithara*, *chelys*, *phorminx*, etc. *Lyra* appears to have implied instruments of this class in general, and also the lyre with a body oval at the base and held in the arms of the performer; while the *kithara* had a square base and was held against the side by a sash around it. The *chelys* was a small lyre with the body made of the shell of a tortoise, or of wood in imitation of the tortoise. The *phorminx* was a large lyre, and, like the *kithara*, was used at an early period singly, for accompanying recitations. It is recorded that the *kithara* was employed for solo performances as early as B.C. 700.



**FIG. 6.—A Muse with a HARP, and two others with LYRES.
From a Greek vase in the Munich Museum.**

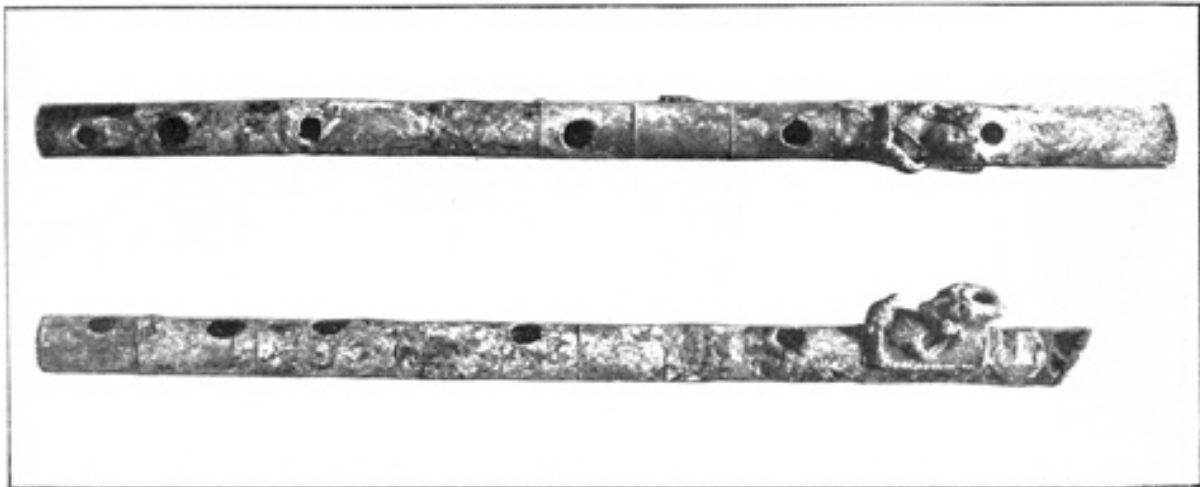
The design on the Greek vase at Munich (already alluded to) represents the nine Muses, of whom three are given in the engraving ([Fig. 6](#)), viz., one with the harp, and two others with lyres. Some of the lyres were provided with a bridge, while others were without it. The largest was held probably on or between the knees, or were attached to the left arm by means of a band, to enable the performer to use his hands without impediment. The strings, made of catgut or sinew, were more usually twanged with a *plektron* than merely with the fingers. The *plektron* was a short stem of ivory or metal pointed at both ends.

A fragment of a Greek lyre which was found in a tomb near Athens is deposited in the British Museum. The two pieces constituting its frame are of wood. Their length is about 18 inches, and the length of the cross-bar at

the top is about 9 inches. The instrument is unhappily in a condition too dilapidated and imperfect to be of any essential use to the musical inquirer.

The *trigonon* consisted originally of an angular frame, to which the strings were affixed. In the course of time a third bar was added to resist the tension of the strings, and its triangular frame resembled in shape the Greek delta. Subsequently it was still further improved, the upper bar of the frame being made slightly curved, whereby the instrument obtained greater strength and more elegance of form.

The *magadis*, also called *pektis*, had twenty strings which were tuned in octaves, and therefore produced only ten tones. It appears to have been some sort of dulcimer, but information respecting its construction is still wanting. There appears to have been also a kind of bagpipe in use called *magadis*, of which nothing certain is known. Possibly, the same name may have been applied to two different instruments.



**FIG. 7.—PAIR OF BRONZE FLUTES, with mouthpiece in the form of the bust of a Mænad holding a bunch of grapes. Greek.
British Museum.**

The *barbitos* was likewise a stringed instrument of this kind. The *sambyke* is traditionally said to have been invented by Ibykos, about 560 B.C. The *simikon* had thirty-five strings, and derived its name from its inventor, Simos, who lived about 600 B.C. It was perhaps a kind of

dulcimer. The *nabla* had ten, or according to Josephus, twelve strings, and probably resembled the *nevel* of the Hebrews, of which but little is known with certainty. The *pandoura* is supposed to have been a kind of lute with three strings. Several of the instruments just noticed were used in Greece, chiefly by musicians who had immigrated from Asia; they can therefore hardly be considered as national musical instruments of the Greeks. The *monochord* had (as its name implies) only a single string, and was used as a tuning string.

The *aulos*, of which there were many varieties, was a highly popular instrument, and differed in construction from the flutes and pipes of the ancient Egyptians. Instead of being blown through a hole at the side near the top it was held like a flageolet, and a vibrating reed was inserted into the mouth-piece, so that it might be more properly described as a kind of oboe or clarinet. The Greeks were accustomed to designate by the name of *aulos* all wind instruments of the flute and oboe kind, some of which were constructed like the flageolet or like our antiquated *flûte à bec*. The single flute was called *monaulos* (Fig. 7), and the double one *diaulos* (Fig. 8). A *diaulos*, which was found in a tomb at Athens, is in the British Museum. The wood of which it is made seems to be cedar, and the tubes are fifteen inches in length. Each tube has a separate mouth-piece and six finger-holes, five of which are at the upper side and one is underneath.



FIG. 8.—A Muse playing the DIAULOS.

The *syrinx*, or Pandean pipe, had from three to nine tubes, but seven was the usual number. The straight trumpet, *salpinx*, and the curved horn, *keras*, made of brass, were used exclusively in war. The small hand-drum, called *tympanon*, resembled in shape our tambourine, and was covered with parchment at the back as well as at the front. The *kymbala* were made of

metal, and resembled our small cymbals. The *krotala* were almost identical with our castanets, and were made of wood or metal.

THE ETRUSCANS AND ROMANS.

The Romans are recorded to have derived some of their most popular instruments originally from the Etruscans, a people which at an early period excelled all other Italian nations in the cultivation of the arts as well as in social refinement, and which possessed musical instruments similar to those of the Greeks. It must, however, be remembered that many of the vases and other specimens of art which have been found in Etruscan tombs, and on which delineations of lyres and other instruments occur, are supposed to be productions of Greek artists whose works were obtained from Greece by the Etruscans, or who were induced to settle in Etruria.

The flutes of the Etruscans were not unfrequently made of ivory; those used in religious sacrifices were of box-wood, of a species of the lotus, of ass' bone, bronze and silver. A bronze flute, somewhat resembling our flageolet, has been found in a tomb; likewise a huge trumpet of bronze. An Etruscan *cornu* is deposited in the British Museum, and measures about four feet in length.

To the Etruscans is also attributed by some the invention of the hydraulic organ. The Greeks possessed a somewhat similar contrivance which they called *hydraulis*, i.e., water-flute and which probably was identical with the *organum hydraulicum* of the Romans. The instrument ought more properly to be regarded as a pneumatic organ, for the sound was produced by the current of air through the pipes; the water applied serving merely to give the necessary pressure to the bellows and to regulate their action. The pipes were probably caused to sound by means of stops, perhaps resembling those on our organ, which were drawn out or pushed in. The construction was evidently but a primitive contrivance, contained in a case which could be carried by one or two persons and which was placed on a table. The highest degree of perfection which the hydraulic organ obtained with the ancients is perhaps shown in a representation on a coin of the Emperor Nero, in the British Museum. Only ten pipes are given to it, and there is no indication of any keyboard, which would probably have been shown had it existed. The

man standing at the side and holding a laurel leaf in his hand is surmised to represent a victor in the exhibitions of the circus or the amphitheatre. The hydraulic organ probably was played on such occasions; and the medal containing an impression of it may have been bestowed upon the victor.

During the time of the Republic, and especially subsequently under the reign of the Emperors, the Romans adopted many new instruments from Greece, Egypt, and even from western Asia; without essentially improving any of their importations.

Their most favourite stringed instrument was the lyre, of which they had various kinds, called, according to their form and arrangement of strings, *lyra*, *cithara*, *chelys*, *testudo*, and *fidis* (or *fides*). The name *cornu* was given to the lyre when the sides of the frame terminated at the top in the shape of two horns. The *barbitos* was a kind of lyre with a large body, which gave the instrument somewhat the shape of the Welsh *crwth*. The *psalterium* was a kind of lyre of an oblong square shape. Like most of the Roman lyres, it was played with a rather large plectrum. The *trigonum* was the same as the Greek *trigonon*. It is recorded that a certain musician of the name of Alexander Alexandrinus was so admirable a performer upon it that when exhibiting his skill in Rome he created the greatest *furor*. Less common, and derived from Asia, were the *sambuca* and *nablia*, the exact construction of which is unknown.

The flute, *tibia*, was originally made of the shin bone, and had a mouth-hole and four finger-holes. Its shape was retained even when, at a later period, it was constructed of other substances than bone. The *tibia gingrina* consisted of a long and thin tube of reed with a mouth-hole at the side of one end. The *tibia obliqua* and *tibia vasca* were provided with mouth-pieces affixed at a right angle to the tube; a contrivance somewhat similar to that on our bassoon. The *tibia longa* was especially used in religious worship. The *tibia curva* was curved at its broadest end. The *tibia ligula* appears to have resembled our flageolet. The *calamus* was nothing more than a simple pipe cut off the kind of reed which the ancients used as a pen for writing.

The Romans had double flutes as well as single flutes. The double flute consisted of two tubes united, either so as to have a mouth-piece in common or to have each a separate mouth-piece. If the tubes were exactly

alike the double flute was called *tibiae pares*; if they were different from each other, *tibiae impares*. Little plugs, or stoppers, were inserted into the finger-holes to regulate the order of intervals. The *tibia* was made in various shapes. The *tibia dextra* was usually constructed of the upper and thinner part of a reed; and the *tibia sinistra*, of the lower and broader part. The performers used also the *capistrum*,—a bandage round the cheeks identical with the *phorbeia* of the Greeks.



**FIG. 9.—WALL PAINTING of a youth wearing a myrtle wreath and playing on the Double Pipes. Restored in places. Said to have been found in a columbarium in the Vigna Ammendola on the Appian Way near Rome, about 1823.
British Museum.**

The British Museum contains a wall painting ([Fig. 9](#)) representing a Roman youth playing the double pipes, which is stated to have been

disinterred in the year 1823 on the Via Appia. Here the *holmos* or mouth-piece, somewhat resembling the reed of our oboe, is distinctly shown. The finger-holes, probably four, are not indicated, although they undoubtedly existed on the instrument.

Furthermore, the Romans had two kinds of Pandean pipes viz., the *syrinx* and the *fistula*. The bagpipe, *tibia utricularis*, is said to have been a favourite instrument of the Emperor Nero.

The *cornu* was a large horn of bronze, curved. The performer held it under his arm with the broad end upwards over his shoulder. It is represented in the engraving (Fig. 10), with the *tuba* and the *lituus*.

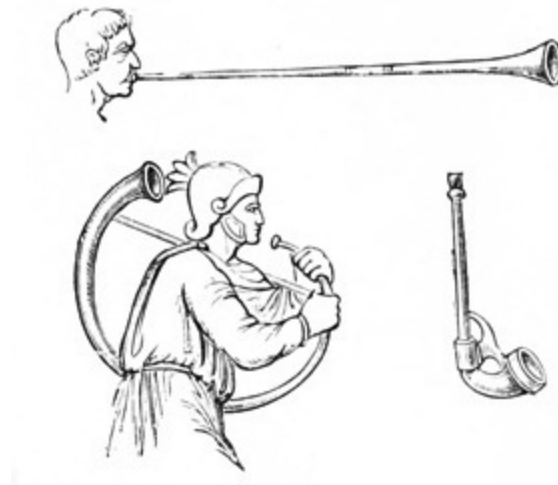


FIG. 10.—TUBA CORNU and LITUUS.

The *tuba* was a straight trumpet. Both the *cornu* and the *tuba* were employed in war to convey signals. The same was the case with the *buccina*,—originally perhaps a conch shell, and afterwards a simple horn of an animal,—and the *lituus*, which was bent at the broad end but otherwise straight. The *tympanum* resembled the tambourine, and was beaten like the latter with the hands. Among the Roman instruments of percussion the *scabellum*, which consisted of two plates combined by means of a sort of hinge, deserves to be noticed; it was fastened under the foot and trodden in time, to produce certain rhythmical effects in musical performances. The *cymbalum* consisted of two metal plates similar to our cymbals. The *crotala*

and the *crusmata* were kinds of castanets, the former being oblong and of a larger size than the latter. The Romans had also a *triangulum*, which resembled the triangle occasionally used in our orchestra. The *sistrum* they derived from Egypt with the introduction of the worship of Isis. Metal bells, arranged according to a regular order of intervals and placed in a frame, were called *tintinnabula*. The *crepitaculum* appears to have been a somewhat similar contrivance on a hoop with a handle.

Through the Greeks and Romans we have the first well-authenticated proof of musical instruments having been introduced into Europe from Asia. The Romans in their conquests undoubtedly made their musical instruments known, to some extent, also in western Europe. But the Greeks and Romans are not the only nations which introduced Eastern instruments into Europe. The Phœnicians at an early period colonized Sardinia, and traces of them are still to be found on that island. Among these is a peculiarly constructed double-pipe, called *lionedda* or *launedda*. Again, at a much later period the Arabs introduced several of their instruments into Spain, from which country they became known in France, Germany, and England. Also the crusaders, during the eleventh and twelfth centuries, may have helped to familiarize the western European nations with instruments of the East.

V.

ORIENTAL.

THE CHINESE.

ALLOWING for any exaggeration as to chronology, natural to the lively imagination of Asiatics, there is no reason to doubt that the Chinese possessed long before our Christian era musical instruments to which they attribute a fabulously high antiquity. There is an ancient tradition, according to which they obtained their musical scale from a miraculous bird, called *fêng-huang*, which appears to have been a sort of phoenix. When Confucius, who lived about B.C. 551-479, happened to hear on a certain occasion some Chinese music, he is said to have become so greatly enraptured that he could not take any food for three months afterwards. The sounds which produced this effect were those of K'uei, the Orpheus of the Chinese, whose performance on the *ch'ing*—a kind of harmonicon constructed of slabs of sonorous stone—would draw wild animals around him and make them subservient to his will. As regards the invention of musical instruments the Chinese have other traditions. In one of these we are told that the origin of some of their most popular instruments dates from the period when China was under the dominion of heavenly spirits, called Ch'i. Another assigns the invention of several stringed instruments to the great Fu-hsi who was the founder of the empire and who lived about B.C. 3000, which was long after the dominion of the Ch'i, or spirits. Again, another tradition holds that the most important instruments and systematic arrangements of sounds are an invention of Nü-wa, sister and successor of Fu-hsi.

According to their records, the Chinese possessed their much-esteemed *ch'ing* 2200 years before our Christian era, and employed it for

accompanying songs of praise. It was regarded as a sacred instrument. During religious observances at the solemn moment when the *ch'ing* was sounded sticks of incense were burnt. It was likewise played before the emperor early in the morning when he awoke. The Chinese have long since constructed various kinds of the *ch'ing*, by using different species of stones. Their most famous stone selected for this purpose is called *yü*. *Yü* includes the two varieties of jade, nephrite and jadeite. It is not only very sonorous but also beautiful in appearance. It is found in mountain streams and crevices of rocks. The largest known specimens measure from two to three feet in diameter, but examples of this size rarely occur. The *yü* is very hard and heavy. Some European mineralogists, to whom the missionaries transmitted specimens for examination, pronounce it to be a species of agate (*ma-nao*). It is found of different colours, and the Chinese appear to have preferred in different centuries particular colours for the *ch'ing*.

The Chinese consider the *yü* especially valuable for musical purposes, because it always retains exactly the same pitch. All other musical instruments, they say, are in this respect doubtful; but the tone of the *yü* is influenced neither by cold nor heat, nor by humidity, nor dryness.

The stones used for the *ch'ing* have been cut from time to time in various grotesque shapes. Some represent animals: as, for instance, a bat with outstretched wings; or two fishes placed side by side: others are in the shape of an ancient Chinese bell. The angular shape appears to be the oldest form and is still retained in the ornamental stones of the *pien-ch'ing*, which is a more modern instrument than the *ch'ing*. The tones of the *pien-ch'ing* are attuned according to the Chinese intervals called *lü*, of which there are twelve in the compass of an octave. The same is the case with the other Chinese instruments of this class. They vary, however, in pitch. The pitch of the *sung-ch'ing*, for instance, is four intervals lower than that of the *pien-ch'ing*.

Sonorous stones have always been used by the Chinese also singly, as rhythmical instruments. Such a single stone is called *t'ê-ch'ing*.

The ancient Chinese had several kinds of bells, frequently arranged in sets so as to constitute a musical scale. The Chinese name for the bell is *chung*. At an early period they had a somewhat square-shaped bell called

t'ê-chung. Like other ancient Chinese bells it was made of copper alloyed with tin, the proportion being one part of tin to six of copper. The *t'ê-chung*, which is also known by the name of *piao*, was principally used to indicate the time and divisions in musical performances. It had a fixed pitch of sound, and several of these bells attuned to a certain order of intervals were not unfrequently ranged in a regular succession, thus forming a musical instrument which was called *pien-chung*. The musical scale of the sixteen bells which the *pien-chung* contained was the same as that of the *ch'ing* before mentioned.

The *hsüan-chung* was, according to popular tradition, included with the antique instruments at the time of Confucius, and came into popular use during the Han dynasty (from B.C. 200 until A.D. 200). It was of a peculiar oval shape and had nearly the same quaint ornamentation as the *t'ê-chung*; this consisted of symbolical figures, in four divisions, each containing nine mammals. The mouth was crescent-shaped. Every figure had a deep meaning referring to the seasons and to the mysteries of the Buddhist religion. The largest *hsüan-chung* was about twenty inches in length; and, like the *t'ê-chung*, was sounded by means of a small wooden mallet with an oval knob. None of the bells of this description had a clapper. It would, however, appear that the Chinese had at an early period some kind of bell provided with a wooden tongue: this was used for military purposes as well as for calling the people together when an imperial messenger promulgated his sovereign's commands. An expression of Confucius is recorded to the effect that he wished to be "A wooden-tongued bell of Heaven," *i.e.*, a herald of heaven to proclaim the divine purposes to the multitude.

The *fang-hsiang* was a kind of wood-harmonicon. It contained sixteen wooden slabs of an oblong square shape, suspended in a wooden frame elegantly decorated. The slabs were arranged in two tiers, one above the other, and were all of equal length and breadth but differed in thickness. The *ch'un-tu* consisted of twelve slips of bamboo, and was used for beating time and for rhythmical purposes. The slips being banded together at one end could be expanded somewhat like a fan. The Chinese state that they used the *ch'un-tu* for writing upon before they invented paper.

The *yü*, likewise an ancient Chinese instrument of percussion and still in use, is made of wood in the shape of a crouching tiger. It is hollow, and

along its back are about twenty small pieces of metal, pointed, and in appearance not unlike the teeth of a saw. The performer strikes them with a sort of plectrum resembling a brush, or with a small stick called *chên*. Occasionally the *yü* is made with pieces of metal shaped like reeds.

The ancient *yü* was constructed with only six tones which were attuned thus—*f*, *g*, *a*, *c*, *d*, *f*. The instrument appears to have deteriorated in the course of time; for, although it has gradually acquired as many as twenty-seven pieces of metal, it evidently serves at the present day more for the production of rhythmical noise than for the execution of any melody. The modern *yü* is made of a species of wood called *k'iu* or *ch'iu*; and the tiger rests generally on a hollow wooden pedestal about three feet six inches long, which serves as a sound-board.

The *chu*, likewise an instrument of percussion, was made of the wood of a tree called *ch'iu-mu*, the stem of which resembles that of the pine and whose foliage is much like that of the cypress. It was constructed of boards about three-quarters of an inch in thickness. In the middle of one of the sides was an aperture into which the hand was passed for the purpose of holding the handle of a wooden hammer, the end of which entered into a hole situated in the bottom of the *chu*. The handle was kept in its place by means of a wooden pin, on which it moved right and left when the instrument was struck with a hammer. The Chinese ascribe to the *chu* a very high antiquity, as they almost invariably do with any of their inventions when the date of its origin is unknown to them.

The *po-fu* was a drum, about one foot four inches in length, and seven inches in diameter. It had a parchment at each end, which was prepared in a peculiar way by being boiled in water. The *po-fu* used to be partly filled with a preparation made from the husk of rice, in order to mellow the sound. The Chinese name for the drum is *ku*.

The *chin-ku*, a large drum fixed on a pedestal which raises it above six feet from the ground, is embellished with symbolical designs. A similar drum on which natural phenomena are depicted is called *lei-ku*; and another of the kind, with figures of certain birds and beasts which are regarded as symbols of long life, is called *ying-ku*, and also *tsu-ku*.

The flutes, *ti*, *yüeh*, and *ch'ih* were generally made of bamboo. The *kuan-tzũ* was a Pandean pipe containing twelve tubes of bamboo. The *hsiao*, likewise a Pandean pipe, contained sixteen tubes. The *p'ai-hsiao* differed from the *hsiao* inasmuch as the tubes were inserted into an oddly-shaped case highly ornamented with grotesque designs and silken appendages.

The Chinese are known to have constructed at an early period a curious wind-instrument, called *hsüan* (the "Chinese ocarina") (Fig. 11). It was made of baked clay and had five finger-holes, three of which were placed on one side and two on the opposite side, as in the cut. Its tones were in conformity with the pentatonic scale. The reader unacquainted with the pentatonic scale may ascertain its character by playing on the pianoforte the scale of C major with the omission of *f* and *b* (the *fourth* and *seventh*); or by striking the black keys in regular succession from *f*-sharp to the next *f*-sharp above or below.

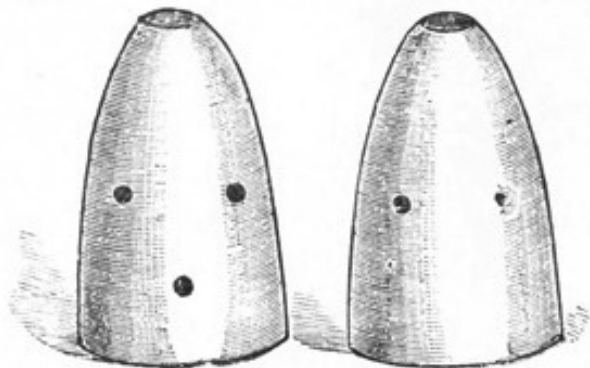
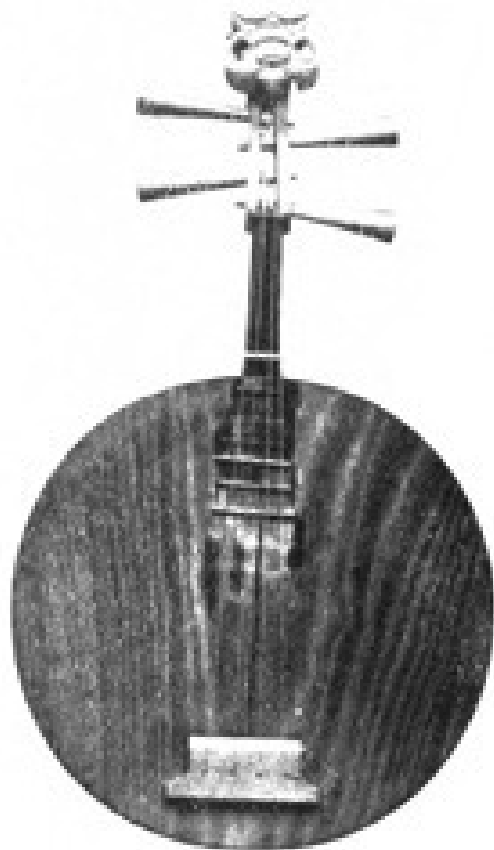
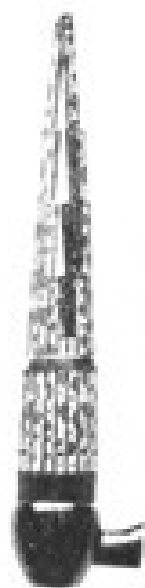
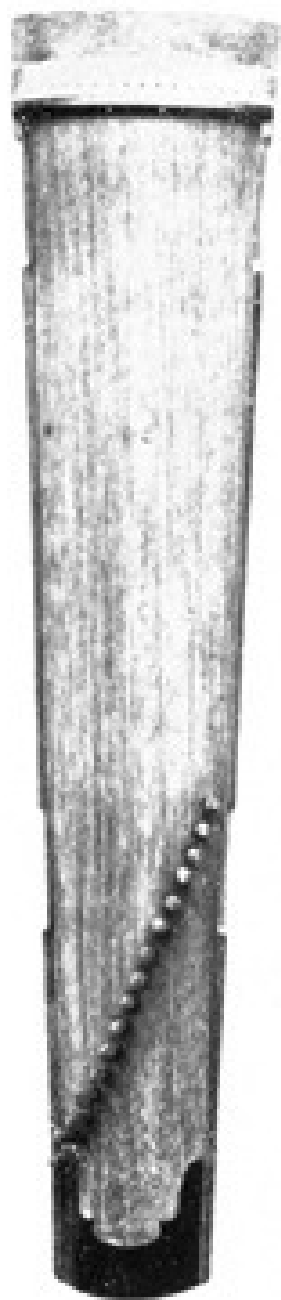


FIG. 11.—HSÜAN.

The *shêng* (Fig. 12*b*) is one of the oldest instruments of the Chinese still in use, and may be regarded as the most ancient species of organ with which we are exactly acquainted. Formerly it had either thirteen, nineteen, or twenty-four tubes placed in a calabash; and a long curved tube served as a mouth-piece. A similarly-constructed instrument, though different in outward appearance, is the *ken* of Siam and Burmah. The Siamese call the *ken* "The Laos organ," and it is principally used by the inhabitants of the Laos states. Moreover, there deserves to be noticed another Chinese instrument of this kind, simple in construction, which probably represents

the *shêng* in its most primitive condition. It is to be found among the Miao-tsze, or mountaineers, who are supposed to be the aboriginal inhabitants of China. They call it *sang*. This species has no bowl, or air-chest; it rather resembles the Panpipe, but is sounded by means of a common mouthpiece consisting of a tube, which is placed at a right angle across the pipes. The Chinese assert that the *shêng* was used in olden time in the religious rites performed in honour of Confucius. Tradescant Lay, in his account of the Chinese, calls it "Jubal's organ," and remarks, "this seems to be the embryo of our multiform and magnificent organ."



- FIG. 12.—a. CH'IN (a species of Lute). Modern Chinese.**
No. 9-'70. L. 38½ in., W. 8½ in.
b. SHÊNG (Mouth Organ). Chinese, 19th century. No. 977-'72.
L. 17 in., W. 4¼ in.
c. YUEH-CH'IN (Moon Guitar). Chinese. 19th Century.
No. 256-'82.
Victoria and Albert Museum.

The ancient stringed instruments, the *ch'in* (Fig. 12a) and *sê*, were of the dulcimer kind, they are still in use, and specimens of them are in the Museum.

The *yueh-ch'in* (Fig. 12c) is a favourite instrument of the Chinese. The Canton pronunciation of *yueh-ch'in* is *yuet-kum*, and this may be the reason why some European travellers in China have called the instrument *gut-komm*. The wood of which it is made is called by the Chinese *shwan-che*. The strings are twanged with a plectrum, or with the nails, which, it will be remembered, are grown by the Chinese to an extravagant length.

The Buddhists introduced from Tibet into China their god of music, who is represented as a rather jovial-looking man with a moustache and an imperial, playing the *p'i-p'a*, a kind of lute with four silken strings. Perhaps some interesting information respecting the ancient Chinese musical instruments may be gathered from the famous ruins of the Buddhist temples *Angcor-Wat* and *Angcor-Thom*, in Cambodia. These splendid ruins are supposed to be above two thousand years old: and, at any rate, the circumstance of their age not being known to the Cambodians suggests a high antiquity. On the bas-reliefs with which the temples were enriched are figured musical instruments, which European travellers describe as “flutes, organs, trumpets, and drums, resembling those of the Chinese.” Faithful sketches of these representations, might, very likely, afford valuable hints to the student of musical history.

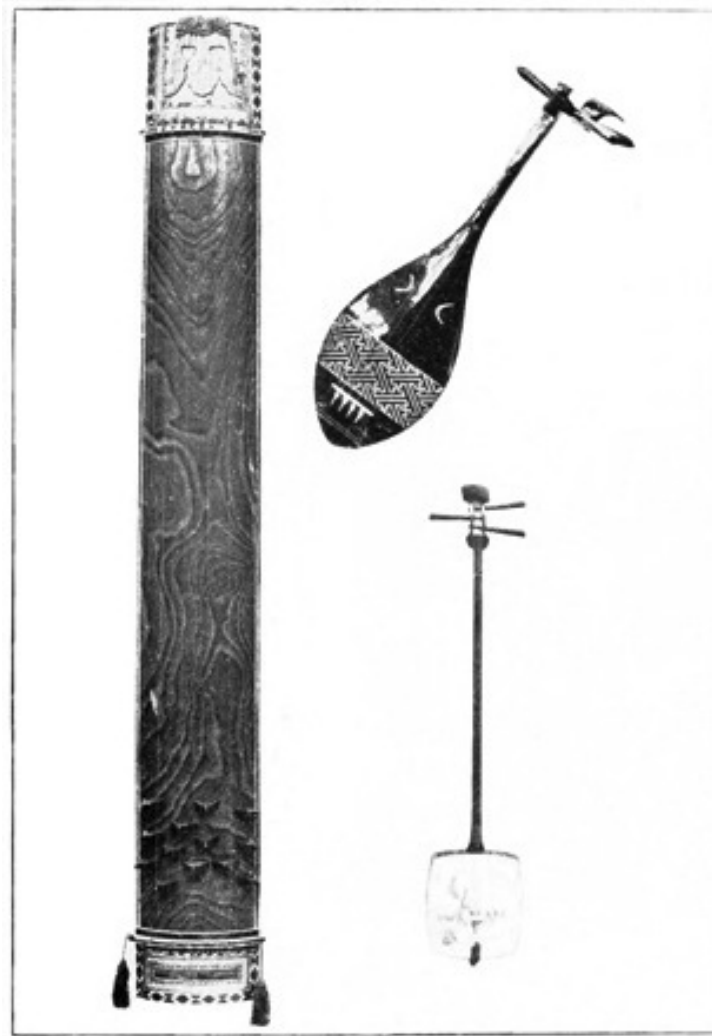
THE JAPANESE.

The Japanese musical instruments are in the main derived from those of China, and their names consequently represent the Japanese pronunciation of the Chinese sounds.

The *biwa* (Fig. 13b) is almost identical with the Chinese *p'i-p'a*. The example illustrated is of wood, lacquered black and ornamented with a band of Japanese design in gold lacquer. It has four silken strings, and two very small sound holes.

The *samisen* (the Chinese *san-hsien* or "three-stringed guitar") is played especially by the Japanese ladies, and is as great a favourite with them as the lute was formerly with us. An example in the Museum (Fig. 13c) has three strings of silk. Both the *biwa* and the *samisen* are played with a wooden plectrum. The *ko-kiū* is the Japanese violin, and resembles a small *samisen*, but has four strings. It is held head upwards and played with a loose-strung bow.

The Japanese have several instruments of the dulcimer class, called *koto* (the Chinese *ch'in*) (Fig. 13a). Some species of the *koto* are played with *plectra* affixed to the fingers; and there are different successions of intervals adopted in the tuning of the several species.



**FIG. 13.—a. KOTO (a species of Lute).
Japanese. 19th century.**

L. 75 $\frac{3}{8}$ in., W. 9 $\frac{1}{2}$ in. No. 439-'91.

b. BIWA (a species of Guitar).

Modern Japanese.

H. 32 $\frac{1}{2}$ in., diam. 11 in. No. 838-'6c.

c. SAMISEN. Japanese.

L. 37 $\frac{1}{2}$ in. No. 229-'82.

Victoria and Albert Museum.

The *ikuta-goto* is provided with thirteen movable bridges, by means of which the pitch of the strings is regulated. The bridges are of wood, and about 2 $\frac{1}{2}$ inches in height. The *ikuta-goto* is learnt chiefly by Japanese ladies moving in the upper circles of society. It is a rather expensive

instrument, and requires much practice. The performer places it on the floor, and, sitting in the usual Japanese attitude, bends over it and twangs the strings with her fingers, the tips of which are encased in *plectra*, resembling thimbles, which terminate in a little projecting piece of ivory in size and form like the finger nail.

Of wind instruments the Japanese use three principal kinds:—(1) The *fuye*, like our flute, with six or seven finger-holes; (2) the *hichiriki*, a reed-flageolet, with seven finger-holes and two thumb-holes; (3) the *shakuhachi*, a bamboo pipe 20 inches high.

The *shêng* (described on [p. 42](#)) is also popular in Japan. The Japanese name for it is *shō*. The general name in Japanese for the drum is *taiko* (= Chinese *ta ku*, “large drum”). The Japanese have a great variety of drums, some of which are used at religious ceremonies in the temples. The *shime-daiko* is a shallow drum hung obliquely before the player in a low wooden frame. It is beaten with two plain sticks, and is used to accompany singers. The *tsudzumi* is a small hand-drum with hour-glass-shaped body.

The Japanese have different kinds of gongs (*dora* = Chinese *t’ung-lo*, “copper gong”), which are used in the service of the temple, in processions, at funerals, and on several other solemn occasions. The *dōhachi* (= Chinese *t’ung po*, “copper bowl”) resembles a copper basin. Another consists of two metal basins suspended by cords on a frame composed of a pole and two cross-sticks.

The Japanese, as well as the Chinese, possess superbly ornamented gongs (*kei*) raised on a stand. Those of the former are perhaps the more magnificent.

The Japanese employ large bells (*kane* or *tsuri-gane* = Chinese *chung*) in their Buddhist worship. There is a famous bell, richly decorated, near the Daibutsu at Kiōto, which is struck, at different hours of the day, with a heavy wooden mallet; and its sound is said to be particularly sonorous, mellow, and far-reaching. Another celebrated Japanese bell is placed on a high hill near the town of Nara. It is suspended in a wooden shed, close to the Tōdaiji Temple. A thick pole, affixed to the rafters, is drawn backwards, and then, by being let loose, is made to rebound so as to hit the bell sideways in the usual manner. This bell is admired throughout the country,

and pictures representing it are sold on the spot to the visitors, who have to ascend a long flight of narrow steps before they reach its station on the summit of the hill. Small bells (*rin*) are used by the Buddhist priests in Japan while officiating in the temple, just as is the case in China, Thibet and other districts of the Asiatic continent.

THE HINDUS.

In the Brahmin mythology of the Hindus the demi-god Nareda is the inventor of the *vina*, the principal national instrument of Hindustan. His mother, Saraswati, the consort of Brahma, may be regarded as the Minerva of the Hindus. She is the goddess of music as well as of speech. To her is attributed the invention of the systematic arrangement of the sounds into a musical scale. She is represented seated on a peacock and playing either on the southern *vina* or the *bîn*, stringed instruments of the lute kind. Brahma himself we occasionally find depicted as a vigorous man with four handsome heads, beating with his hands upon a small drum; and Vishnu, in his incarnation as Krishna, is represented as a beautiful youth playing upon a flute. The Hindus construct a peculiar kind of flute, the *bansi*, which they consider as the favourite instrument of Krishna.

The *sankha*, or conch-shell trumpet of victory, one of the important attributes of Vishnu the preserver, and his consort Lakshmi, is occasionally represented in the possession of Siva, and other deities. Siva the destroyer, and his consort Parvati, also carry the *budbudika*, or *damaru*, a rattle-drum shaped like an hour-glass.

It is a suggestive fact that we find among several nations in different parts of the world an ancient tradition, according to which their most popular stringed instrument was originally derived from the water. Thus with Nareda and the *vina*, the latter has also the name *kach'-hapi*, signifying a tortoise (*testudo*), whilst *nara* denotes in Sanskrit water, and *narada*, or *nareda*, the giver of water. Like Nareda, Nereus and his fifty daughters, the Nereïdes, were much renowned for their musical accomplishments; and Hermes (it will be remembered) made his lyre, the *chelys*, of a tortoise-shell. The Scandinavian god Odin, the originator of magic songs, is mentioned as the ruler of the sea, and as such he had the

name of *Nikarr*. In the depth of the sea he played the harp with his subordinate spirits, who occasionally came up to the surface of the water to teach some favoured human being their wonderful instrument. Wäinämöinen, the divine player on the Finnish *kantele* (according to the Kalewala, the old national epic of the Finns) constructed his instrument of fish-bones. The frame he made out of the bones of the pike; and the teeth of the pike he used for the tuning-pegs.

Jacob Grimm in his work on German mythology points out an old tradition, preserved in Swedish and Scotch national ballads, of a skilful harper who constructs his instrument out of the bones of a young girl drowned by a wicked woman. Her fingers he uses for the tuning screws, and her golden hair for the strings. The harper plays, and his music kills the murderess. A similar story is told in the old Icelandic national songs; and the same tradition has been preserved in the Faroe islands, as well as in Norway and Denmark.

May not the agreeable impression produced by the rhythmical flow of the waves and the soothing murmur of running water have led various nations, independently of each other, to the widespread conception that they obtained their favourite instrument of music from the water? Or is the notion traceable to a common source dating from a pre-historic age, perhaps from the early period when the Aryan race is surmised to have diffused its lore through various countries? Or did it originate in the old belief that the world, with all its charms and delights, arose from a chaos in which water constituted the predominant element?

Howbeit, Nareda, the giver of water, was the offspring of Brahma the creator; and Odin had his throne in the skies. Indeed, many of the musical water-spirits appear to have been originally considered as rain deities. Their music may, therefore, be regarded as derived from the clouds rather than from the sea. In short, the traditions respecting spirits and water are not in contradiction to the opinion of the ancient Hindus that music is of heavenly origin, but rather tend to support it.

The earliest musical instruments of the Hindus on record have, almost all of them, remained in popular use until the present day scarcely altered. Besides these, the Hindus possess several Arabic and Persian instruments

which are of comparatively modern date in Hindustan: evidently having been introduced into that country scarcely 1,000 years ago, at the time of the Muhammadan irruption. There are several treatises on music extant, written in Sanskrit, which contain descriptions of the ancient instruments.

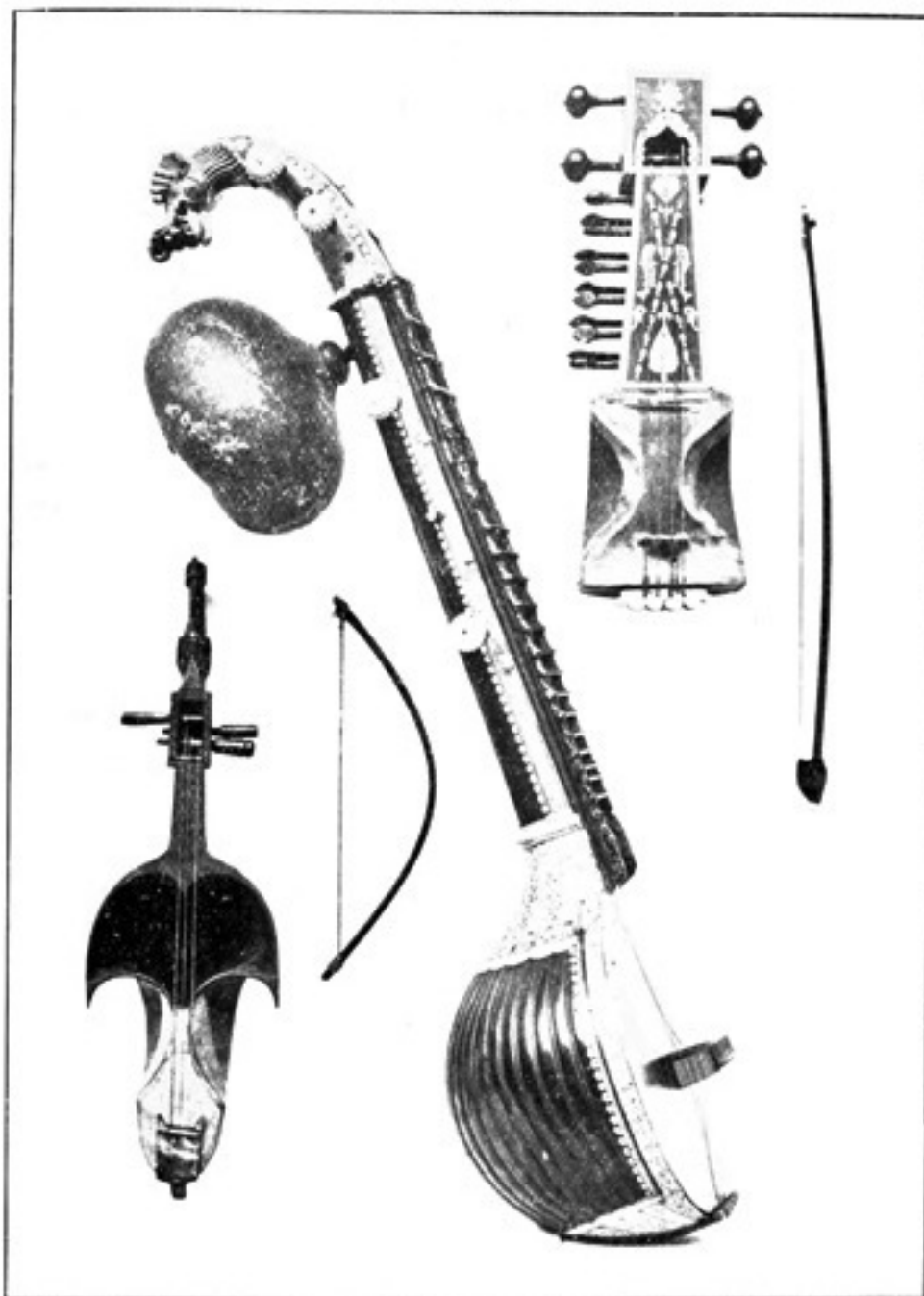


FIG. 14.—a. SÂRINDA AND BOW. Indian (Bengal). 19th century.

L. 25 in.; bow 15¾ in. No. 180. 180^a-’82.
b. RUDRA VINA. Southern Indian (Madras).
19th century. L. 45 in. No. 02130. I.S.
c. SÂRANGI AND BOW. Southern Indian.
19th century. L. 22 in. No. 02118. I.S.
Victoria and Albert Museum.

Of these the Bhârata Nâtya S’astra by Bhârata Muni (period: B.C. 200 to A.D. 100), and the Sangita Ratnâkara, are probably the oldest and most valuable. The latter, according to information supplied by the late Major C. R. Day, is an exhaustive work, consisting of seven âdhyayas, compiled by Sarnga Deva, son of Sotala Deva, King of Karnata, and grandson of Bhaskara, a Kashmirian (period: so far undetermined).

The *vina* is undoubtedly of high antiquity. It has seven wire strings, and movable frets which are generally fastened with wax. Gourds, often tastefully ornamented, are affixed for the purpose of increasing the sonorousness. There are several kinds of the *vina* in different districts.

Concerning the two principal present-day derivations from the ancient *vina*, the following abbreviated descriptions of the *rudra vina* of Southern India and the *bîn* or *mahati vina* of Northern India, are obtained from “The Music and Musical Instruments of Southern India,” by the late Major C. R. Day (London, 1891).

The *rudra vina* (see [Fig. 14b](#)) is composed of a pear-shaped body of thin wood, hollowed out of the solid; wooden belly; four principal metal strings passing over twenty-four frets and three shorter wires placed at the side of the finger-board; also a single detachable *burra*, or hollow gourd, fastened to the under-side of the neck, near the head, to increase the volume of sound. In the method of playing it differs from that of other Indian musical instruments, the left hand being employed to stop the strings on the frets, whilst the fingers, or rather the finger-nails, of the right hand are used, without plectra, for striking. The *bîn*, or *mahali vina*, differs from the *rudra vina* in shape and in method of playing. Two large gourd-resonators replace the wooden body with its small *burra*; the side strings are placed two on the left side and one upon the right; the frets vary from nineteen to twenty-two

in number; and in playing, the two first fingers of the right hand are armed with wire plectra.

The *sârangi*, or the common fiddle of Southern India (Fig. 14c) has a wooden body hollowed out of a single block, a parchment belly, three strings of thick gut, and usually fifteen sympathetic strings of wire, tuned chromatically. Sometimes a fourth principal string of wire, called *luruj*, is added. It is played with a bow, the instrument being held vertically, head uppermost; the tone resembling that of the viola. The *sârangi* of Northern India, usually carved with a conventional swan-shaped head, has a rounded body, and possesses a lesser number of sympathetic wires.

The *sârinda*, or Bengal fiddle (Fig. 14a), another of the few bowed instruments of India, consists of a hollow wooden body, usually decorated with carving, a curious parchment belly covering only the lower half of the body, and three strings either of gut or silk.

The Hindus divided their musical scale into several intervals smaller than our modern semitones. They adopted twenty-two intervals called *s'ruti* in the compass of an octave, which may therefore be compared to our chromatic intervals. As the frets of the *vina* are movable the performer can easily regulate them according to the scale, or mode, which he requires for his music.

The harp has long been obsolete. If some Hindu drawings of it can be relied upon, it had at an early time a triangular frame and was in construction as well as in shape and size almost identical with the Assyrian harp.

The Hindus claim to have invented the violin bow. They maintain that the *ravanastra*, one of their old instruments played with the bow, was invented about 5,000 years ago by Ravana, a mighty king of Ceylon. However this may be, there is a great probability that the fiddle-bow originated in Hindustan; because Sanskrit scholars inform us that there are names for it in works which cannot be less than from 1,500 to 2,000 years old. The non-occurrence of any instrument played with a bow on the monuments of the nations of antiquity is by no means so sure a proof as has generally been supposed, that the bow was unknown. The fiddle in its primitive condition must have been a poor contrivance. It probably was

despised by players who could produce better tones with greater facility by twanging the strings with their fingers, or with a plectrum. Thus it may have remained through many centuries without experiencing any material improvement. It must also be borne in mind that the monuments transmitted to us chiefly represent historical events, religious ceremonies, and royal entertainments. On such occasions instruments of a certain kind only were used, and these we find represented; while others, which may have been even more common, never occur. In 2,000 years' time people will possibly maintain that some highly perfected instrument popular with them was entirely unknown to us, because it is at present in so primitive a condition that no one hardly notices it.

"What the *ravanastra*, or *râbanastra*, was like is rather doubtful, but at the present time there exists in Ceylon a primitive instrument played with a bow, called *vinavah*, which has two strings of different kinds, one made of a species of flax, and the other of horsehair, which is the material also of the string of the bow.... The hollow part of this instrument is half a cocoa-nut shell polished, covered with the dried skin of a lizard, and perforated below." (Day, p. 102.)

This instrument again is almost identical with the Chinese fiddle called *ur-heen*, which also has two strings, and a body consisting of a small block of wood, hollowed out and covered with the skin of a serpent. The *ur-heen* has not been mentioned among the most ancient instruments of the Chinese, since there is no evidence of its having been known in China before the introduction of the Buddhist religion into that country. From indications, which to point out would lead too far here, it would appear that several instruments found in China originated in Hindustan. They seem to have been gradually diffused from Hindustan and Thibet, more or less altered in the course of time, through the East as far as Japan.

Another curious Hindu instrument, probably of very high antiquity, is the *pungi*, or *jinagovi*, also called *toumrie* and *magoudi*. It consists of a gourd or of the *cuddos* nut, hollowed, into which two reed-pipes are inserted. The *pungi* therefore, somewhat resembles in appearance a bagpipe. It is generally used by the *saperá* or snake-charmer, who plays upon it when exhibiting the antics of the cobra. The name *magoudi*, given in certain districts to this instrument, rather tends to corroborate the opinion of some

musical historians that the *magadis* of the ancient Greeks was a sort of double-pipe, or bagpipe.

Many instruments of Hindustan are known by different names in different districts, and there are many varieties. On the whole, the Hindus possess about fifty instruments. To describe them properly would fill a volume. Some, which are in the Museum, will be found well described and illustrated in the previously mentioned work by the late Major C. R. Day, which, in addition to affording much valuable information to the student and collector, contains a lengthy bibliography of Indian music and musical instruments.

THE PERSIANS AND ARABS.

Of the musical instruments of the ancient Persians, before the Christian era, scarcely anything is known. It may be surmised that they closely resembled those of the Assyrians, and probably also those of the Hebrews.

The harp, *chang*, in olden time a favourite instrument of the Persians, has gradually fallen into desuetude. A small harp is represented in the celebrated sculptures which exist on a stupendous rock, called Tak-i-Bostan, in the vicinity of the town of Kermanshah. These sculptures are said to have been executed during the lifetime of the Persian monarch Chosroes II. (591-628). They form the ornaments of two lofty arches, and consist of representations of field sports and aquatic amusements. In one of the boats is seated a man in an ornamental dress, with a halo round his head, who is receiving an arrow from one of his attendants; while a female, who is sitting near him, plays on a Trigonon. Towards the top of the bas-relief is represented a stage, on which are performers on small straight trumpets and little hand drums; six harpers; and four other musicians, apparently females—the first of whom plays a flute; the second, a sort of Pandean pipe; the third, an instrument which is too much defaced to be recognisable; and the fourth, a bagpipe. Two harps of a peculiar shape were copied by Sir Gore Ouseley from Persian manuscripts about four hundred years old, resembling, in the principle on which they are constructed, all other oriental harps. There existed evidently various kinds of the *chang*. It may be remarked here that the instrument *tschenk* (or *chang*) in use at the present day in Persia, is

more like a dulcimer than a harp. The Arabs adopted the harp from the Persians, and called it *junk*.

The Persians appear to have adopted, at an early period, smaller musical intervals than semitones. When the Arabs conquered Persia (A.D. 641) the Persians had already attained a higher degree of civilisation than their conquerors. The latter found in Persia the cultivation of music considerably in advance of their own, and the musical instruments superior also. They soon adopted the Persian instruments, and there can be no doubt that the musical system exhibited by the earliest Arab writers whose works on the theory of music have been preserved was based upon an older system of the Persians. In these works the octave is divided in seventeen *one-third-tones*—intervals which are still made use of in the East. Some of the Arabian instruments are constructed so as to enable the performer to produce the intervals with exactness. The frets on the lute and tamboura, for instance, are regulated with a view to this object.

The Arabs had to some extent become acquainted with many of the Persian instruments before the time of their conquest of Persia. An Arab musician of the name of Nadr Ben el-Hares Ben Kelde is recorded as having been sent to the Persian King Chosroes II., in the sixth century, for the purpose of learning Persian singing and performing on the lute. Through him, it is said, the lute was brought to Mekka. Saib Chatir, the son of a Persian, is spoken of as the first performer on the lute in Medina, A.D. 682; and of an Arab lutist, Ebn Soreidsch from Mekka, A.D. 683, it is especially mentioned that he played in the Persian style; evidently the superior one. The lute, *el-ood*, had before the tenth century only four strings, or four pairs producing four tones, each tone having two strings tuned in unison. About the tenth century a string for a fifth tone was added. The strings were made of silk neatly twisted. The neck of the instrument was provided with frets of string, which were carefully regulated according to the system of seventeen intervals in the compass of an octave before mentioned. Other favourite stringed instruments were the *tamboura*, a kind of lute with a long neck, and the *quanūn*, a kind of dulcimer strung with lamb's gut strings (generally three in unison for each tone) and played upon with two little plectra which the performer had fastened to his fingers. The *quanūn* is likewise still in use in countries inhabited by Muhammadans. The Persian *santir*, the prototype of our dulcimer, is mounted with wire strings and played with two slightly

curved sticks. The musician depicted in the left-hand corner of [Fig. 15c](#) is playing a *santir*.



FIG. 15.—a. KEMÁNGEII, SITÂRA or FIDDLE. Persian.
About 1800. No. 939-'73.
L. 36½ in.; diam. 8 in.
b. NUY (Flute). Persian. 19th century.
L. 17¾ in. No. 959-'86.
c. SANTIR (Dulcimer) CASE. Persian.

**L. 33 in.; W. 11½ in. No. 779-'76.
Victoria and Albert Museum.**

Al-Farabi, one of the earliest Arabian musical theorists known, who lived in the beginning of the tenth century, does not allude to the fiddle-bow. This is noteworthy inasmuch as it seems in some measure to support the opinion maintained by some historians that the bow originated in England or Wales. Unfortunately we possess no exact descriptions of the Persian and Arabian instruments between the tenth and fourteenth centuries, otherwise we should probably have earlier accounts of some instrument of the violin kind in Persia. Ash-shakandi, who lived in Spain about A.D. 1200, mentions the *rabôb*, which may have been in use for centuries without having been thought worthy of notice on account of its rudeness. Persian writers of the fourteenth century speak of two instruments of the violin class, viz., the *rabôb* and the *kemáneh*. As regards the *kemáneh*, the Arabs themselves assert that they obtained it from Persia, and their statement appears all the more worthy of belief from the fact that both names, *rabôb* and *kemáneh*, are originally Persian.

The *nuy*, a flute ([Fig. 15b](#)), and the *surnai*, a species of oboe, are still popular in the East.

The *sitâra* is a Persian three stringed instrument with a wooden body and a parchment belly ([Fig. 15a](#)).

The Arabs must have been indefatigable constructors of musical instruments. Kiesewetter gives a list of above two hundred names of Arabian instruments, and this does not include many known to us through Spanish historians. A careful investigation of the musical instruments of the Arabs during their sojourn in Spain is particularly interesting to the student of mediæval music, inasmuch as it reveals the Eastern origin of many instruments which are generally regarded as European inventions. Introduced into Spain by the Saracens and the Moors they were gradually diffused towards northern Europe. The English, for instance, adopted not only the Moorish dance (morris dance) but also the *kuitra* (gittern), the *el-ood* (lute), the *rabôb* (rebec), the *nakḳârah* (naker), and several others. In an old Cornish sacred drama, supposed to date from the fourteenth century, we have in an enumeration of musical instruments the *nakrys*, designating

“kettle-drums.” It must be remembered that the Cornish language, which has now become obsolete, was nearly akin to the Welsh. Indeed, names of musical instruments derived from the Moors in Spain occur in almost every European language.

Not a few fanciful stories are traditionally preserved among the Arabs testifying to the wonderful effects they ascribed to the power of their instrumental performances. One example will suffice. Al-Farabi had acquired his proficiency in Spain, in one of the schools at Cordova which flourished as early as towards the end of the ninth century, and his reputation became so great that ultimately it extended to Asia. The mighty Caliph of Bagdad himself desired to hear the celebrated musician, and sent messengers to Spain with instructions to offer rich presents to him and to convey him to the court. But Al-Farabi feared that if he went he should be retained in Asia, and should never again see the home to which he felt deeply attached. At last he resolved to disguise himself, and ventured to undertake the journey which promised him a rich harvest. Dressed in a mean costume, he made his appearance at the court just at the time when the caliph was being entertained with his daily concert. Al-Farabi, unknown to everyone, was permitted to exhibit his skill on the lute. Scarcely had he commenced his performance in a certain musical mode when he set all his audience laughing aloud, notwithstanding the efforts of the courtiers to suppress so unbecoming an exhibition of mirth in the royal presence. In truth, even the caliph himself was compelled to burst out into a fit of laughter. Presently the performer changed to another mode, and the effect was that immediately all his hearers began to sigh, and soon tears of sadness replaced the previous tears of mirth. Again he played in another mode, which excited his audience to such a rage that they would have fought each other if he, seeing the danger, had not directly gone over to an appeasing mode. After this wonderful exhibition of his skill Al-Farabi concluded in a mode which had the effect of making his listeners fall into a profound sleep, during which he took his departure.

It will be seen that this incident is almost identical with one recorded as having happened about twelve hundred years earlier at the court of Alexander the Great, and which forms the subject of Dryden’s “Alexander’s Feast.” The distinguished flutist Timotheus successively aroused and

subdued different passions by changing the musical modes during his performance, exactly in the same way as did Al-Farabi.

VI.

AMERICAN INDIAN.

IF the preserved antiquities of the American Indians, dating from a period anterior to our discovery of the western hemisphere, possess an extraordinary interest because they afford trustworthy evidence of the degree of progress which the aborigines had attained in the cultivation of the arts and in their social condition before they came in contact with Europeans, it must be admitted that the ancient musical instruments of the American Indians are also worthy of examination. Several of them are constructed in a manner which, in some degree, reveals the characteristics of the musical system prevalent among the people who used the instruments. And although most of these interesting relics, which have been obtained from tombs and other hiding-places, may not be of great antiquity, it has been satisfactorily ascertained that they are genuine contrivances of the Indians before they were influenced by European civilisation.

Some account of these relics is therefore likely to prove of interest also to the ethnologist, especially as several facts may perhaps be found of assistance in elucidating the still unsolved problem as to the probable original connection of the American with Asiatic races.

Among the instruments of the Aztecs in Mexico and of the Peruvians none have been found so frequently, and have been preserved in their former condition so unaltered, as pipes and flutes. They are generally made of pottery or of bone, substances which are unsuitable for the construction of most other instruments, but which are remarkably well qualified to withstand the decaying influence of time. There is, therefore, no reason to conclude from the frequent occurrence of such instruments that they were

more common than other kinds of which specimens have rarely been discovered.



FIG. 16.—POTTERY WHISTLES. Ancient Mexican. British Museum.

The Mexicans possessed a small whistle formed of baked clay, a considerable number of which have been found. Some specimens ([Fig. 16](#)) are singularly grotesque in shape, representing caricatures of the human face and figure, birds, beasts, and flowers. Some were provided at the top with a finger-hole which, when closed, altered the pitch of the sound, so that two different tones were producible on the instrument. Others had a little ball of baked clay lying loose inside the air-chamber. When the instrument was blown the current of air set the ball in a vibrating motion, thereby causing a shrill and whirring sound. A similar contrivance is

sometimes made use of by Englishmen for conveying signals. The Mexican whistle most likely served principally the same purpose, but it may possibly have been used also in musical entertainments. In the Russian horn band each musician is restricted to a single tone; and similar combinations of performers—only, of course, much more rude—have been witnessed by travellers among some tribes in Africa and America.

Rather more complete than the above specimens are some of the whistles and small pipes which have been found in graves of the Indians of Chiriqui in Central America.

The pipe of the Aztecs, which is called by the Mexican Spaniards *pito*, somewhat resembled our flageolet: the material was a reddish pottery, and it was provided with four finger holes. Although among about half a dozen specimens which the writer has examined some are considerably larger than others, they all have, singularly enough, the same pitch of sound. The smallest is about six inches in length, and the largest about nine inches. Several *pitos* have been found in a remarkably well-preserved condition. They are easy to blow, and their order of intervals is in conformity with the

pentatonic scale, thus:  The usual shape of the *pito* is that here represented (Fig. 17a & c). A specimen of a less common shape, is given in Fig. 17b. They are all in the British Museum. Indications suggestive of the popular estimation in which the flute (or perhaps, more strictly speaking, the pipe) was held by the Aztecs are not wanting. It was played in religious observances, and we find it referred to allegorically in orations delivered on solemn occasions. For instance, at the religious festival which was held in honour of Tezcatlepoca—a divinity depicted as a handsome youth, and considered second only to the supreme being—a young man was sacrificed who, in preparation for the ceremony, had been instructed in the art of playing the flute. Twenty days before his death four young girls, named after the principal goddesses, were given to him as companions; and when the hour arrived in which he was to be sacrificed he observed the established symbolical rite of breaking a flute on each of the steps, as he ascended the temple.

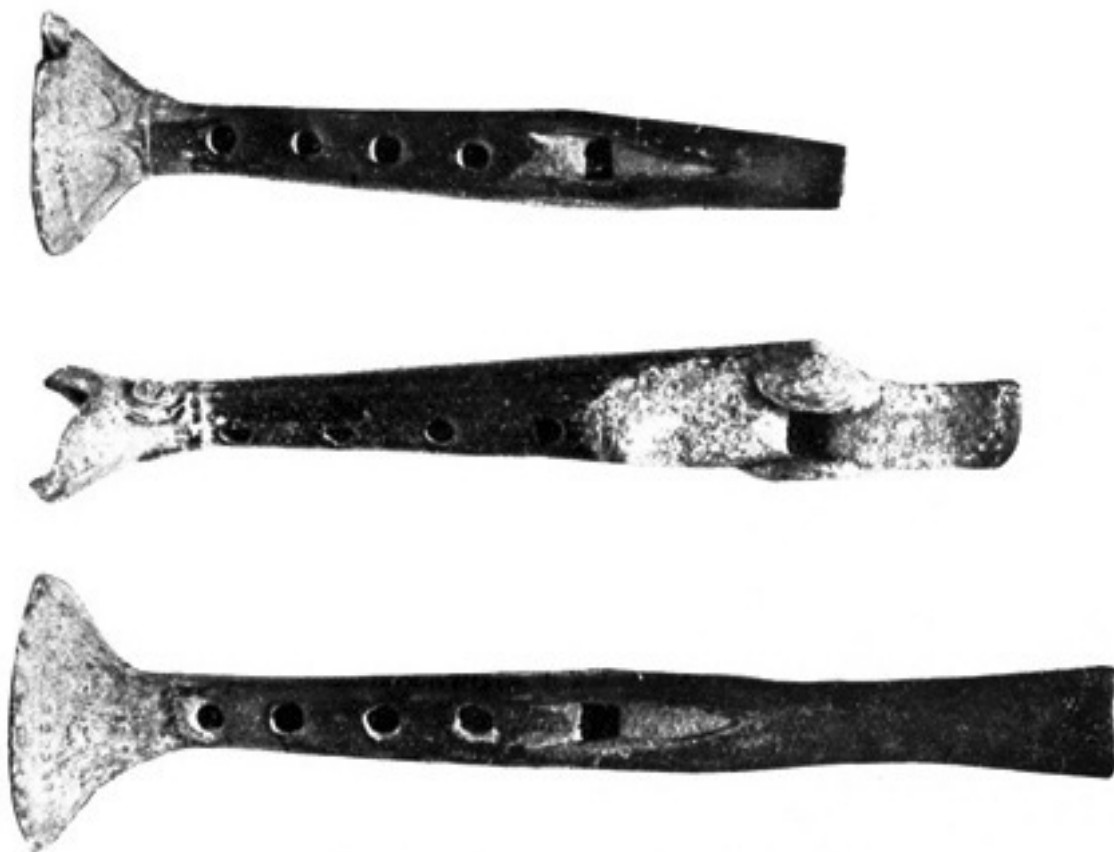
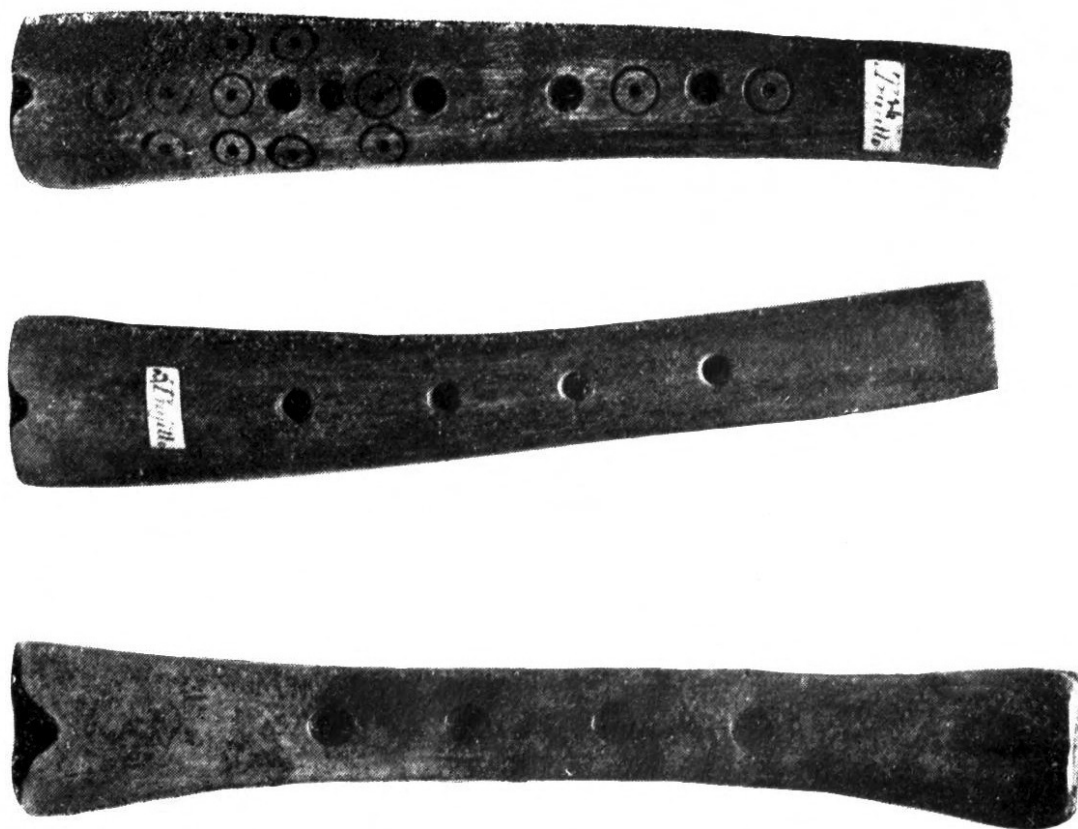


FIG. 17.—PITOS (flageolets of pottery).
a. and c. Ancient Mexican.
b. From the Island of Sacrificios.
British Museum.



**FIG. 18.—BONE FLUTES. Ancient Peruvian.
a. and b. Truxillo. c. Lima.
British Museum.**

Again, at the public ceremonies which took place on the accession of a prince to the throne the new monarch addressed a prayer to the god, in which occurred the following allegorical expression:—"I am thy flute; reveal to me thy will; breathe into me thy breath like into a flute, as thou hast done to my predecessors on the throne. As thou hast opened their eyes, their ears, and their mouth to utter what is good, so likewise do to me. I resign myself entirely to thy guidance." Similar sentences occur in the orations addressed to the monarch. In reading them one can hardly fail to be reminded of Hamlet's reflections addressed to Guildenstern, when the servile courtier expresses his inability to "govern the ventages" of the pipe and to make the instrument "discourse most eloquent music," which the prince bids him to do.

M. de Castelnau, in his “Expédition dans l’Amérique,” gives among the illustrations of objects discovered in ancient Peruvian tombs a flute made of a human bone. It has four finger holes at its upper surface and appears to have been blown into at one end. Two bone flutes (Figs. 18b & c), in appearance similar to the engraving given by M. de Castelnau, which have been disinterred at Truxillo, are deposited in the British Museum. They are about six inches in length, and each is provided with five finger holes. One of these has all the holes at its upper side, and one of the holes is considerably smaller than the rest. The specimen which we illustrate (Fig. 18a) is ornamented with some simple designs in black.

The other has four holes at its upper side and one underneath, the latter being placed near to the end at which the instrument evidently was blown. In the aperture of this end some remains of a hardened paste, or resinous substance, are still preserved. This substance probably was inserted for the purpose of narrowing the end of the tube, in order to facilitate the producing of the sounds. The same contrivance is still resorted to in the construction of the bone flutes by some Indian tribes in Guiana. The bones of slain enemies appear to have been considered especially appropriate for such flutes. The Araucanians having killed a prisoner, made flutes of his bones, and danced and “thundered out their dreadful war songs, accompanied by the mournful sounds of these horrid instruments.” Alonso de Ovalle says of the Indians in Chili: “Their flutes, which they play upon in their dances, are made of the bones of the Spaniards and other enemies whom they have overcome in war. This they do by way of triumph and glory for their victory. They make them likewise of bones of animals; but the warriors dance only to the flutes made of their enemies.” The Mexicans and Peruvians obviously possessed a great variety of pipes and flutes, some of which are still in use among certain Indian tribes. Those which were found in the famous ruins at Palenque are deposited in the museum in Mexico. They are:—The *cuyvi*, a pipe on which only five tones were producible; the *huayllaca*, a sort of flageolet; the *pincullu*, a flute; and the *chayna*, which is described as “a flute whose lugubrious and melancholy tones filled the heart with indescribable sadness, and brought involuntary tears into the eyes.” It was perhaps a kind of oboe.

The Peruvians had the syrinx, which they called *huayra-puhura*. Some clue to the proper meaning of this name may perhaps be gathered from the

word *huayra*, which signifies “air.” The *huayra-puhura* was made of cane, and also of stone. Sometimes an embroidery of needlework was attached to it as an ornament. One specimen which has been disinterred is adorned with twelve figures precisely resembling Maltese crosses. The cross is a figure which may readily be supposed to suggest itself very naturally; and it is therefore not so surprising, as it may appear at a first glance, that the American Indians used it not unfrequently in designs and sculptures before they came in contact with Christians.

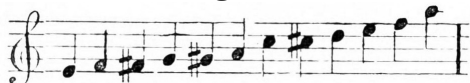
The British Museum possesses a *huayra-puhura* consisting of fourteen reed pipes of a brownish colour, tied together in two rows by means of thread, so as to form a double set of seven reeds. Both sets are almost exactly of the same dimensions and are placed side by side. The shortest of these reeds measure three inches, and the longest six and a half. In one set they are open at the bottom, and in the other they are closed. Consequently octaves are produced. The reader is probably aware that the closing of a pipe at the end raises its pitch an octave. Thus, in our organ, the so-called stopped diapason, a set of closed pipes, requires tubes of only half the length of those which constitute the open diapason, although both these stops produce tones in the same pitch; the only difference between them being the quality of sound, which in the former is less bright than in the latter.

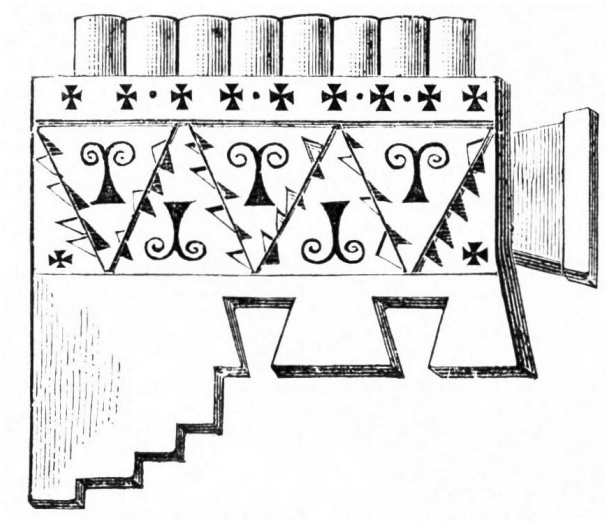
The tones yielded by the *huayra-puhura* in question are as follows:



The highest octave is indistinct, owing to some injury done to the shortest tubes; but sufficient evidence remains to show that the intervals were purposely arranged according to the pentatonic scale. This interesting relic was brought to light from a tomb at Arica.

Another *huayra-puhura* (Fig. 19), likewise still yielding sounds, was discovered placed over a corpse in a Peruvian tomb, and was procured by the French general, Paroissien. This instrument is made of soapstone, and contains eight pipes. It now belongs to the Rev. Canon J. H. Rawdon.^[4] In the Museum may be seen a good plaster cast taken from this curious relic. The height is five and three-quarter inches, and its width six and a quarter inches. Four of the tubes have small lateral finger-holes, which, when closed, lower the pitch a semitone. These holes are on the second, fourth,

six, and seventh pipe, as shown in the engraving. When the holes are open, the tones are:  and when they are closed:  The other tubes have unalterable tones. The following notation exhibits all the tones producible on the instrument: 



**FIG. 19.—HUAYRA-PUHURA,
discovered in a Peruvian tomb.
The property of the Rev. Canon Rawdon.**

The musician is likely to speculate what could have induced the Peruvians to adopt so strange a series of intervals: it seems rather arbitrary than premeditated.

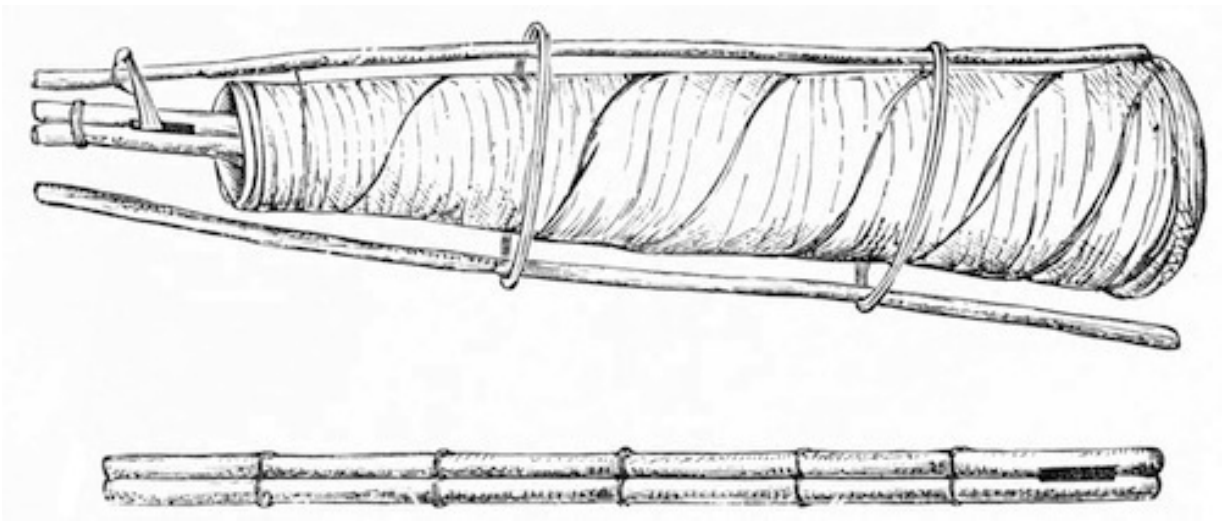
If (and this seems not to be improbable) the Peruvians considered those tones which are produced by closing the lateral holes as additional intervals only, a variety of scales or kinds of *modes* may have been contrived by the admission of one or other of these tones among the essential ones. If we may conjecture from some remarks of Garcilasso de la Vega, and other historians, the Peruvians appear to have used different orders of intervals for different kinds of tunes, in a way similar to what we find to be the case with certain Asiatic nations. We are told, for instance, “Each poem, or song, had its appropriate tune, and they could not put two different songs to one tune; and this was why the enamoured gallant, making music at night on his

flute, with the tune which belonged to it, told the lady and all the world the joy or sorrow of his soul, the favour or ill-will which he possessed; so that it might be said that he spoke by the flute.” Thus also the Hindus have certain tunes for certain seasons and fixed occasions, and likewise a number of different modes or scales used for particular kinds of songs.



**FIG. 20. WOODEN TRUMPET, used by
Indians near the Orinoco.**

Trumpets are often mentioned by writers who have recorded the manners and customs of the Indians at the time of the discovery of America. There are, however, scarcely any illustrations to be relied on of these instruments transmitted to us. The Conch was frequently used as a trumpet for conveying signals in war.



**FIG. 21.—JURUPARIS, with and without cover, used by Indians on the Rio Haupés.
In the Museum at Kew Gardens.**

Fig. 20 represents a kind of trumpet made of wood, and nearly seven feet in length, which Gumilla found among the Indians in the vicinity of the Orinoco. It somewhat resembles the *juruparis* (Fig. 21), a mysterious instrument of the Indians on the Rio Haupés, a tributary of the Rio Negro, South America. The *juruparis* is regarded as an object of great veneration. Women are never permitted to see it. So stringent is this law that any woman obtaining a sight of it is put to death—usually by poison. No youths are allowed to see it until they have been subjected to a series of initiatory fastings and scourgings. The *juruparis* is usually kept hidden in the bed of some stream, deep in the forest; and no one dares to drink out of that sanctified stream, or to bathe in its water. At feasts the *juruparis* is brought out during the night, and is blown outside the houses of entertainment. The inner portion of the instrument consists of a tube made of slips of the Paxiaba palm (*Triarteia exorrhiza*). When the Indians are about to use the instrument they nearly close the upper end of the tube with clay, and also tie above the oblong square hole (shown in the engraving) a portion of the leaf of the Uaruma, one of the arrow-root family. Round the tube are wrapped long strips of the tough bark of the Jébaru (*Parivoea grandiflora*). This covering descends in folds below the tube. The length of the instrument is from four to five feet. The illustration (Fig. 21), which exhibits the *juruparis* with its cover and without it, has been taken from a specimen in

the museum at Kew gardens. The mysteries connected with this trumpet are evidently founded on an old tradition from prehistoric Indian ancestors. *Jurupari* means "demon"; and with several Indian tribes on the Amazon customs and ceremonies still prevail in honour of Jurupari.

The Caroados, an Indian tribe in Brazil, have a war trumpet which closely resembles the *juruparis*. With this people it is the custom for the chief to give on his war trumpet the signal for battle, and to continue blowing as long as he wishes the battle to last. The trumpet is made of wood, and its sound is described by travellers as very deep but rather pleasant. The sound is easily produced, and its continuance does not require much exertion; but a peculiar vibration of the lips is necessary which requires practice. Another trumpet, the *turé*, is common with many Indian tribes on the Amazon who use it chiefly in war. It is made of a long and thick bamboo, and there is a split reed in the mouthpiece. It therefore partakes rather of the character of an oboe or clarinet. Its tone is described as loud and harsh. The *turé* is especially used by the sentinels of predatory hordes, who, mounted on a lofty tree, give the signal of attack to their comrades.

Again, the aborigines in Mexico had a curious contrivance of this kind, the *acocotl*, now more usually called *clarin*. The former word is its old Indian name, and the latter appears to have been first given to the instrument by the Spaniards. The *acocotl* consists of a very thin tube from eight to ten feet in length, and generally not quite straight but with some irregular curves. This tube, which is often not thicker than a couple of inches in diameter, terminates at one end in a sort of bell, and has at the other end a small mouthpiece resembling in shape that of a clarinet. The tube is made of the dry stalk of a plant which is common in Mexico, and which likewise the Indians call *acocotl*. The most singular characteristic of the instrument is that the performer does not blow into it, but inhales the air through it; or rather, he produces the sound by sucking the mouthpiece. It is said to require strong lungs to perform on the *acocotl* effectively according to Indian notions of taste.

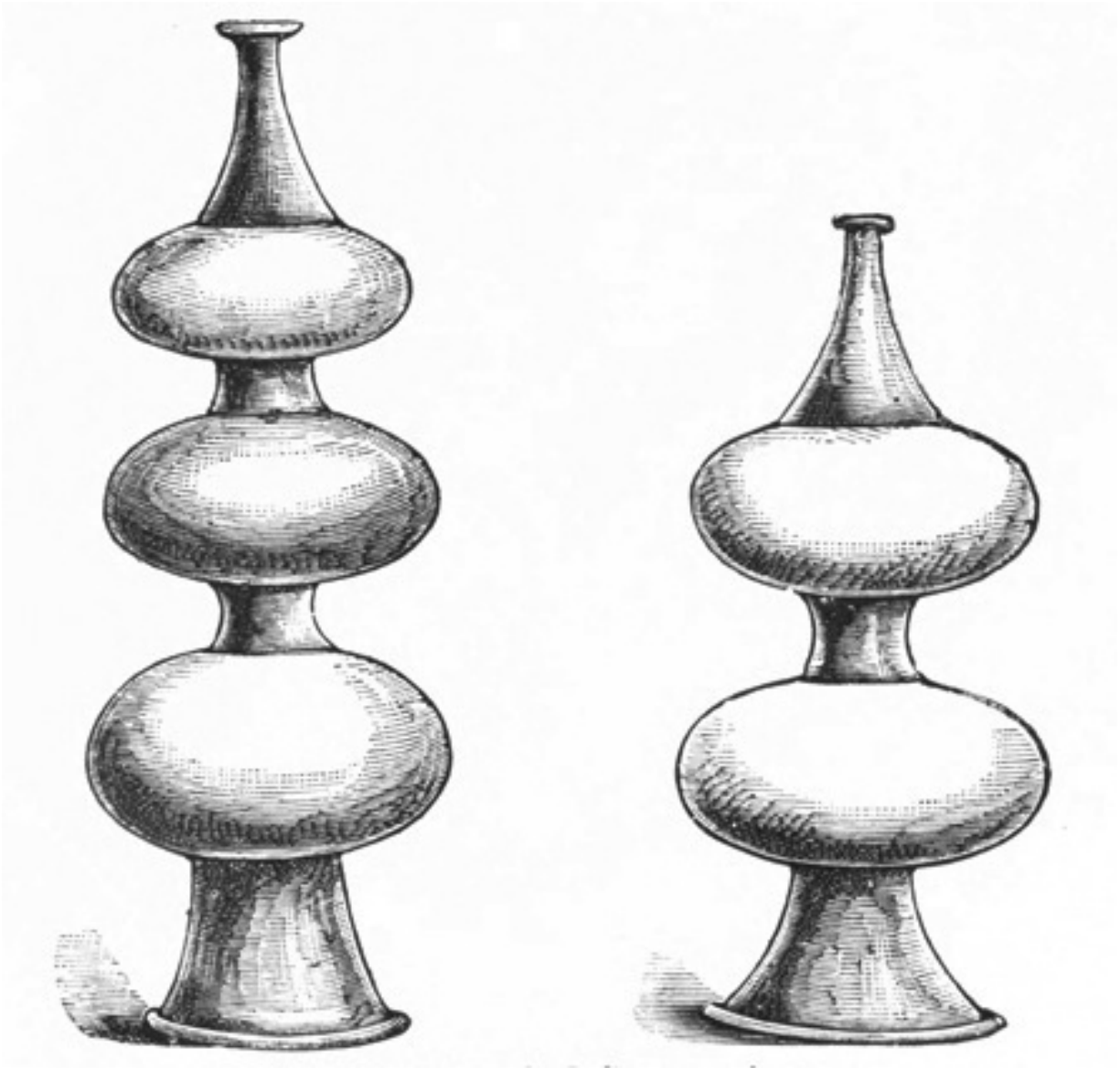


FIG. 22.—BOTUTO, used by Indians near the Orinoco.

The *botuto*, which Gumilla saw used by some tribes near the river Orinoco ([Fig. 22](#)), was evidently an ancient Indian contrivance, but appears to have fallen almost into oblivion during the last two centuries. It was made of baked clay and was commonly from three to four feet long; but some trumpets of this kind were of enormous size. The *botuto* with two bellies was usually made thicker than that with three bellies and emitted a deeper sound, which is described as having been really terrific. These trumpets were used on occasions of mourning and funeral dances.

Alexander von Humboldt saw the *botuto* among some Indian tribes near the river Orinoco.

Besides those which have been noticed, other antique wind instruments of the Indians are mentioned by historians; but the descriptions given of them are too superficial to convey a distinct notion as to their form and purport. Several of these barbarous contrivances scarcely deserve to be classed with musical instruments. This may, for instance, be said of certain musical jars or earthen vessels producing sounds, which the Peruvians constructed for their amusement. These vessels were made double; and the sounds imitated the cries of animals or birds. A similar contrivance of the Indians in Chili, preserved in the museum at Santiago, is described by the traveller S. S. Hill as follows:—"It consists of two earthen vessels in the form of our india-rubber bottles, but somewhat larger, with a flat tube from four to six inches in length, uniting their necks near the top and slightly curved upwards, and with a small hole on the upper side one third of the length of the tube from one side of the necks. To produce the sounds the bottles were filled with water and suspended to the bough of a tree, or to a beam, by a string attached to the middle of the curved tube, and then swung backwards and forwards in such a manner as to cause each end to be alternately the highest and lowest, so that the water might pass backwards and forwards from one bottle to the other through the tube between them. By this means soothing sounds were produced which, it is said, were employed to lull to repose the drowsy chiefs who usually slept away the hottest hours of the day. In the meantime, as the bottles were porous, the water within them diminished by evaporation, and the sound died gradually away."

As regards instruments of percussion, a kind of drum deserves special notice on account of the ingenuity evinced in its construction. The Mexicans called it *teponaztli*. They generally made it of a single block of very hard wood, somewhat oblong square in shape, which they hollowed, leaving at each end a solid piece about three or four inches in thickness, and at its upper side a kind of sound-board about a quarter of an inch in thickness. In this sound-board, if it may be called so, they made three incisions; namely, two running parallel some distance lengthwise of the drum, and a third running across from one of these to the other just in the centre. By this means they obtained two vibrating tongues of wood which,

when beaten with a stick, produced sounds as clearly defined as are those of our kettle drums. By making one of the tongues thinner than the other they ensured two different sounds, the pitch of which they were enabled to regulate by shaving off more or less of the wood. The bottom of the drum they cut almost entirely open. The traveller, M. Nebel, was told by archæologists in Mexico that these instruments always contained the interval of a third, but on examining several specimens which he saw in museums he found some in which the two sounds stood towards each other in the relation of a fourth; while in others they constituted a fifth, in others a sixth, and in some even an octave. This is noteworthy in so far as it points to a conformity with our diatonic series of intervals, excepting the seventh.

The *teponaztli* was generally carved with various fanciful and ingenious designs. It was beaten with two drumsticks covered at the end with an elastic gum, called *ule*, which was obtained from the milky juice extracted from the ule-tree. Some of these drums were small enough to be carried on a string or strap suspended round the neck of the player; others, again, measured upwards of 5 feet in length, and their sound was so powerful that it could be heard at a distance of three miles. In some rare instances a specimen of the *teponaztli* is still preserved by the Indians in Mexico, especially among tribes who have been comparatively but little affected by intercourse with their European aggressors. Herr Heller saw such an instrument in the hands of the Indians of Huatusco—a village near Mirador in the Tierra Templada, or temperate region, occupying the slopes of the Cordilleras. Its sound is described as so very loud as to be distinctly audible at an incredibly great distance. This circumstance, which has been noticed by several travellers, may perhaps be owing in some measure to the condition of the atmosphere in Mexico.

Instruments of percussion constructed on a principle more or less similar to the *teponaztli* were in use in several other parts of America, as well as in Mexico.

The largest kind of Mexican *teponaztli* appears to have been generally of a cylindrical shape. Clavigero gives a drawing of such an instrument. Drums, also constructed of skin or parchment in combination with wood were not unknown to the Indians. Of this description was, for instance, the *huehuetl* of the Aztecs in Mexico, which consisted, according to Clavigero,

of a wooden cylinder somewhat above 3 feet in height, curiously carved and painted and covered at the top with carefully prepared deer-skin. And, what appears the most remarkable, the parchment (we are told) could be tightened or slackened by means of cords in nearly the same way as with our own drum. The *huehuetl* was not beaten with drumsticks but merely struck with the fingers, and much dexterity was required to strike it in the proper manner. Oviedo states that the Indians in Cuba had drums which were stretched with human skin. And Bernal Diaz relates that when he was with Cortés in Mexico they ascended together the *Teocalli* ("House of God"), a large temple in which human sacrifices were offered by the aborigines; and there the Spanish visitors saw a large drum which was made, Diaz tells us, with skins of great serpents. This "hellish instrument," as he calls it, produced, when struck, a doleful sound which was so loud that it could be heard at a distance of two leagues.

The name of the Peruvian drum was *huanca*; they had also an instrument of percussion, called *chhilchiles*, which appears to have been a sort of tambourine.

The rattle was likewise popular with the Indians before the discovery of America. The Mexicans called it *ajacaxtli*. In construction it was similar to the rattle at the present day commonly used by the Indians. It was oval or round in shape, and appears to have been usually made of a gourd into which holes were pierced, and to which a wooden handle was affixed. A number of little pebbles were enclosed in the hollowed gourd. They were also made of pottery. The little balls in the *ajacaxtli* of pottery, enclosed as they are, may at a first glance appear a puzzle. Probably, when the rattle was being formed they were attached to the inside as slightly as possible; and after the clay had been baked they were detached by means of an implement passed through the holes.

The Tezcucans (or Acolhuans) belonged to the same race as the Aztecs, whom they greatly surpassed in knowledge and social refinement. Nezahualcoyotl, a wise monarch of the Tezcucans, abhorred human sacrifices, and erected a large temple which he dedicated to "The unknown god, the cause of causes." This edifice had a tower nine storeys high, on the top of which were placed a number of musical instruments of various kinds which were used to summon the worshippers to prayer. Respecting these

instruments especial mention is made of a sonorous metal which was struck with a mallet. This is stated in a historical essay written by Ixtlilxochitl, a native of Mexico and of royal descent, who lived in the beginning of the seventeenth century, and who may be supposed to have been familiar with the musical practices of his countrymen. But whether the sonorous metal alluded to was a gong or a bell is not clear from the vague record transmitted to us. That the bell was known to the Peruvians appears to be no longer doubtful, since a small copper specimen has been found in one of the old Peruvian tombs. This interesting relic is now deposited in the museum at Lima. M. de Castelnau has published a drawing of it. The Peruvians called their bells *chanrares*; but it remains questionable whether this name did not designate rather the so-called horse bells, which were certainly known to the Mexicans, who called them *yotl*. It is noteworthy that these *yotl* are found figured in the picture-writings representing the various objects which the Aztecs used to pay as tribute to their sovereigns. The collection of Mexican antiquities in the British Museum contains a cluster of *yotl*-bells. Being nearly round, they closely resemble the *Schellen* which the Germans are in the habit of affixing to their horses, particularly in the winter when they are driving their noiseless sledges.

Again, in South America sonorous stones are not unknown, and were used in olden time for musical purposes. The traveller G. T. Vigne saw among the Indian antiquities preserved in the town of Cuzco, in Peru, “a musical instrument of green sonorous stone, about a foot long, and an inch and a half wide, flat-sided, pointed at both ends, and arched at the back, where it was about a quarter of an inch thick, whence it diminished to an edge, like the blade of a knife.... In the middle of the back was a small hole, through which a piece of string was passed; and when suspended and struck by any hard substance a singularly musical note was produced.” Humboldt mentions the Amazon-stone, which on being struck by a hard substance yields a metallic sound. It was formerly cut by the American Indians into very thin plates, perforated in the centre and suspended by a string. These plates were remarkably sonorous. This kind of stone is not, as might be conjectured from its name, found exclusively near the Amazon. The name was given to it as well as to the river by the first European visitors to America, in allusion to the female warriors respecting whom strange stories are told. The natives pretending, according to an ancient tradition, that the

stone came from the country of "Women without husbands," or "Women living alone."

As regards the ancient stringed instruments of the American Indians our information is indeed but scanty. Clavigero says that the Mexicans were entirely unacquainted with stringed instruments; a statement the correctness of which is questionable, considering the stage of civilisation to which these people had attained. At any rate, we generally find one or other kind of such instruments with nations whose intellectual progress and social condition are decidedly inferior. The Aztecs had many claims to the character of a civilised community and (as before said) the Tezcucans were even more advanced in the cultivation of the arts and sciences than the Aztecs. "The best histories," Prescott observes, "the best poems, the best code of laws, the purest dialect, were all allowed to be Tezcucan. The Aztecs rivalled their neighbours in splendour of living, and even in the magnificence of their structures. They displayed a pomp and ostentatious pageantry, truly Asiatic." Unfortunately historians are sometimes not sufficiently discerning in their communications respecting musical questions. J. Ranking, in describing the grandeur of the establishment maintained by Montezuma, says that during the repasts of this monarch "there was music of fiddle, flute, snail-shell, a kettle-drum, and other strange instruments." But as this waiter does not indicate the source whence he drew his information respecting Montezuma's orchestra including the fiddle, the assertion deserves scarcely a passing notice.

The Peruvians possessed a stringed instrument, called *tinya*, which was provided with five or seven strings. To conjecture from the unsatisfactory account of it transmitted to us, the *tinya* appears to have been a kind of guitar. Considering the fragility of the materials of which such instruments are generally constructed, it is perhaps not surprising that we do not meet with any specimens of them in the museums of American antiquities.

A few remarks will not be out of place here referring to the musical performances of the ancient Indians, since an acquaintance with the nature of the performances is likely to afford additional assistance in appreciating the characteristics of the instruments. In Peru, where the military system was carefully organised, each division of the army had its trumpeters, called *cqueppacamayo*, and its drummers, called *huancarcamayo*. When the Inca

returned with his troops victorious from battle his first act was to repair to the temple of the Sun in order to offer up thanksgiving; and after the conclusion of this ceremony the people celebrated the event with festivities, of which music and dancing constituted a principal part. Musical performances appear to have been considered indispensable on occasions of public celebrations; and frequent mention is made of them by historians who have described the festivals annually observed by the Peruvians.

About the month of October the Peruvians celebrated a solemn feast in honour of the dead, at which ceremony they executed lugubrious songs and plaintive instrumental music. Compositions of a similar character were performed on occasion of the decease of a monarch. As soon as it was made known to the people that their Inca had been “called home to the mansions of his father the sun” they prepared to celebrate his obsequies with becoming solemnity. Prescott, in his graphic description of these observances, says: “At stated intervals, for a year, the people assembled to renew the expressions of their sorrow; processions were made displaying the banner of the departed monarch; bards and minstrels were appointed to chronicle his achievements, and their songs continued to be rehearsed at high festivals in the presence of the reigning monarch—thus stimulating the living by the glorious example of the dead.” The Peruvians had also particular agricultural songs, which they were in the habit of singing while engaged in tilling the lands of the Inca; a duty which devolved upon the whole nation. The subject of these songs, or rather hymns, referred especially to the noble deeds and glorious achievements of the Inca and his dynasty. While thus singing, the labourers regulated their work to the rhythm of the music, thereby ensuring a pleasant excitement and a stimulant in their occupation, like soldiers regulating their steps to the music of the military band. These hymns pleased the Spanish invaders so greatly that they not only adopted several of them but also composed some in a similar form and style. This appears, however, to have been the case rather with the poetry than with the music.

The name of the Peruvian elegiac songs was *haravi*. Some tunes of these songs, pronounced to be genuine specimens, have been published in recent works; but their genuineness is questionable. At all events they must have been much tampered with, as they exhibit exactly the form of the Spanish *bolero*. Even allowing that the melodies of these compositions have been

derived from Peruvian *harivaris*, it is impossible to determine with any degree of certainty how much in them has been retained of the original tunes, and how much has been supplied besides the harmony, which is entirely an addition of the European arranger. The Peruvians had minstrels, called *haravecs* (i.e., “inventors”), whose occupation it was to compose and to recite the *haravis*.

The Mexicans possessed a class of songs which served as a record of historical events. Furthermore they had war-songs, love-songs, and other secular vocal compositions, as well as sacred chants, in the practice of which boys were instructed by the priests in order that they might assist in the musical performances of the temple. It appertained to the office of the priests to burn incense, and to perform music in the temple at stated times of the day. The commencement of the religious observances which took place regularly at sunrise, at mid-day, at sunset, and at midnight, was announced by signals blown on trumpets and pipes. Persons of high position retained in their service professional musicians whose duty it was to compose ballads, and to perform vocal music with instrumental accompaniment. The nobles themselves, and occasionally even the monarch, not infrequently delighted in composing ballads and odes.

Especially to be noticed is the institution termed “Council of music,” which the wise monarch Nezahualcoyotl founded in Tezcuco. This institution was not intended exclusively for promoting the cultivation of music; its aim comprised the advancement of various arts, and of sciences such as history, astronomy, etc. In fact, it was an academy for general education. Probably no better evidence could be cited testifying to the remarkable intellectual attainments of the Mexican Indians before the discovery of America than this council of music. Although in some respects it appears to have resembled the board of music of the Chinese, it was planned on a more enlightened and more comprehensive principle. The Chinese “board of music,” called *Yoh Pu*, is an office connected with the *Li Pu* or “board of rites,” established by the imperial government at Peking. The principal object of the board of rites is to regulate the ceremonies on occasions of sacrifices offered to the gods; of festivals and certain court solemnities; of military reviews; of presentations, congratulations, marriages, deaths, burials—in short, concerning almost every possible event in social and public life.

The reader is probably aware that in one of the various hypotheses which have been advanced respecting the Asiatic origin of the American Indians China is assigned to them as their ancient home. Some historians suppose them to be emigrants from Mongolia, Thibet, or Hindustan; others maintain that they are the offspring of Phœnician colonists who settled in Central America. Even more curious are the arguments of certain inquirers who have no doubt whatever that the ancestors of the American Indians were the lost ten tribes of Israel, of whom since about the time of the Babylonian captivity history is silent. Whatever may be thought as to which particular one of these speculations hits the truth, they certainly have all proved useful, in so far as they have made ethnologists more exactly acquainted with the habits and predilections of the American aborigines than would otherwise have been the case. For, as the advocates of each hypothesis have carefully collected and adduced every evidence they were able to obtain tending to support their views, the result is that (so to say) no stone has been left unturned. Nevertheless, any such hints as suggest themselves from an examination of musical instruments have hitherto remained unheeded. It may therefore perhaps interest the reader to have his attention drawn to a few suggestive similarities occurring between instruments of the American Indians and of certain nations inhabiting the eastern hemisphere.

We have seen that the Mexican pipe and the Peruvian syrinx were purposely constructed so as to produce the intervals of the pentatonic scale only. There are some additional indications of this scale having been at one time in use with the American Indians. For instance, the music of the Peruvian dance *cachua* is described as having been very similar to some Scotch national dances; and the most conspicuous characteristics of the Scotch tunes are occasioned by the frequently exclusive employment of intervals appertaining to the pentatonic scale. We find precisely the same series of intervals adopted on certain Chinese instruments, and evidences are not wanting of the pentatonic scale having been popular among various races in Asia at a remote period. The series of intervals appertaining to the Chiriqui pipe, mentioned on [p. 60](#), consisted of a semitone and two whole tones, like the *tetrachord* of the ancient Greeks.

In the Peruvian *huayra-puhura* made of soapstone some of the pipes possess lateral holes. This contrivance, which is rather unusual, occurs on the Chinese *shêng*. The *chayna*, mentioned on [p. 62](#), seems to have been

provided with a reed, like the oboe: and in Hindustan we find a species of oboe called *shehna*. The *turé* of the Indian tribes on the Amazon, mentioned on p. 67, reminds us of the trumpets *turi*, or *tuturi*, of the Hindus. The name appears to have been known also to the Arabs; but there is no indication whatever of its having been transmitted to the peninsula by the Moors, and afterwards to South America by the Portuguese and Spaniards.

The wooden tongues in the drum *teponaztli* may be considered as a contrivance exclusively of the ancient American Indians. Nevertheless a construction nearly akin to it may be observed in certain drums of the Tonga and Fiji islanders, and of the natives of some islands in Torres Strait. Likewise some negro tribes in Western and Central Africa have certain instruments of percussion which are constructed on a principle somewhat reminding us of the *teponaztli*. The method of bracing the drum by means of cords, as exhibited in the *huehuetl* of the Mexican Indians, is evidently of very high antiquity in the East. It was known to the ancient Egyptians.

Rattles, Pandean pipes made of reed, and conch trumpets, are found almost all over the world, wherever the materials of which they are constructed are easily obtainable. Still, it may be noteworthy that the Mexicans employed the conch trumpet in their religious observances apparently in much the same way as it is used in the Buddhist worship of the Tibetans and Kalmuks.

As regards the sonorous metal in the great temple at Tezcuco some inquirers are sure that it was a gong: but it must be borne in mind that these inquirers detect everywhere traces proving an invasion of the Mongols, which they maintain to have happened about six hundred years ago. Had they been acquainted with the little Peruvian bell they would have had more tangible musical evidence in support of their theory than the supposed gong; for this bell certainly bears a suggestive resemblance to the little hand-bell which the Buddhists use in their religious ceremonies.

The Peruvians interpolated certain songs, especially those which they were in the habit of singing while cultivating the fields, with the word *hailli* which signified "Triumph." As the subject of these compositions was principally the glorification of the Inca, the burden *hailli* is perhaps all the more likely to remind Europeans of the Hebrew *hallelujah*. Moreover,

Adair, who lived among the Indians of North America during a period of about forty years, speaks of some other words which he found used as burdens in hymns sung on solemn occasions, and which appeared to him to correspond with certain Hebrew words of a sacred import.

As regards the musical accomplishments of the Indian tribes at the present day they are far below the standard which we have found among their ancestors. A period of three hundred years of oppression has evidently had the effect of subduing the melodious expressions of happiness and contentedness which in former times appear to have been quite as prevalent with the Indians as they generally are with independent and flourishing nations. The innate talent for music evinced by those of the North American Indians who were converted to Christianity soon after the emigration of the Puritans to New England is very favourably commented on by some old writers. In the year 1661 John Elliot published a translation of the psalms into Indian verse. The singing of these metrical psalms by the Indian converts in their places of worship appears to have been actually superior to the sacred vocal performances of their Christian brethren from Europe; for we find it described by several witnesses as “excellent” and “most ravishing.

In other parts of America the priests from Spain did not neglect to turn to account the susceptibility of the Indians for music. Thus, in central America the Dominicans composed as early as in the middle of the sixteenth century a sacred poem in the Guatemalian dialect containing a narrative of the most important events recorded in the Bible. This production they sang to the natives, and to enhance the effect they accompanied the singing with musical instruments. The alluring music soon captivated the heart of a powerful cazique, who was thus induced to adopt the doctrines embodied in the composition, and to diffuse them among his subjects, who likewise delighted in the performances. In Peru a similar experiment, resorted to by the priests who accompanied Pizarro’s expedition, proved equally successful. They dramatised certain scenes in the life of Christ and represented them with music, which so greatly fascinated the Indians that many of them readily embraced the new faith. Nor are these entertainments dispensed with even at the present day by the Indian Christians, especially in the village churches of the Sierra in Peru; and as several religious ceremonies have been retained by these people from their heathen

forefathers, it may be conjectured that their sacred musical performances also retain much of their ancient heathen character.

Most of the musical instruments found among the American Indians at the present day are evidently genuine old Indian contrivances as they existed long before the discovery of America. Take, for example, the peculiarly-shaped rattles, drums, flutes, and whistles of the North American Indians, of which some specimens in the Museum are described in the large catalogue. A few African instruments, introduced by the negro slaves, are now occasionally found in the hands of the Indians, and have been by some travellers erroneously described as genuine Indian inventions. This is the case with the African *marimba*, which has become rather popular with the natives of Guatemala in central America; but such adaptations are very easily discernible.

VII.

EUROPEAN INSTRUMENTS OF THE MIDDLE AGES.

MANY representations of musical instruments of the middle ages have been preserved in manuscripts, as well as in sculptures and paintings forming ornamental portions of churches and other buildings. Valuable facts and hints are obtainable from these evidences, provided they are judiciously selected and carefully examined. The subject is, however, so large that only a few observations on the most interesting instruments can be offered here. Unfortunately there still prevails much uncertainty respecting several of the earliest representations as to the precise century from which they date, and there is reason to believe that in some instances the archæological zeal of musical investigators has assigned a higher antiquity to such discoveries than can be satisfactorily proved.

It appears certain that the most ancient European instruments known to us were in form and construction more like the Asiatic than was the case with later ones. Before a nation has attained to a fairly high degree of civilisation its progress in the cultivation of music, as an art, is very slow indeed. The instruments found at the present day in Asia are scarcely superior to those which were in use among oriental nations about three thousand years ago. It is, therefore, perhaps not surprising that no material improvement is perceptible in the construction of the instruments of European countries during the lapse of nearly a thousand years. True, evidences to be relied on referring to the first five or six centuries of the Christian era are but scanty; although indications are not wanting which may help the reflecting musician.

There are some early monuments of Christian art dating from the fourth century in which the lyre is represented. In one of them Christ is depicted as Apollo touching the lyre. This instrument occurs at an early period in western Europe as used in popular pastimes. In an Anglo-Saxon manuscript of the ninth century in the British Museum (Cleopatra C. VIII.) are the figures of two gleemen, one playing the lyre and the other a double-pipe. M. de Coussemaker has published in the “*Annales Archéologiques*” the figure of a crowned personage playing the lyre, which he found in a manuscript of the ninth or tenth century in the library at Angers. The player twangs the strings with his fingers, while the Anglo-Saxon gleeman before mentioned uses a plectrum.

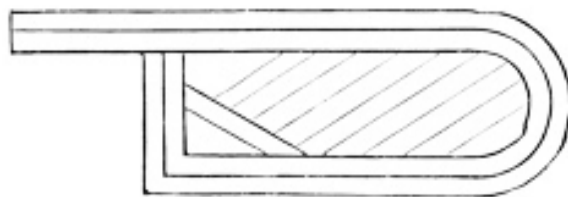


FIG. 23.—CITHARA. From a 9th century MS. formerly in the monastery of St. Blasius in the Black Forest.

Cithara was a name applied to several stringed instruments greatly varying in form, power of sound, and compass. The illustration (Fig. 23) represents a cithara from a manuscript of the ninth century, formerly in the library of the great monastery of St. Blasius in the Black Forest. When in the year 1768 the monastery was destroyed by fire, this valuable book perished in the flames; fortunately the celebrated Abbot Gerbert possessed tracings of the illustrations, which were saved from destruction. He published them, in the year 1774, in his work “*De cantu et musica sacra.*” As the older works on music were generally written in Latin we do not learn from them the popular names of the instruments; the writers merely adopted such Latin names as they thought the most appropriate. Thus, for instance, a very simple stringed instrument of a triangular shape, and a somewhat similar one of a square shape (Fig. 24), were designated by the name of *psalterium*.

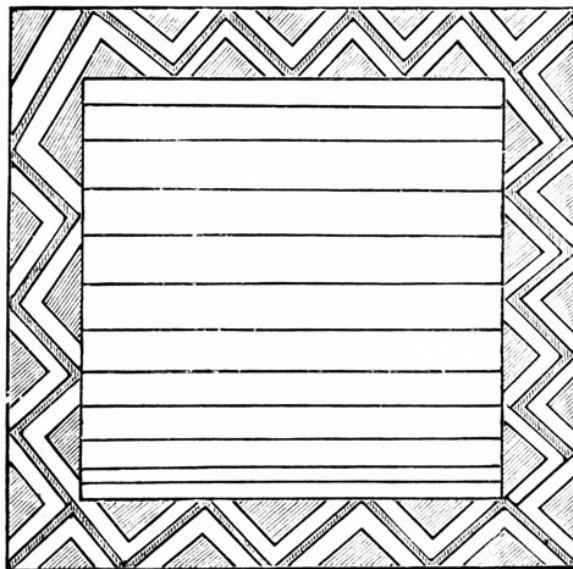


FIG. 24.—PSALTERIUM. From a MS. of the 9th century, formerly in the monastery of St. Blasius in the Black Forest.

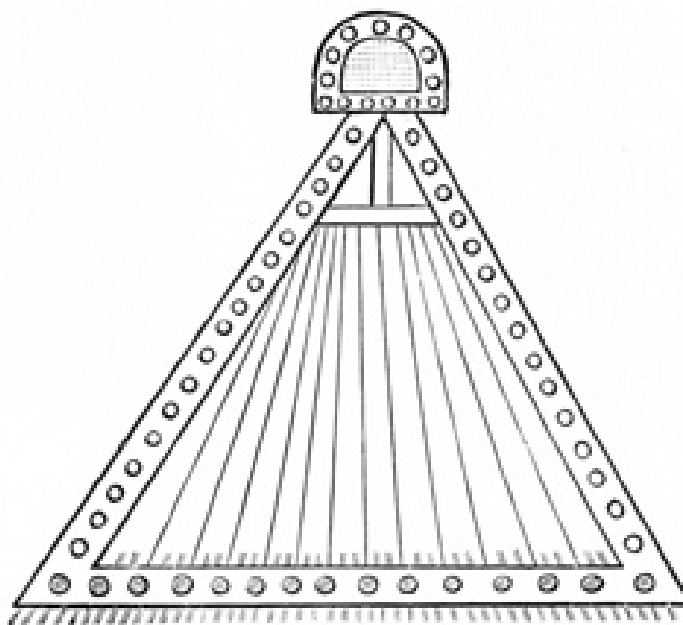


FIG. 25.—CITHARA. From a MS. of the 9th century, formerly in the monastery of St. Blasius in the Black Forest.

The *cithara* here illustrated (Fig. 25) is evidently an improvement upon the triangular psalterium (Fig. 26), because it has a sort of small sound-board at the top. Scarcely better, with regard to acoustics, appears to have been the instrument designated as *nablum*, which is engraved (Fig. 27) from a manuscript of the ninth century at Angers.



FIG. 26.—KING PLAYING PSALTERY. After an engraving in N. X. Willemin's *Monuments Français Inédits*, Vol. I., pl. 19, taken from *Hortus Deliciarum*, a MS. of the 12th century.



FIG. 27.—NABLUM. From a 9th century MS. at Angers.



FIG. 28.—Female playing a species of CITOLE. From a 9th century MS. formerly in the monastery of St. Blasius, in the Black Forest.

A small psalterium with strings placed over a sound-board was apparently the prototype of the *citole*, a kind of dulcimer which was played with the fingers ([Fig. 28](#)). The names were not only often vaguely applied by the mediæval writers, but they changed also in almost every century. The psalterium, or psalterion (Italian *salterio*, English *psaltery*), of the fourteenth century and later had the trapezium shape of the dulcimer.

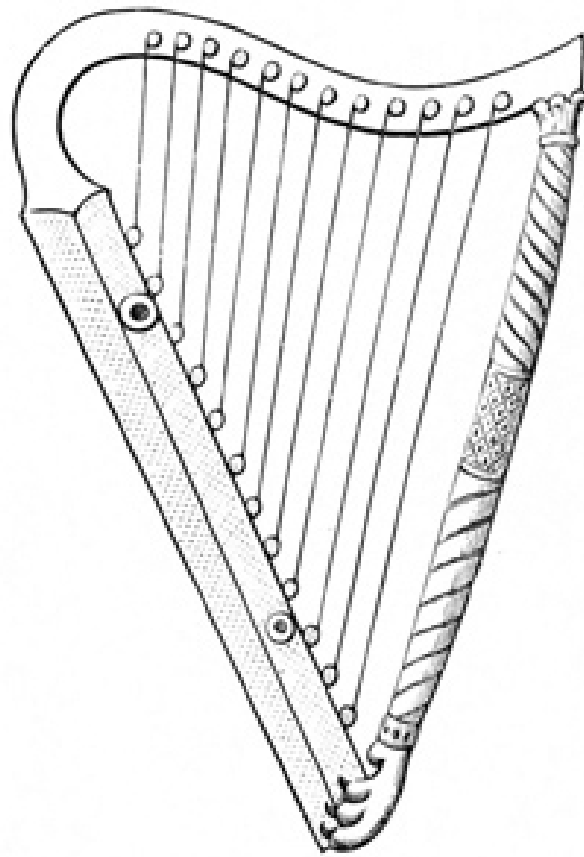


FIG. 29.—HARP. From a 9th century MS. formerly in the monastery of St. Blasius in the Black Forest.

The Anglo-Saxons frequently accompanied their vocal effusions with a harp, more or less triangular in shape, an instrument which may be considered rather as constituting the transition of the lyre into the harp. The harp was especially popular in central and northern Europe, and was the favourite instrument of the German and Celtic bards and of the Scandinavian skalds. In the next illustration ([Fig. 29](#)) from the manuscript of the monastery of St. Blasius twelve strings and two sound-holes are given to it. A harp similar in form and size, but without the front pillar, was known to the ancient Egyptians. Perhaps the addition was also non-existent in the earliest specimens appertaining to European nations; and a sculptured figure of a small harp constructed like the ancient eastern harp has been discovered in the old church of Ullard in the county of Kilkenny. This

curious relic, which is said to date from a period anterior to the year 800, is illustrated in Bunting's "Ancient Music of Ireland." As Bunting was the first who drew attention to this sculpture his account of it may interest the reader. "The drawing," he says, "is taken from one of the ornamental compartments of a sculptured cross, at the old church of Ullard. From the style of the workmanship, as well as from the worn condition of the cross, it seems older than the similar monument at Monasterboice which is known to have been set up before the year 830. The sculpture is rude; the circular rim which binds the arms of the cross together is not pierced in the quadrants, and many of the figures originally in relieve are now wholly abraded. It is difficult to determine whether the number of strings represented is six or seven; but, as has been already remarked, accuracy in this respect cannot be expected either in sculptures or in many picturesque drawings." The Finns had a harp (*harpu*, *kantele*) with a similar frame, devoid of a front pillar, still in use until the commencement of the last century.

One of the most interesting stringed instruments of the middle ages is the *rotta* (German, *Rotte*; English, *rote*). It was sounded by twanging the strings, and also by the application of the bow. The first method was, of course, the elder one. There can hardly be a doubt that when the bow came into use it was applied to certain popular instruments which previously had been treated like the *cithara* or the *psalterium*. The Hindus at the present day use their *suroda* sometimes as a lute and sometimes as a fiddle. In some measure we do the same with the violin by playing occasionally *pizzicato*. The *rotta* from the manuscript of St. Blasius is called in Gerbert's work *cithara teutonica*, while the harp is called *cithara anglica*; from which it would appear that the former was regarded as pre-eminently a German instrument. Possibly its name may have been originally *chrotta* and the continental nations may have adopted it from the Celtic races of the British isles, dropping the guttural sound. This hypothesis is, however, one of those which have been advanced by some musical historians without any satisfactory evidence.

In the *rotta* the ancient Asiatic lyre is easily to be recognized. An illumination of king David playing the *rotta* forms the frontispiece of a manuscript of the eighth century preserved in the cathedral library of Durham; it is musically interesting inasmuch as it represents a *rotta* of an oblong square shape like that just noticed and resembling the Welsh *crwth*.

It has only five strings which the performer twangs with his fingers. Again, a very interesting representation of the Psalmist with a kind of *rotta* occurs in a manuscript of the tenth century, in the British Museum (Vitellius F.XI.). The manuscript was much injured by a fire in the year 1731; but Professor Westwood has succeeded, with great care, and with the aid of a magnifying glass, in making out the lines of the figure. As it has been ascertained that the psalter is written in the Irish semiuncial character it is highly probable that the kind of *rotta* represents the Irish *cionar cruit*, which was played by twanging the strings and also by the application of a bow. Unfortunately we possess no well-authenticated representation of the Welsh *crwth* of an early period; otherwise we should in all probability find it played with the fingers, or with a plectrum. Venantius Fortunatus, an Italian who lived in the second half of the sixth century, mentions in a poem the “Chrotta Britanna.” He does not, however, allude to the bow, and there is no reason to suppose that it existed in England. Howbeit, the Welsh *crwth* (Anglo-Saxon, *crudh*; English, *crowd*) is only known as a species of fiddle closely resembling the *rotta*, but having a fingerboard in the middle of the open frame and being strung with only a few strings; while the *rotta* had sometimes above twenty strings. As it may interest the reader to examine the form of the modern *crwth* we give an illustration of it (Fig. 30). Edward Jones, in his “Musical and poetical relicks of the Welsh bards,” records that the Welsh had before this kind of *crwth* a three-stringed one called “Crwth Trithant,” which was, he says, “a sort of violin, or more properly a rebeck.” The three-stringed *crwth* was chiefly used by the inferior class of bards; and was probably the Moorish fiddle which is still the favourite instrument of the itinerant bards of the Bretons in France, who call it *rébek*. The Bretons, it will be remembered, are close kinsmen of the Welsh.

A player on the *crwth* or *crowd* (a crowder) from a bas-relief on the under part of the seats of the choir in Worcester cathedral dates from the latter part of the fourteenth century.^[5] It was probably identical with the *rotta* of the same century on the continent.

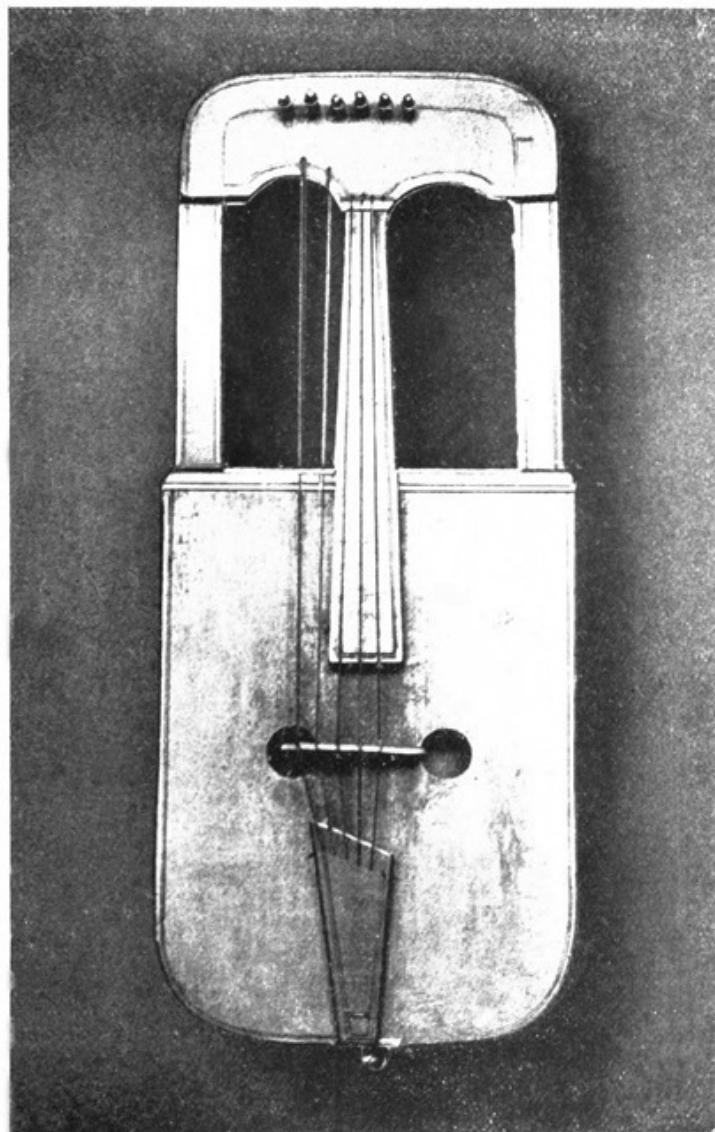


FIG. 30.—CRWTH. Welsh. 13th century.
L. 22 in., W. 9½ in. No. 175-'82.
Victoria and Albert Museum.

An interesting drawing of an Anglo-Saxon fiddle—or *fithle*, as it was called—is given in a manuscript of the eleventh century in the British Museum (Cotton, Tiberius, c. 6). The instrument is of a pear shape, with four strings, and the bridge is not indicated. A German fiddle of the ninth century, called *lyra*, copied by Gerbert from the manuscripts of St. Blasius, has only one string. Other records of the employment of the fiddle-bow in Germany in the twelfth and thirteenth centuries are not wanting. For instance, in the famous “Nibelungenlied” Volker is described as wielding

the fiddle-bow not less dexterously than the sword. And in “Chronicon picturatum Brunswicense” of the year 1203, the following miraculous sign is recorded as having occurred in the village of Ossemer: “On Wednesday in Whitsunweek, while the parson was fiddling to his peasants who were dancing, there came a flash of lightning and struck the parson’s arm which held the fiddle-bow, and killed twenty-four people on the spot.”

Among the oldest representations of performers on instruments of the violin kind found in England those deserve to be noticed which are painted on the interior of the roof of Peterborough Cathedral. They are said to date from the twelfth century. One of these figures is particularly interesting on account of the surprising resemblance which his instrument bears to our present violin. Not only the incurvations on the sides of the body but also the two sound-holes are nearly identical in shape with those made at the present day. Respecting the reliance to be placed on such evidence, it is necessary to state that the roof, originally constructed between the years 1177 and 1194, was thoroughly repaired in the year 1835. Although we find it asserted that “the greatest care was taken to retain every part, or to restore it to its original state, so that the figures, even where retouched, are in effect the same as when first painted,” it nevertheless remains a debatable question whether the restorers have not admitted some slight alterations, and have thereby somewhat modernised the appearance of the instruments. A slight touch with the brush at the sound-holes, the screws, or the curvatures would suffice to produce modifications which might to the artist appear as being only a renovation of the original representation, but which to the musical investigator greatly impair the value of the evidence. Sculptures are, therefore, more to be relied upon in evidence than frescoes.

VIII.

EUROPEAN INSTRUMENTS OF THE MIDDLE AGES.

(Continued.)

THE construction of the *organistrum* ([Fig. 31](#)) requires but little explanation. A glance at the finger-board reveals at once that the different tones were obtained by raising the keys placed on the neck under the strings, and that the keys were raised by means of the handles at the side of the neck. Of the two bridges shown on the body, the one situated nearest the middle was formed by a wheel in the inside, which projected through the sound-board. The wheel which slightly touched the strings vibrated them by friction when turned by the handle at the end. The order of intervals was *c*, *d*, *e*, *f*, *g*, *a*, *b-flat*, *b-natural*, *c*, and were obtainable on the highest string. There is reason to suppose that the other two strings were generally tuned a fifth and an octave below the highest. The *organistrum* may be regarded as the predecessor of the hurdy-gurdy, and was rather a cumbrous contrivance. Two persons seem to have been required to sound it, one to turn the handle and the other to manage the keys. Thus it is generally represented in mediæval concerts.

The *monochord* was mounted with a single string stretched over two bridges which were fixed on an oblong box. The string could be tightened or slackened by means of a turning screw inserted into one end of the box. The intervals of the scale were marked on the side, and were regulated by a sort of movable bridge placed beneath the string when required. As might be expected, the *monochord* was chiefly used by theorists; for any musical performance it was but little suitable. About a thousand years ago when this monochord was in use the musical scale was diatonic, with the exception of

the interval of the seventh, which was chromatic inasmuch as both *b-flat* and *b-natural* formed part of the scale.

This ought to be borne in mind in examining the representations of musical instruments transmitted to us from that period.

As regards the wind instruments popular during the Middle Ages, some were of quaint form as well as of rude construction.

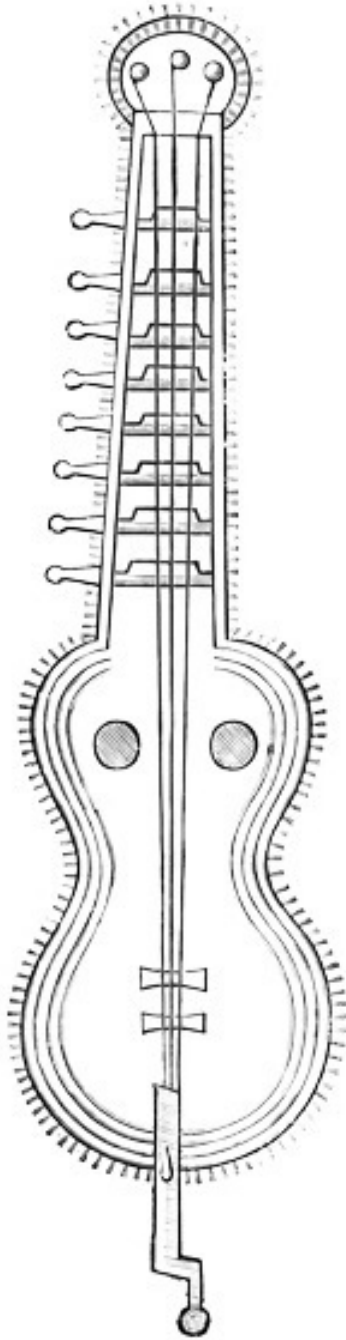


FIG. 31.—ORGANISTRUM.

The *chorus*, or *choron*, had either one or two tubes. There were several varieties of this instrument; sometimes it was constructed with a bladder into which the tube is inserted; this kind of *chorus* resembled the bagpipe; another kind resembled the *pungi* of the Hindus, mentioned on page 52. The name *chorus* was also applied to certain stringed instruments. One of these

had much the form of the *cithara*, page 84. It appears, however, probable that *chorus* or *choron* originally designated a horn (Hebrew, *keren*; Greek, *keras*; Latin, *cornu*).

The flutes of the Middle Ages were blown at the end, like the flageolet. Of the *syrinx* there are extant some illustrations of the ninth and tenth centuries, which exhibit the instrument with a number of tubes tied together, just like the Pandean pipe still in use. In one specimen,^[6] from a manuscript of the eleventh century, the tubes were inserted into a bowl-shaped box. This is probably the *frestele*, *fretel*, or *fretian*, which in the twelfth and thirteenth centuries was in favour with the French ménétriers.

Some large Anglo-Saxon trumpets may be seen in a manuscript of the eighth century in the British Museum. The largest kind of trumpet was placed on a stand when blown. Of the *oliphant*, or hunting horn, some fine specimens are in the Victoria and Albert Museum collection. The *sackbut* (Fig. 32), probably made of metal, could be drawn out to alter the pitch of sound. The sackbut of the ninth century had, however, a very different shape to that in use about three centuries ago, and much more resembled the present *trombone*. The name *sackbut* is supposed to be a corruption of *sambuca*. The French, about the fifteenth century, called it *sacqueboute* and *saquebutte*.

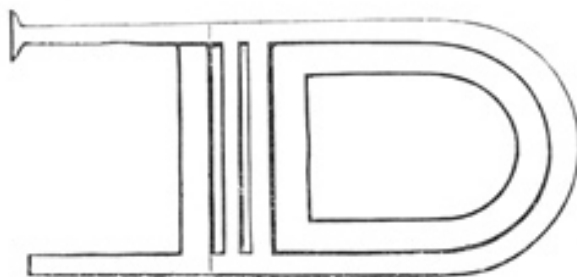


FIG. 32.—SACKBUT.

The most important wind instrument—in fact, the king of all the musical instruments—is the organ.

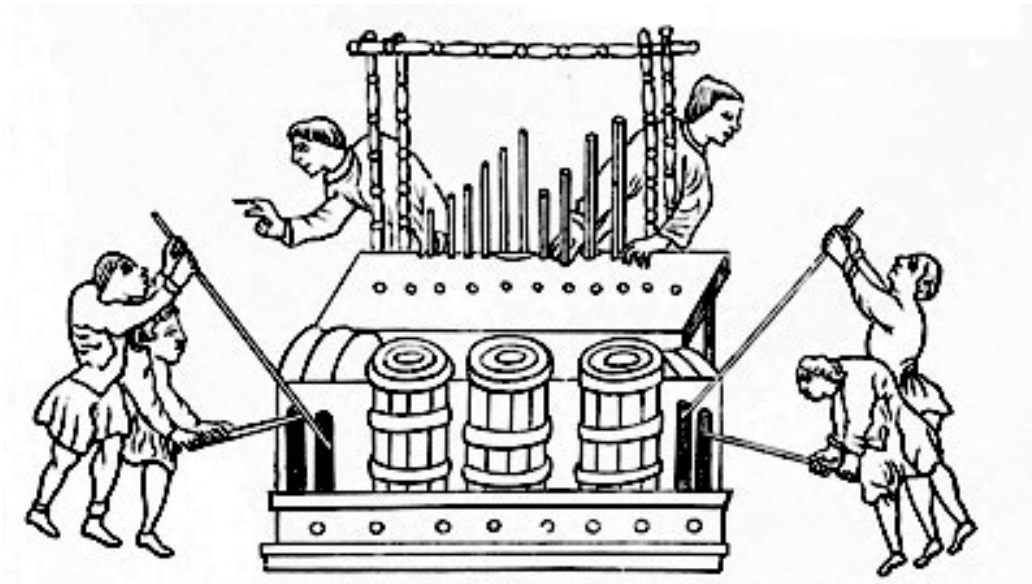


FIG. 33.—ORGAN. From a 12th century psalter in the Library of Trinity College, Cambridge.

The *pneumatic organ* is sculptured on the base of an obelisk FIG. 33.—
 ORGAN. From a 12th century psalter in the Library of Trinity College, Cambridge.] which was erected in Constantinople under Theodosius the Great towards the end of the fourth century. The bellows were pressed by men standing on them. This interesting monument also exhibits performers on the double flute. The *hydraulic organ*, which is recorded to have been already known about two hundred years before the Christian era, was according to some statements occasionally employed in churches during the earlier centuries of the Middle Ages. Probably it was more frequently heard in secular entertainments, for which it was more suitable; and at the beginning of the fourteenth century it appears to have been entirely supplanted by the pneumatic organ. The earliest organs had only about a dozen pipes. The largest, which were made about nine hundred years ago, had only three octaves, in which the chromatic intervals did not occur. Some progress in the construction of the organ is shewn in a psalter of Eadwine, in the library of Trinity College, Cambridge (Fig. 33). The instrument has ten pipes, or perhaps fourteen, as four of them appear to be double pipes. It required four men exerting all their power to produce the necessary wind, and two men to play the instrument. Moreover, both players seem also to be busily engaged in directing the blowers about the proper supply of wind. Six men and only fourteen pipes!

Another illustration is given of an organ of the 14th century ([Fig. 34](#)).

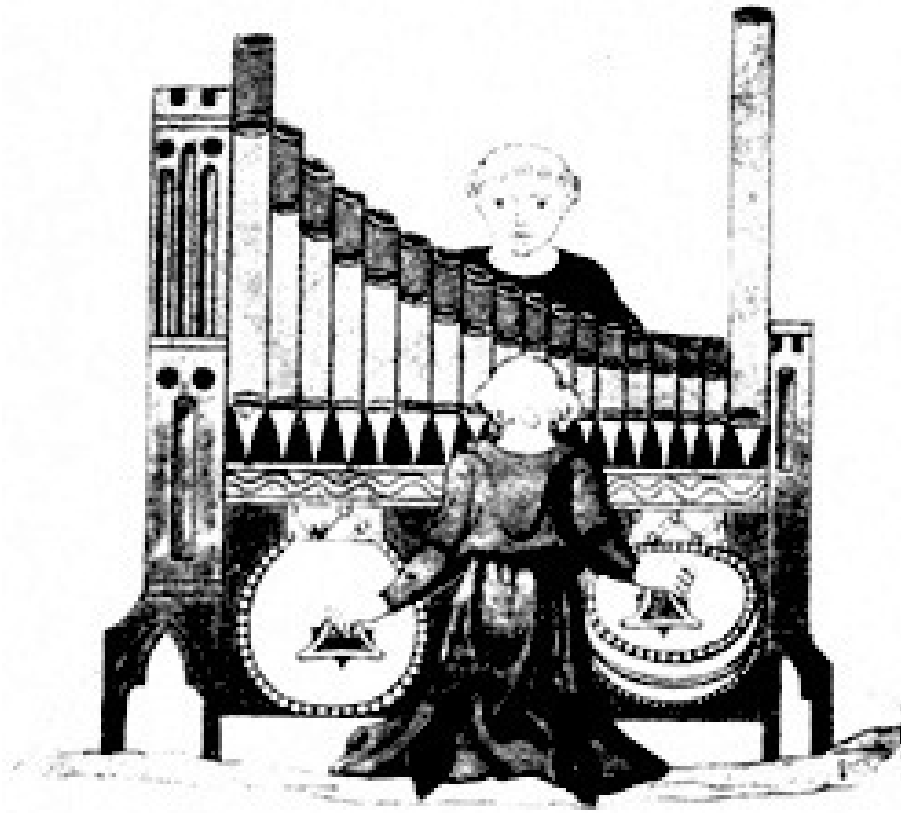


FIG. 34.—ORGAN (Grand Orgue), after an engraving in N. X. Willemin's *Monuments Français Inédits*, Vol. I., pl. 133, taken from a psalter of the 14th century.

The pedal is generally believed to have been invented by Bernhard, a German, who lived in Venice about the year 1470. There are, however, indications extant pointing to an earlier date of its invention. Perhaps Bernhard was the first who, by adopting a more practicable construction, made the pedal more generally known. On the earliest organs the keys of the finger-board were of enormous size, compared with those of the present day; so that a finger-board with only nine keys had a breadth of from four to five feet. The organist struck the keys down with his fist, as is done in playing the *carillon* still in use on the Continent, of which presently some account will be given.

Of the little portable organ, known as the *regal* or *regals*, often tastefully shaped and embellished, some interesting sculptured representations are still extant in the old ecclesiastical edifices of England and Scotland. There is, for instance, in Beverley Minster a figure of a man playing on a single regal, or a regal provided with only one set of pipes; and in Melrose Abbey the figure of an angel holding in his arms a double regal, the pipes of which are in two sets. The regal generally had keys like those of the organ but smaller. A painting in the National Gallery, attributed to Melozzo da Forlì (1438-1494) contains a regal which has keys of a peculiar shape, rather resembling the pistons of certain brass instruments. (Fig. 1, *Frontispiece*.) To avoid misapprehension, it is necessary to mention that the name *regal* (or *regals*, *rigols*) was also applied to an instrument of percussion with sonorous slabs of wood. This contrivance was, in short, a kind of harmonica, resembling in shape as well as in the principle of its construction the little glass harmonica, a mere toy, in which slips of glass are arranged according to our musical scale. In England it appears to have been still known in the beginning of the eighteenth century. Grassineau describes the “Rigols” as “a kind of musical instrument consisting of several sticks bound together, only separated by beads. It makes a tolerable harmony, being well struck with a ball at the end of a stick.” In the earlier centuries of the Middle Ages there appear to have been some instruments of percussion in favour, to which Grassineau’s expression “a tolerable harmony” would scarcely have been applicable. Drums, of course, were known; and their rhythmical noise must have been soft music, compared with the shrill sounds of the *cymbalum* (a contrivance consisting of a number of metal plates suspended on cords, so that they would be clashed together simultaneously) or with the clangour of the *cymbalum* constructed with bells instead of plates; or with the piercing noise of the *bunibulum*, or *bombulom*; an instrument which consisted of an angular frame to which were loosely attached metal plates of various shapes and sizes. The lower part of the frame constituted the handle; and to produce the noise it evidently was shaken somewhat like the sistrum of the ancient Egyptians.^[7]

The *triangle* nearly resembled the instrument of this name in use at the present day; it was more elegant in shape and had some metal ornamentation in the middle.

The *tintinnabulum* consisted of a number of bells arranged in regular order and suspended in a frame.



FIG. 35.—BAS RELIEF, representing a group of Musicians, formerly at the Abbey of St. Georges de Boscherville. Late 11th century(?). After an engraving in N.N. Willemin's *Monuments Français Inédits*, Vol. I., pl. 52.

Museum of Rouen.

IX.

EUROPEAN INSTRUMENTS OF THE MIDDLE AGES.

(Continued).

RESPECTING the orchestras, or musical bands, represented on monuments of the Middle Ages, there can hardly be a doubt that the artists who sculptured them were not unfrequently led by their imagination rather than by an adherence to actual fact. It is, however, not likely that they introduced into such representations instruments that were never admitted in the orchestras, and which would have appeared inappropriate to the contemporaries of the artists. An examination of one or two of the orchestras may therefore find a place here, especially as they throw some additional light upon the characteristics of the instrumental music of mediæval time.

A very interesting group of music performers, dating, it is said, from the end of the eleventh century, is preserved in a bas-relief which formerly ornamented the abbey of St. Georges de Boscherville and which is now removed to the museum of Rouen ([Fig. 35](#)). The orchestra comprises twelve performers, most of whom wear a crown. The first of them plays upon a viol, which he holds between his knees as the violoncello is held. His instrument is scarcely as large as the smallest viola da gamba. By his side are a royal lady and her attendant, the former playing on an *organistrum* of which the latter is turning the wheel. Next to these is represented a performer on a *syrinx*; and next to him a performer on a stringed instrument resembling a lute, which, however, is too much dilapidated to be recognisable. Then we have a musician with a small stringed instrument

resembling the *nablum* (see p. 86). The next musician, also represented as a royal personage, plays on a small species of harp. Then follows a crowned musician playing the *viol* which he holds in almost precisely the same manner as the violin is held. Again, another, likewise crowned, plays upon a harp, using with the right hand a plectrum and with the left hand merely his fingers. The last two performers, apparently a gentleman and a gentlewoman, are engaged in striking the *tintinnabulum*—a set of bells in a frame.

In this group of crowned minstrels the sculptor has introduced a tumbler standing on his head, perhaps the vocalist of the company, as he has no instrument to play upon. Possibly the sculptor desired to symbolise the hilarious effects which music is capable of producing, as well as its elevating influence upon the devotional feelings.

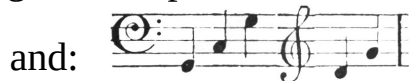
The two positions in which we find the *viol* held is worthy of notice, inasmuch as it refers the inquirer further back than might be expected for the origin of our peculiar method of holding the violin, and the violoncello, in playing. There were several kinds of the *viol* in use, differing in size and in compass of sound. The most common number of strings was five, and it was tuned in various ways. One kind had a string tuned to the note



running at the side of the finger-board instead of over it; this string was, therefore, only capable of producing a single tone. The four

other strings were tuned thus:  Two other species, on

which all the strings were placed over the finger-board, were tuned:



and:  A very beautiful *vielle* is represented in Fig. 36. It is of French workmanship of about 1550, with monograms of Henri II., and is preserved in the Museum.

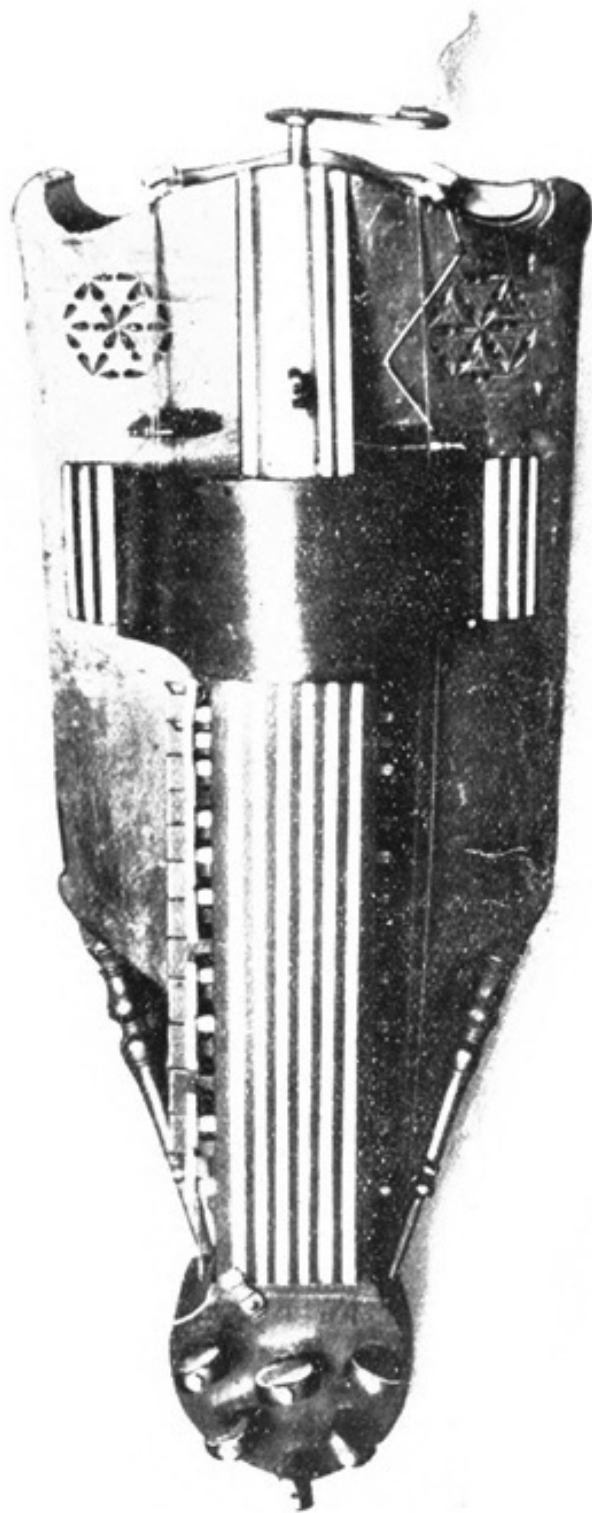


FIG. 36.—HURDY-GURDY (Vielle). With

arms of France and crowned monogram of
Henry II. on back and front. Near the
handle are monograms of Catherine de
Médicis. About 1550. L. 22½ in., W. 8¼ in.
No. 220-'66.
Victoria and Albert Museum.



FIG. 37.—TYMPANUM of the Glory Gate of the Cathedral of Santiago de Compostella. Dated 1188.

From a plaster cast in the Victoria and Albert Museum.

The contrivance of placing a string or two at the side of the finger-board is evidently very old, and was also gradually adopted on other instruments of the violin class of a somewhat later period than that of the *vielle*; for instance, on the *lira di braccio* of the Italians. It was likewise adopted on the lute, to obtain a fuller power in the bass; and hence arose the *theorbo*, the *archlute*, and other varieties of the old lute.

A grand assemblage of musical performers is represented on the Portico della Gloria of the famous pilgrimage church of Santiago de Compostella, in Spain. This triple portal, which is stated by an inscription on the lintel to have been executed in the year 1188, consists of a large semi-circular arch with a smaller arch on either side. The central arch is filled by a tympanum,

round which are twenty-four life-sized seated figures, in high relief, representing the twenty-four elders seen by St. John in the Apocalypse, each with an instrument of music. These instruments are carefully represented, and are of great interest as showing those in use in Spain about the twelfth century. A cast of this sculpture is in the Museum ([Fig. 37](#)).

In examining the group of musicians on this sculpture the reader will probably recognise several instruments in their hands which are identical with those already described in the preceding pages. The *organistrum*, played by two persons, is placed in the centre of the group, perhaps owing to its being the largest of the instruments rather than that it was distinguished by any superiority in sound or musical effect. Besides the small harp seen in the hands of the eighth and nineteenth musicians (in form nearly identical with the Anglo-Saxon harp) we find a small triangular harp, without a front-pillar, held on the lap by the fifth and eighteenth musicians. The *salterio* on the lap of the tenth and seventeenth musicians resembles the dulcimer, but seems to be played with the fingers instead of with hammers. The most interesting instrument in this orchestra is the *vihuela*, or Spanish viol, of the twelfth century. The first, second, third, sixth, seventh, ninth, twentieth, twenty-second, twenty-third, and twenty-fourth musicians are depicted with a *vihuela* which bears a close resemblance to the *rebec*. The instrument is represented with three strings, although in one or two instances five tuning-pegs are indicated. A large species of *vihuela* is given to the eleventh, fourteenth, fifteenth, and sixteenth musicians. This instrument differs from the *rebec* in so far as its body is broader and has incurvations at the sides. Also the sound-holes are different in form and position. The bow does not occur with any of these viols. But, as will be observed, the musicians are not represented in the act of playing; they are tuning and preparing for the performance, and the second of them is adjusting the bridge of his instrument.

The minstrel gallery of Exeter Cathedral ([Fig. 38](#)) dates from the fourteenth century. The front is divided into twelve niches, each of which contains a winged figure or an angel playing on an instrument of music. There is a cast also of this famous sculpture at South Kensington. The instruments are so much dilapidated that some of them cannot be clearly recognised; but, as far as may be ascertained, they appear to be as follows:—(1) The *lute* or possibly *cittern*; (2) the *bagpipe*; (3) the *clarion* or the

shalm; (4) the *rebec*; (5) the *psaltery* or the *harp*; (6) the *jew's harp* (?); (7) the *sackbut* or the *clarion*; (8) the *regals*; (9) the *gittern*, a small guitar strung with catgut; (10) the *shalm* (?); (11) the *timbrel*, resembling our present tambourine, with a double row of gingles; (12) *cymbals*. Most of these instruments have been already noticed in the preceding pages. The *shalm*, or *shawm*, was a pipe with a reed in the mouth-hole. The *wait* was an English wind instrument of the same construction. If it differed in any respect from the *shalm*, the difference consisted probably in the size only. The *wait* obtained its name from being used principally by watchmen, or *wrights*, to proclaim the time of night. Such were the poor ancestors of our fine oboe and clarinet.



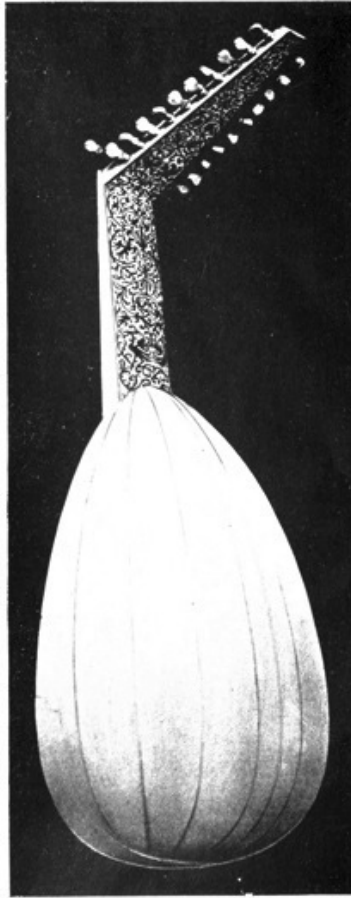
**FIG 38.—MINSTREL GALLERY, Exeter Cathedral. 14th century.
From a plaster cast in the Victoria and Albert Museum.**

X.

POST-MEDIÆVAL INSTRUMENTS.

ATTENTION must now be drawn to some instruments which originated during the Middle Ages, but which attained their highest popularity at a somewhat later period.

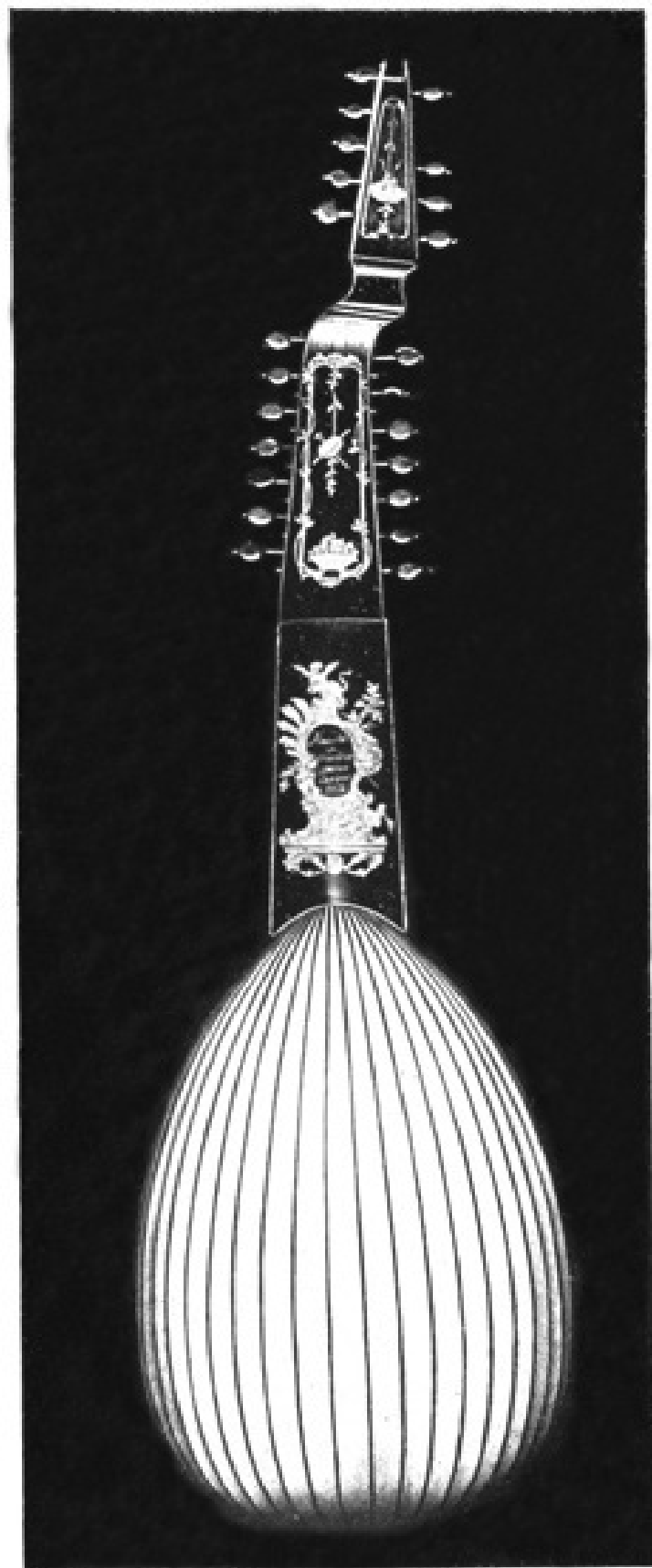
About 300 years ago the *lute* (Fig. 39) was almost as popular as is the pianoforte at the present day. Originally it had eight thin catgut strings arranged in four pairs, each pair being tuned in unison; so that its open strings produced four tones; but in the course of time more strings were added. Until the sixteenth century twelve was the largest number, or rather, six pairs. Eleven appears for some centuries to have been the most usual number of strings; these produced six tones, since they were arranged in five pairs and a single string. The latter, called the *chanterelle*, was the highest. According to Thomas Mace, the English lute in common use during the seventeenth century had twenty-four strings, arranged in twelve pairs, of which six pairs ran over the finger-board and the other six by the side of it. This lute was therefore, more properly speaking, a theorbo. The neck of the lute, and also of the theorbo, had frets consisting of catgut strings tightly fastened round it at the proper distances required for ensuring a chromatic succession of intervals. The illustration (Fig. 40) represents a lute-player of the late fifteenth century. The order of tones adopted for the open strings varied in different centuries and countries; and this was also the case with the notation of lute music. The most common practice was to write the music on six lines, the upper line representing the first string; the second line, the second string, etc., and to mark with letters on the lines the frets at which the fingers ought to be placed—*a* indicating the open string, *b* the first fret, *c* the second fret, and so on.



**FIG. 39.—LUTE. Italian (Venetian).
Beginning of 17th century.
L. 32½ in., W. 12 in. No. 1125-'69.
Victoria and Albert Museum.**



**FIG. 40.—Angel playing a Lute, after an oil painting by Ambrogio da Predis. Late 15th century.
National Gallery.**



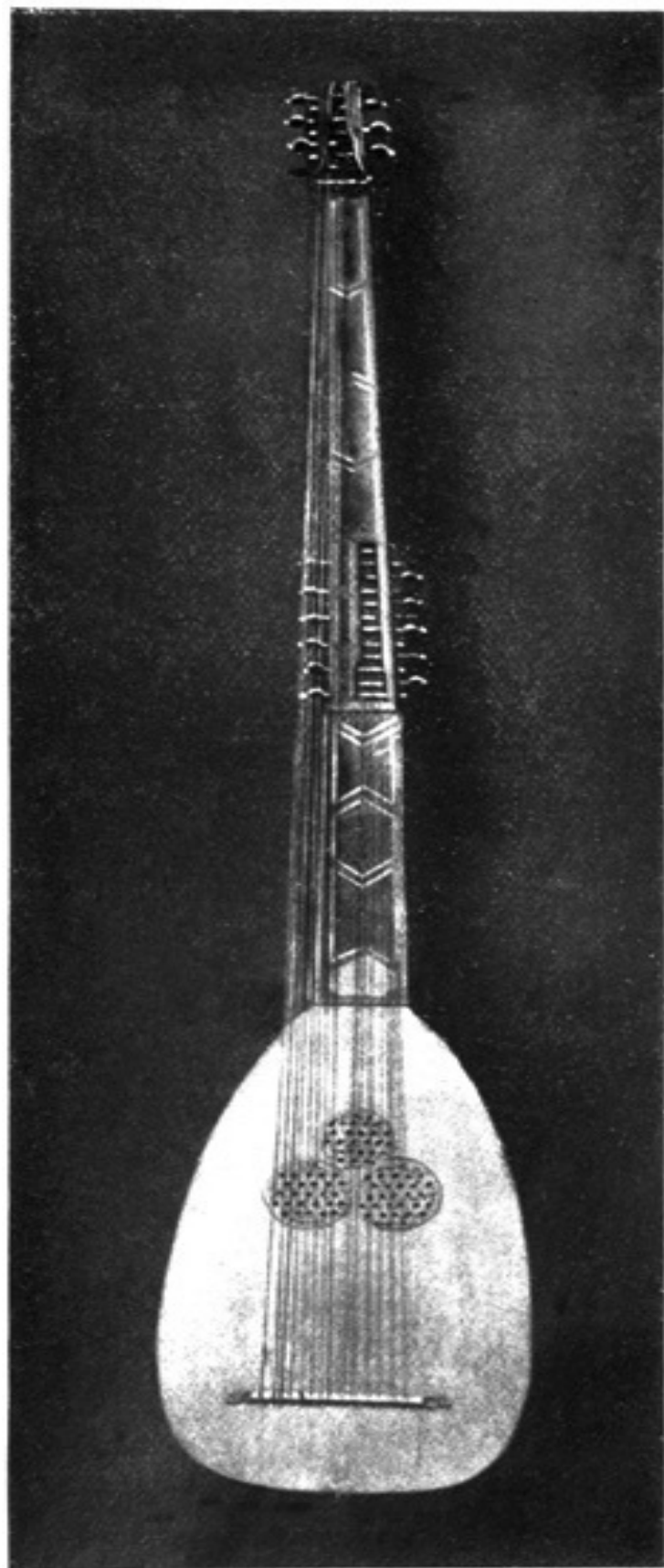
**FIG. 41.—ARCHLUTE. Inscribed “Rauche
in Chandos Street, London, 1762.”
L. 49½ in., W. 14½ in. No. 9-’71.
Victoria and Albert Museum.**

The lute was made of various sizes, according to the purpose for which it was intended in performance. The treble-lute was of the smallest dimensions, and the bass-lute of the largest. The *theorbo*, or double-necked lute which appears to have come into use during the sixteenth century, had in addition to the strings situated over the finger-board a number of others running at the left side of the finger-board which could not be shortened by the fingers, and which produced the bass tones.

The *archlute* is a large theorbo with a peculiar arrangement of the strings (Fig. 41). Several of them were doubled, the additional string being tuned an octave higher than the other. The process of tuning such instruments was evidently troublesome and tedious. Mattheson, the quaint contemporary of Handel, in his “*Das Neu-eröffnete Orchestre*,” Hamburg, 1713, remarks:—“If a lutenist attains the age of eighty, you may be sure he has tuned sixty years; and the worst of it is that among a hundred players, especially of the amateurs, scarcely two are capable of tuning with accuracy. Now there is something amiss with the strings; now with the frets; and now again with the screws; so that I have been told that in Paris it costs as much money to keep a lute as to keep a horse.” Also Mace, an enthusiastic admirer of the lute, testifies to the difficulty of keeping the instrument in proper condition; for his treatise on the lute and theorbo (contained in “*Musick’s Monument*,” London, 1676) is replete with rules for stringing, tuning, cleaning, repairing, etc. And, as regards preserving the instrument, he gives the advice—“You shall do well, ever when you lay it by in the day-time, to put it into a bed that is constantly used, between the rug and blanket.”

The *chitarrone* is a theorbo with an extraordinarily long neck, by which the length of the eight bass strings is considerably increased (Fig. 42). The largest instruments of this kind were made some centuries ago, in Rome. They were used in the theatre for accompanying the voice, before the Clavicembalo, or Harpsichord, was introduced for this purpose. The finest instruments of the lute kind were made in Italy, especially at Bologna, Rome, Venice, and Padua. Many of the manufacturers in Italy were,

however, foreigners. Evelyn, in his Diary (May 21, 1645), speaking of Bologna, says, "This place has also been celebrated for lutes made by the old masters, Mollen [Maler ?], Hans Frey, and Nicholas Sconvelt, which were of extraordinary price; the workmen were chiefly Germans." One of the earliest and most celebrated of these makers was Lucas Maler (or "Laux Maler" as he inscribed his name on his instruments). He lived at Bologna about 1415.



**FIG. 42.—CHITARRONE. Italian. Made by
Buchenberg in Rome, anno 1614.
L. 74 in. No. 190-'82.
Victoria and Albert Museum.**

Other celebrated lute-makers^[8] were:—

Ludwig Porgt, Regensburg, 1525.

Hanns Gerle, Nuremberg, b. about 1505, d. 1599.

Hans Neusedler, Nuremberg, d. 1563.

Sebastian Rauser, Verona, working about 1590 to 1605.

Mattheus Buchenberg, Rome, working about 1592-1619.

Hanns Fichtholdt, Ingoldstadt (?), about 1612; his lutes, the backs of which are made with narrow strips of wood, in the Italian manner, were formerly much prized by connoisseurs.

Paolo Belami, Paris, about 1612, probably an Italian. His lutes were highly valued.

Joachim Tielke, Hamburg, b. 1641, d. 1719.

Antonio Castaro, Rome, about 1615.

Christofilo Rochi, Padua, about 1620.

Sebastian Rochi, Venice, about 1620.

Clays von Pommersbach, Cologne, probably during the sixteenth century.

Magnus Tieffenbrucker, Venice, latter half of seventeenth century.

Wendelin Tieffenbrucker, Padua, working about 1572-1611, and Leonhard Tieffenbrucker, Padua (?), during the sixteenth century; their lutes were rather flat and long in body.

Michael Hartung, Padua, working about 1602 to 1624; he was a pupil of Leonhard Tieffenbrucker.

Raphael Mest, Füssen, working about 1610 to 1650; said to have been pupil of Michael Hartung.

Johann Christian Hoffmann, Leipzig, working about 1710 to 1750; his lutes were exported to Holland and England.

Martin Schott, Prague, latter half of seventeenth century.

Sebastian Rauch, Prague, working about 1700 to 1724.

Matthias Hummel, Nuremberg, end of seventeenth century.

Sebastian Schelle, Nuremberg, working about 1700 to 1745; his lutes were much valued, not only in Germany, but also in other European

countries.

There used to be in Italy various kinds of mandolines, of which the Milanese and the Neapolitan were the most common. The first-named had usually ten strings, constituting five pairs. The Neapolitan *mandolino* had eight strings, constituting four pairs. The strings were usually twanged with a quill. Mozart, in his “Don Giovanni,” has made use of the Neapolitan *mandolino* in the serenade; but, as the instrument has fallen into disuse, at least in most countries except Italy, the part written for it by Mozart is now generally played on the violin, *pizzicato*. The *mandolino* is now often strung with catgut strings. It resembles a diminutive lute; but its fingerboard has metal frets, and its strings are fastened to little ivory pins at the end of the body, instead of being looped through holes in the bridge. The convex back of the mandoline is deeper than that of the lute. It is one of the handsomest musical instruments.

Besides the mandoline the Italians had various instruments in shape resembling the lute. Of this description are, for instance, the *mandora*, *mandorina*, and the *pandurina*. The mandoline differs from the *pandurina* chiefly in having a rounder and deeper body, and in having the tuning-pegs placed at the back of the head; while the *pandurina* has a sort of scroll, with the tuning-pegs situated sideways, similar to the old English cither (Fig. 43). The *mandora* had usually for each tone two strings, which were of catgut and wire; and there were eight pairs of them. The *mandorina* had four wire strings.

The *guitar* (Fig. 44) is evidently an importation from the East, but it has undergone various modifications since its adoption by European nations. It was an instrument of the Moors in Spain, and became known in France about the 11th century. The French called it formerly *guiterne*, and the English *gittern*, *ghittern*, and *gythorn*. At the time of Henry VIII. we find it occasionally called “the Spanish viol.” At an early period it probably had the oval shape of the *kuitra*, still in use by the Arab musicians in Tunis and Algiers. In Spain it had formerly also the name of *vihuela*.

Instruction books for the old Spanish guitar have been written by:—Ludovico Milan, Valencia, 1534; Sixtus Kargel, Mayence, 1569; Joannes Carolus, Lerida, 1626; Pietro Milioni, Rome, 1638; Lucas Ruiz de Ribayaz,

Madrid, 1672, etc. The number of guitar manuals published during the 18th century is enormous. Germany alone contributed above fifty.



FIG. 43.—PANDURINA. On the back is carved a group consisting of Juno, Minerva and Venus. French. Second half of 16th century. L. 16½ in., W. 4½ in. No. 219-'66. Victoria and Albert Museum.

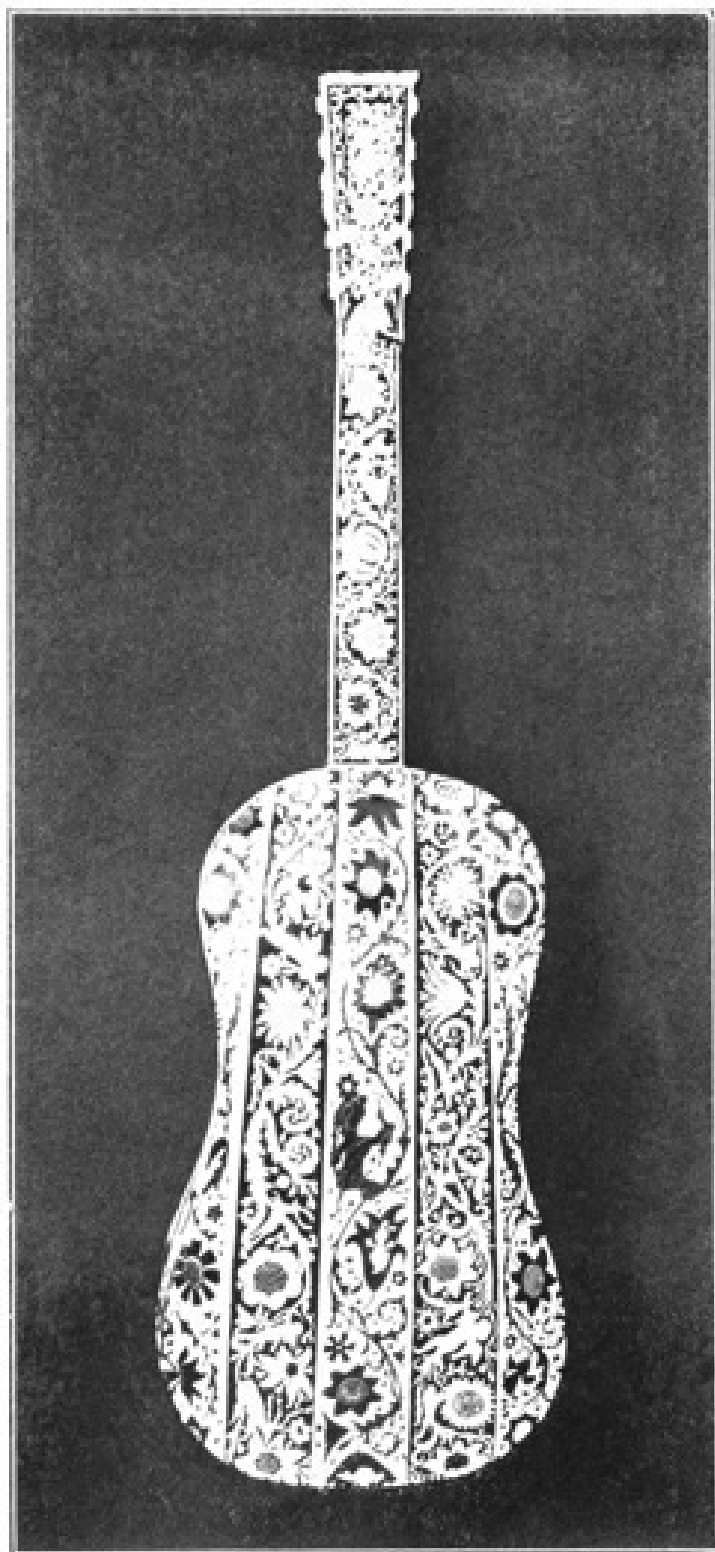


FIG. 44.—GUITAR. French (?). 17th century. L. $40\frac{3}{8}$ in., W. $11\frac{1}{8}$ in. No. 676-'72.

Victoria and Albert Museum.

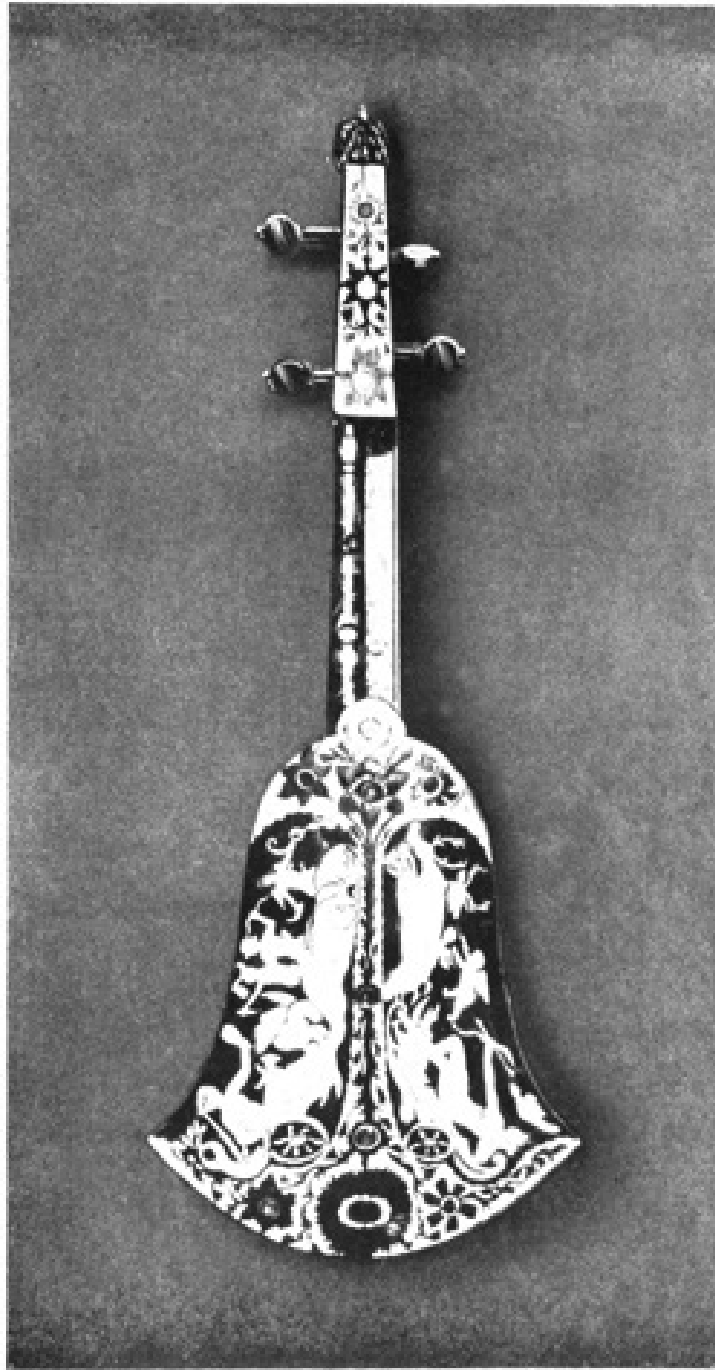
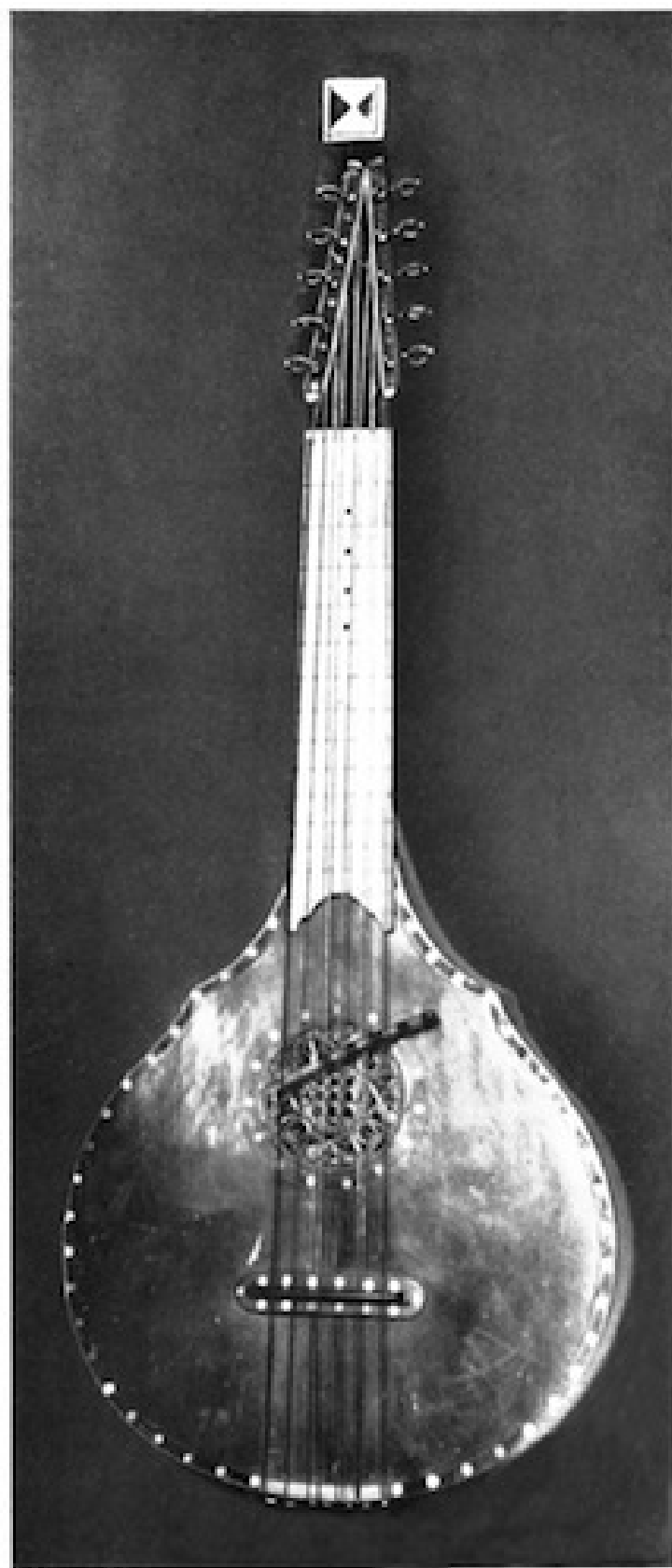


FIG. 45.—QUINTERNA, OR CHITERNA.
Inscribed “Joachim Tielke in Hamburg,
1539,” but of later date. L. 25½ in., W. 9⅜
in. No. 1122-’69.
Victoria and Albert Museum.



**FIG. 46.—CITHER. German. End of 17th
century. L. 31½ in. No. 219-'82.
Victoria and Albert Museum.**

The guitar was a fashionable instrument in England, played by ladies, in the time of Charles II. On the Continent it generally had ten catgut strings, of which two were always tuned in unison. At the present day it has six strings, the two of which are of silk covered with silver wire, and the others are of catgut.

A species of guitar is the *quinterna*, or *chiterna*, somewhat resembling a violin in shape (Fig. 45). It was used about two centuries ago, especially in Italy, by the lower orders of musicians and comedians for accompanying their vocal performances. It was played with the fingers instead of a plectrum.

The *cithern*, *cittern*, or *cither* (Fig. 46), which during the sixteenth and seventeenth centuries was a popular instrument in England, where it was often played in the barbers' shops, had four pairs of wire strings.

Its top generally terminated in a grotesquely-carved human head. The cithers made in England during the eighteenth century have generally at the top some inlaid ornamentation in ivory, mother-of-pearl, or fancy wood.

Although not well suited for the performance of harmonious combinations, since its wire strings are twanged with a quill, and therefore only such chords can be properly produced as are on strings following each other in uninterrupted succession, the cither, nevertheless, possesses considerable charms.

There are several conjectures as to the derivation of the German name *zither* or *zitter*. Some suppose it to be from "*zittern*," on account of the peculiarly trembling sound of the instrument. During the first centuries of the Christian era the word *cythera* (*cithara*) implied almost any stringed instrument, especially if the strings were twanged with a plectrum, or with the fingers. It is also noteworthy, though perhaps only as a singular coincidence, that the Persians and Hindus have a three-stringed species of *zither*, which they call *sitar*, from the Persian word *si*, "three," and *tar*, "a string." The Hindu *sitar* is, however, now usually mounted with five strings.

The *harp-guitar* and *harp-theorbo* (Fig. 47) were manufactured in England with the intention of improving the sound of the guitar and theorbo by adopting for them the body of the harp.

There was also another invention of this kind, called the *harp-lute*.

The *harp-ventura* (Fig. 48) was invented at the beginning of the last century by Signor Angelo Benedetto Ventura, professor of music, and teacher of the guitar and harp-lute to the Princess Charlotte of Wales. The example given has a back of satin wood, and sides of turtle shell; the belly and pillar are painted and gilt. It has nineteen catgut strings, six of which are covered with wire.

The *banduria* (Fig. 49) a lyre-shaped guitar, was often strung with wire instead of catgut, and played with a plectrum generally made of tortoise-shell. The specimen illustrated is made of various woods, has three sound-holes, a machine head, and twelve catgut strings tuned in pairs.

The Spanish peasants call their rustic guitar *vihuela*; and it appears probable that the “gittrons that are called Spanish vials,” mentioned in the list of musical instruments of Henry VIII. (Harl. MSS. 1419, p. 202) were small guitars of this description.

The Irish harp (*clarseth*) illustrated in Fig. 50, belonged formerly to a celebrated Irish harper. A similar one, which is in the possession of the Marquess of Kildare, bears the date 1671.



**FIG. 47. HARP THEORBO. Made by
Harley. English. About 1800. L. 36 in. No.
250-'82.
Victoria and Albert Museum.**

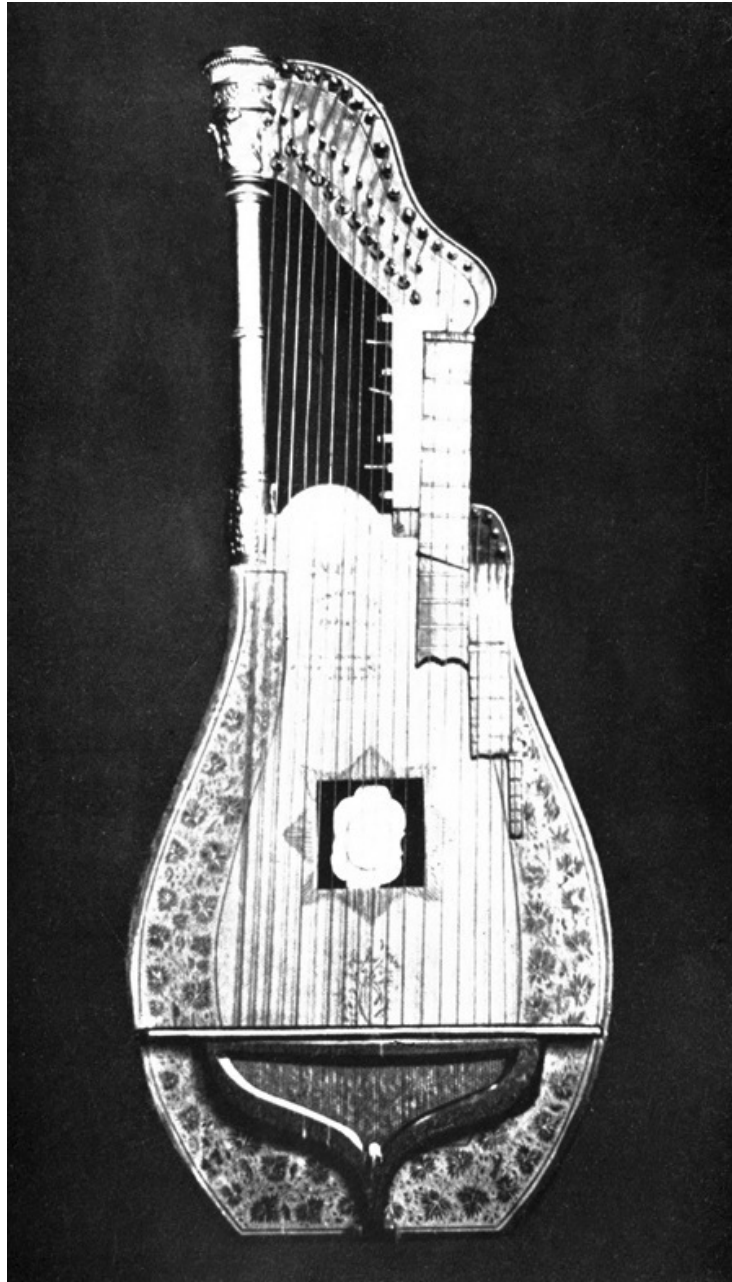


FIG. 48.—HARP VENTURA. So-called from the inventor, Signor Ventura. English. Early 19th century. L. 33 in. No. 248-'82. Victoria and Albert Museum



FIG. 49.—BANDURIA. English. Early 19th century. L. 22¼ in. No. 227-'82. Victoria and Albert Museum.



**FIG. 50.—HARP. Old Irish. H. 52 in., W.
43 in. No. 616-'72.
Victoria and Albert Museum.**

Considering the scarcity of the old Irish *clarseth*, mention may be made of a fine specimen formerly in the collection of Irish antiquities belonging to Thomas Crofton Croker, from which it was purchased, in the year 1854, at an auction in London, by Thomas Bateman, Esq. It bears on its front the inscription, *Made by John Kelly for the Rev. Charles Bunworth Baltdaniel, 1734*. At the contentions or meetings of the bards of Ireland, between the years 1730 and 1750, which were generally held at Bruree, county

Limerick, the Rev. Charles Bunworth was five times chosen umpire, or president. Although this harp is not of high antiquity, it is an interesting example of the ancient form and construction, and likewise of the ancient manner of ornamenting the instrument. A wood engraving of it, from a drawing by Maclise, is given in "A Descriptive Catalogue of the Antiquities and Miscellaneous Objects preserved in the Museum of Thomas Bateman, at Lomberdale House, Derbyshire," Bakewell, 1855. An account of the Irish harps deposited in the Museum of Dublin is to be found in "A Descriptive Catalogue of the Antiquities in the Museum of the Royal Irish Academy," by W. R. Wilde, Dublin, 1863. The illustrations of the Irish harp in the works of Bunting and similar writers may be supposed to be known to musicians.

The number of strings appears to have been greater on the older specimens recorded than on the later ones. Prætorius, in his "Syntagma musicum," etc., vol. ii., Wolfenbüttel, 1619, gives an illustration of the Irish harp, in which it is represented with forty-three strings. He describes the instrument as having a pleasant resonance, and being constructed with a considerable degree of ingenuity. The illustration exhibits the same shape, with the fore-bar bent outwards, which is shown in the present specimen.

Some harps after the model of the old Irish *clarseth*, which are painted and gilt, were made in Dublin in the beginning of the last century.

The small harp of the middle ages of Central and Western Europe, depicted in old sculptures and paintings, generally exhibits the front-bar of its frame somewhat bent outwardly, much as is the case with the Irish *clarseth*. Gradually the number of its strings was increased; and, likewise the strength of the frame for resisting the tension of the strings. The front-bar of our harp is straight, or a front-pillar. Until the seventeenth century only the diatonic series of intervals was properly obtainable on the instrument. The performer had, however, a method of producing occasionally a semitone by pressing the finger against the string towards the end, much in the same manner in which the Burmese produce chromatic intervals on the *soung*. Towards the end of the seventeenth century the Tyrolian harp makers adopted little plates with hooks, which could be moved so as to press upon the strings, and thereby shorten them, for the production of the semitones, more rapidly and unerringly than could be

done by the fingers. A French harp of the period of Louis XVI.. is illustrated ([Fig. 51](#)). It is carved and gilt in the style of Gouthière, and decorated with oak foliage and acorns; at the top of the pillar is a figure of a Cupid.

Students who examine the old instruments above described will probably wish to know something about their quality of tone. “How do they sound? Might they still be made effective in our present state of the art?” are questions which naturally occur to the musical inquirer having such instruments brought before him. A few words bearing on these questions may therefore not be out of place here.



**FIG. 51.—HARP. French. About 1770.
H. 63 in., W. 30 in. No. 4087-'57.
Victoria and Albert Museum.**



FIG. 52.—VIOLIN. Said to have belonged to James I. English. Early 17th century. L. 23¼ in., W. 8 in. No. 34-'69. Victoria and Albert Museum.



FIG. 53.—Angel playing a Viol, after an oil painting by Ambrogio da Predis. Late 15th century. National Gallery.

It is generally and justly admitted that in no other branch of the art of music has greater progress been made during the last century than in the construction of musical instruments. Nevertheless, there are people who think that we have also lost something here which might with advantage be restored. Our various instruments by being more and more perfected are becoming too much alike in quality of sound, or in that character of tone which the French call *timbre*, and the Germans *Klangfarbe*, and which professor Tyndall in his lectures on sound has translated *clang-tint*. Every musical composer knows how much more suitable one *clang-tint* is for the expression of a certain emotion than another. Our old instruments, imperfect though they were in many respects, possessed this variety of *clang-tint* to a high degree. Neither were they on this account less capable of expression than the modern ones. That no improvement has been made during the last two centuries in instruments of the violin class is a well-known fact. As to lutes and cithers the collection at South Kensington contains specimens so rich and mellow in tone as to cause musicians to regret that these instruments have entirely fallen into oblivion.

As regards beauty of appearance our earlier instruments were certainly superior to the modern. Indeed, we have now scarcely a musical instrument which can be called beautiful. The old lutes, cithers, viols, dulcimers, etc., are not only elegant in shape but are also often tastefully ornamented with carvings, designs in marquetry, and painting.

Of the stringed instruments used in our orchestra, the *violin* (Fig. 52) is the one which has been longest preserved entirely unaltered. Its name (Italian, *violino*), a diminutive of *viola*, suggests that our *tenor* (*viola di braccio*) is the older instrument of the two. The *viol* (Fig. 53, facing p. 104) in use about three centuries ago, was however somewhat different in shape. As the oldest-known instruments played with a bow, which in European countries preceded the violin, may be mentioned:—The *rebec*, which, it appears, was first popular in Spain; the *crwth* of the Welsh; the *fidla* of the Norwegian, which, in shape somewhat resembled the *crwth*, and which, with some slight modifications, is still occasionally to be found in Iceland, where it is called *langspiel*; and the *fithele* of the Anglo-Saxons.

Such were the instruments from which our violin has gradually been developed, until it attained, in the seventeenth century, that degree of

perfection which has never since been surpassed. The violin makers whose instruments are still most highly valued are:—Antonio Amati, whose most flourishing period dates between the years 1592 and 1619; Nicolo Amati, the nephew of the preceding, 1662-1692; Giuseppe Guarneri, 1690-1707; Antonio Stradivari, 1700-1725; and Jakob Stainer, 1650-1670. All these celebrated makers, except Jakob Stainer, were Italians, living at Cremona. Jakob Stainer (or Jacobus Steiner) was a native of Absam, a village near Innsbruck in the Tyrol. Few musical instruments have experienced so great an increase in price as the violins of these celebrated makers. Stainer used himself to carry his violins to the monasteries situated in the neighbourhood of Absam, where he lived. He sold them at 40 florins apiece. It was not until after his death that his workmanship was duly appreciated.

The *viola da gamba* (French, *basse de viole*; German, *Kniegeige*) derives its name from its being held between the knees of the performer (Figs. 54 and 55). It was the predecessor of the violoncello, and was made with frets. It was a favourite instrument in England at the time of Queen Elizabeth, and even ladies played it occasionally. In England it was called *base viol*, and also *viol-de-gambo*. Sir Toby Belch, in Shakespeare's "Twelfth Night," says of Sir Andrew Aguecheek:—

"He plays o' the viol-de-gamboys, and speaks three or four languages word for word without book, and hath all the good gifts of nature."



**FIG. 54.—VIOLA DA GAMBA. Italian.
About 1600. H. 48 in., W. 14 in. No.
7360-'61.
Victoria and Albert Museum.**



FIG. 55.—VIOLA DA GAMBA. Italian. 17th century. L. 47¼ in. No. 168-'82.

Victoria and Albert Museum.



**FIG. 56.—VIOLA DI BARDONE, OR
BARYTON, WITH BOW. Inscribed “Jaques
Sainprae, à Berlin.” German. 17th
century. L. 54 in., W. 16½ in. No. 1444,
1444^a-’70.
Victoria and Albert Museum.**

Among the English public performers on the *viola da gamba* are recorded a Mrs. Sarah Ottey, in the year 1723, and a Miss Ford in 1760. Carl Friedrich Abel, a German, who lived in London during the latter half of the eighteenth century, was the last performer of celebrity on this instrument. Johann Sebastian Bach has employed it in his admirable “*Passionsmusik des Matthæus*”; and there are some fine “*Suites*,” still occasionally to be met with, composed for it by M. de Caix d’Herveloix, published in the year 1710. The tone of the *viola da gamba* is rather nasal, but sweet and expressive; indeed, it is to be regretted that this charming instrument has fallen into disuse. There is, however, a *gamba* stop in the organ, which resembles the famous *vox humana* stop, and which has recently been much favoured by organ builders.

The *violoncello* came into competition with the *viola da gamba* at the beginning of the eighteenth century, and has now entirely superseded its predecessor.

A *viola di bardone* in the Museum ([Fig. 56](#)) has a neck of carved and pierced box-wood, terminating in a figure of Apollo playing the lyre; the principal finger-board is of ivory, engraved and inlaid with ebony and tortoiseshell, with figures of Jupiter and Juno, and a lady playing a lute; the second finger-board is also of pierced and engraved ivory. The instrument has four catgut and fourteen metal sympathetic strings, and a double wrest. It was made by Jaques Sainprae, of Berlin, and is said to have belonged to Quanz, music master of Frederick the Great.

The most accomplished performers on the *viola di bardone* were Anton Lidl of Vienna (to whom is sometimes erroneously ascribed the invention of this instrument) and Karl Franz, a musician of the band of Prince Esterhazy, about the middle of the 18th century. Lidl played on the *viola di bardone* in concerts in England during the year 1776. Joachim Tielke of Hamburg, the

manufacturer of a specimen in the Museum, was an instrument maker whose lutes were much esteemed on account of their fine tone, and their elegant ornamentation. He made them of ebony inlaid with ivory, mother-of-pearl, silver, and gold.

Joseph Haydn wrote sixty-three compositions for the *viola di bardone* by order of Prince Esterhazy, who was himself a performer on this instrument, and who admired it greatly. Its tone is soft and very expressive, but rather tremulous; owing to this quality, probably, it was also called *viola di fagotto*. It never became very popular, since its rather complicated construction offered too many difficulties in its treatment. In Germany it was generally called *Baryton*.

The *viola d'amore* (Fig. 57) was often strung entirely with wire. It appears to have been a novelty to Evelyn, for he records in his Diary of November 20th, 1679, "I dined with Mr. Slingsby, Master of the Mint, with my wife, invited to hear music, which was exquisitely performed by four of the most renowned masters: Du Prue, a Frenchman, on the lute; Signor Bartholomeo, an Italian, on the harpsichord; Nicholao, on the violin; but above all, for its sweetness and novelty, the *viol d'amore* of five wire strings played on with a bow, being but an ordinary violin played on lyre-way by a German." Mattheson ("Das Neu-Eröffnete Orchestre," Hamburg, 1713) describes the *viola d'amore* as being mounted with four wire strings, and with one catgut string for the highest tone.



FIG. 57.—VIOLA D'AMORE. Probably

**English. Late 17th century.
L. 27½ in. No. 154-'82.
Victoria and Albert Museum.**



**FIG. 58.—DOUBLE-BASS, WITH BOW.
Known as “The Giant.” Italian. 17th
century. L. 103 in., W. 42 in. No. 487-'72.
Victoria and Albert Museum.**

He praises its sweetness of sound, but does not mention the sympathetic strings. The transformation of the wire-strung *viola d'amore* into the so-called psaltery or sultana, which has no sympathetic strings, is indicated in the following statement by Sir John Graham Dalyell ("Musical Memoirs of Scotland," Edinburgh, 1849), "The instrument was first introduced in public

in London during the year 1715, when it was heard between the acts of an opera. It was known in Scotland in the middle of the century, and a taste for it was probably encouraged by the performance of Passerini, an Italian resident in Edinburgh, in the year 1752, when it was said to be a new instrument called *viola d'amour*. Passerini was manager of the Gentleman's and St. Cecilia Concert, where he and his wife had a permanent engagement as skilled musicians. He played solos and accompanied singing with the instrument. Perhaps the *viola d'amour* underwent several modifications, as its name was changed to *psaltery*, in the belief of its being the ancient instrument so denominated, which is quite different according to most authorities, not belonging to the fidicinal tribe. In 1754 a concert for the new instrument called the *psaltery* was announced for Signor Carusi's benefit concert in Edinburgh, and performed by Pasquali, another Italian musician, also resident there. From its soft and simple nature it was eulogised in 1762 as unequalled for delicacy and sweetness. I knew a lady many years ago in Edinburgh who played melodies with great delicacy on this instrument, which was strung with wire, and had frets on the finger-board." From these accounts it would appear that the *viola d'amore* strung entirely with wire was not much used in England before the year 1700, although it evidently existed in this country in the seventeenth century.

The *double-bass* (Italian, *contrabasso*, *violone*; French, *contrebasse*; German, *grosse Bassgeige*, *Kontrabass*) is either four-stringed or three-stringed. A three-stringed example known as "The Giant" presented by Dragonetti to the Duke of Leinster, and given by the latter to the Museum, is illustrated in [Fig. 58](#).

Dragonetti, the celebrated *virtuoso* on the double-bass, came to England in the year 1794. His favourite instrument, upon which he played in public concerts, was a "Gaspar di Salo," which he obtained from the Convent of St. Pietro at Vicenza, and which he never could be induced to part with, although £800, it is said, was offered him for it by one of his rich and enthusiastic pupils in England. After the death of Dragonetti this bass, and another valuable one by Stradivarius, were sent back to Italy, he having bequeathed them in his will to the town of Venice. Dragonetti died in the year 1846 at his house in Leicester Square, at the age of eighty-three. A year before his death he was still able to assist in the public performances at the Beethoven Festival in Bonn. His friend H. Philipps mentions in his

“Musical Recollections” that the ends of Dragonetti’s fingers had gradually become quite flat and deformed from playing.

Some double-basses of extraordinarily large size are known to have been made in England. William Gardiner (“Music and Friends,” London, 1838, p. 70) mentions such an instrument, made by Martin in Leicester, which he saw in the year 1786, and which, if his statement may be relied upon, “was of such height that Mr. Martin was obliged to cut a hole in the ceiling to let the head through; so that it was tuned by going upstairs into the room above.”

A *sordino* (French, *pochette*; German, *Taschengeige*) is illustrated in Fig. 59. About 300 years ago the *sordino* was kept by gentlemen in a case resembling a pen case, which they put in the pocket when they went to a singing party; and they used the instrument for insuring correct intonation while singing madrigals and catches. Kircher, in his “Musurgia Universalis,” Romæ, 1650, calls it *linterculus*, no doubt from its resemblance to a small boat.

Fig. 60 represents a *bûche* (German, *Scheitholz*) made by Fleurot, of the Val d’Ajol, in the Vosges Mountains, early in the last century.

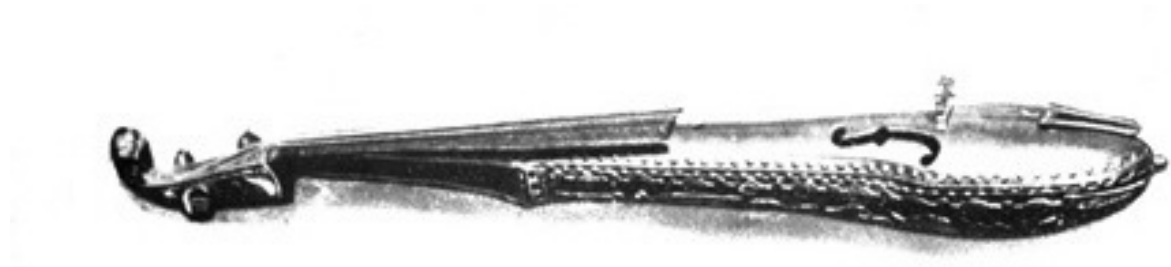


FIG. 59.—SORDINO, OR POCHETTE. Probably German. Late 17th or early 18th century. L. 17½ in. No. 457-'83.

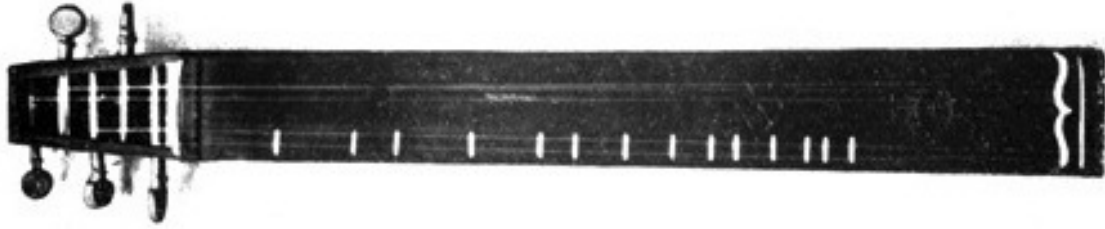


FIG. 60.—BÛCHE, OR SCHEITHOLZ. Made by Fleurot, of the Val d'Ajol in the Vosges Mountains. Early 19th century. L. 27½ in. No. 210-'82. Victoria and Albert Museum.

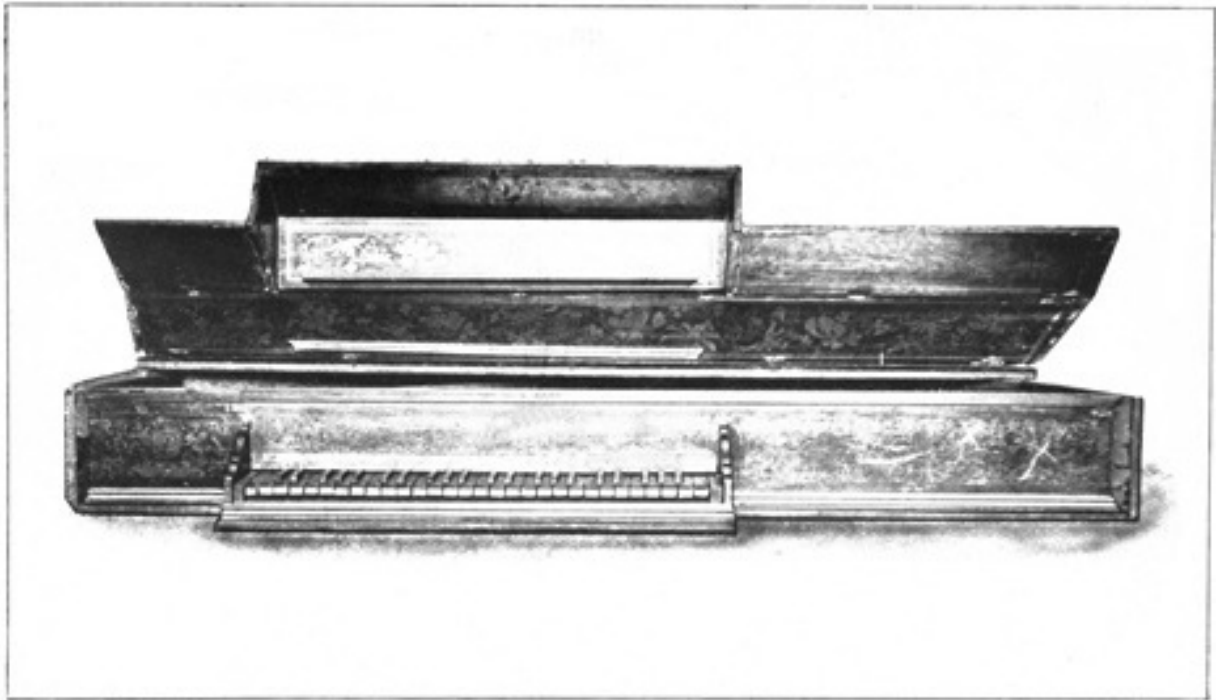


FIG. 61.—VIRGINAL. Formerly belonging to Queen Elizabeth. Italian. Second half of 16th century. H. 8½ in., L. 65 in., D. 23 in. No. 19-'87. Victoria and Albert Museum.



FIG. 62.—VIRGINAL. Bears the arms of William, Duke of Cleves, Berg and Jülich, Count of La Marck and Ravensberg, and originally also Duke of Guelderland (b. 1516, d. 1592). Flemish. Second half of 16th century. H. 16 in., W. 67 in., D. 28 in. No. 447-'96.
Victoria and Albert Museum.

At the present day the people twang the *bûche* with a quill; but in olden time it was played thus:—The performer, having placed the instrument on a table, twanged the strings with the thumb of his right hand, while he used his left hand in pressing down, by means of a little stick, those strings which are placed over the frets, and which, being tuned in unison, serve for producing the melody. The other strings, tuned a *fifth* lower, were occasionally struck as an accompaniment.

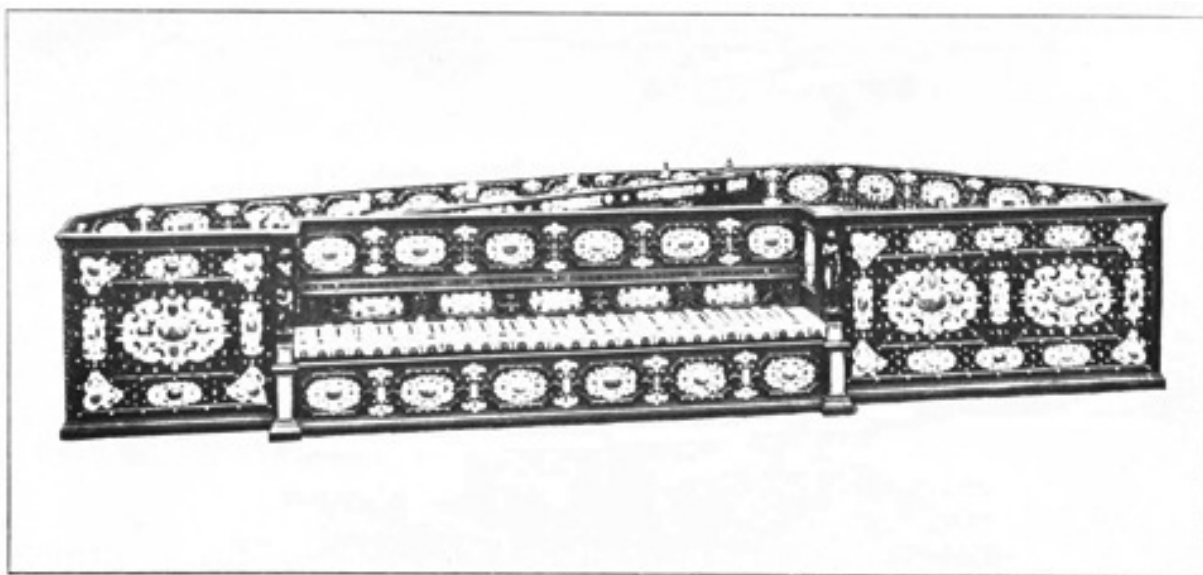
Primitive in construction, and imperfect for our present musical performances as the *Scheitholz* is, it nevertheless is interesting, not only on account of its popularity three centuries ago, but also because it is the prototype of the horizontal cither, which has come somewhat into vogue in the last century.

The most popular instruments played with a bow, in the seventeenth century, were the *treble-viol*, the *tenor-viol*, and the *bass-viol*. It was usual for viol players to have “a chest of viols,” a case containing four or more viols, of different sizes. Thus, Thomas Mace in his directions for the use of the viol, “Musick’s Monument” 1676, remarks, “Your best provision, and most complete, will be a good chest of viols, six in number, viz., two basses, two tenors, and two trebles, all truly and proportionately suited.” The violist, to be properly furnished with his requirements, had therefore to supply himself with a larger stock of instruments than the violinist of the present day.

The *virginal* (Figs. 61 and 62) is said to have obtained its name from having been intended especially to be played by young ladies. The statement of some writers that it was called virginal in compliment to Queen Elizabeth, is refuted by the fact of its being mentioned among the musical instruments of King Henry VIII., in the beginning of the sixteenth century. Probably the name was originally given to it in honour of the Virgin Mary, since the *virginal* was used by the nuns for accompanying their hymns addressed to the Holy Virgin. It was made of various sizes, but generally small in comparison with our square pianoforte. The Italians, about three hundred years ago, constructed a small portable instrument of this kind, which they called *ottavino* (or *octavina*) because its pitch was an octave higher than that of the *clavicembalo*, or harpsichord.

Queen Elizabeth was a performer on the *virginal* (see Fig. 61) as well as on the lute. Sir James Melville, the Scotch ambassador, records in his memoirs an interview with Queen Elizabeth, in the year 1564, in which he heard her play upon the virginal:—“Then sche asked wither the Quen (Mary of Scotland) or sche played best. In that I gaif hir the prayse.” During the Shakesperian age a virginal generally stood in the barbers’ shops for the amusement of the customers. The instrument had evidently retained its popularity at the time of the Great Fire of London; for Pepys (Diary, September 2nd, 1666) records:—“River full of lighters and boats taking in goods, and good goods swimming in the water; and only I observed that hardly one lighter or boat in three that had the goods of a house in, but there was a pair of virginalls in it.”

The instrument has metal strings, one for each tone, which are twanged by means of small portions of quill, attached to slips of wood called "jacks," and provided with thin metal springs. Its construction is therefore similar to that of the spinet and harpsichord. Crowquills were most commonly used in the construction of such instruments; but other materials, as for instance leather, whalebone, and even elastic strips of metal, were occasionally adopted instead.



**FIG. 63.—SPINET. Made by Annibale dei Rossi of Milan. Italian.
Dated 1577. H. 11¼ in., L. 58¼ in., W. 22¼ in. No. 809-'69.
Victoria and Albert Museum.**



FIG. 64.—SPINET. Signed “Johannes Player fecit.” English. About 1700. L. 59 in., W. 22½ in. No. 466-’82. Victoria and Albert Museum.



FIG. 65.—CLAVICHORD. Inscribed “Barthold Fritz fecit, Braunschweig, anno 1751.” German. 18th century. H. 31 in., L. 70½ in., W. 22½ in. No. 339-’82. Victoria and Albert Museum.

There evidently prevailed, some centuries ago, much vagueness in the designation of certain stringed instruments with a key-board. The term *clavichord* seems to have not unfrequently been applied to any stringed instrument with a key-board, no matter what its interior construction might be. Johann Walther, in his “Musicalisches Lexicon,” Leipzig, 1732, describes the *virginal* (or “Virginale,” as he calls it), in these words:—“Ein Clavier vors Frauenzimmer” (*a clavichord for ladies*). The following brief explanation of the difference between the spinet and the clavichord may therefore be of interest to some inquirers.

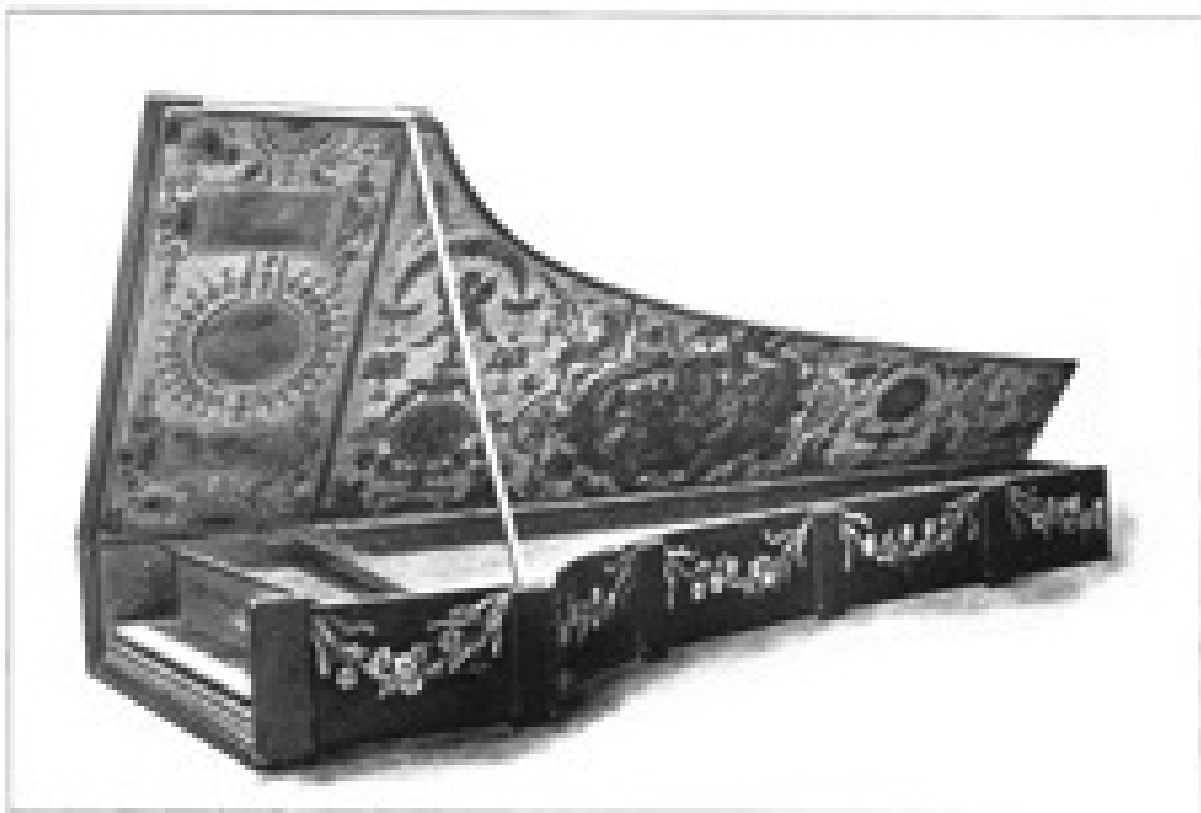
The *spinet* (Italian, *spinetta* or *spinetto*; French, *épinette*) is said to have derived its name from the little quill (*spina*) used in its mechanism, which is the same as that of the harpsichord and the virginal, described before.

The more commonly-known spinet (Figs. 63 and 64) resembles in shape the harpsichord and the grand piano. It is, however, smaller than the harpsichord, and its key-board is placed in a somewhat oblique direction. The tone of the spinet was generally a *fifth* higher than that of the harpsichord.

The *clavichord* (Italian, *clavicordo*; German, *Clavier*, or *Klavier*), differs from the spinet inasmuch as it is of an oblong-square shape (Fig. 65), and especially in its being constructed with so-called *tangents*, *i.e.*, metal pins which press under the strings when the keys are struck. The strings are of thin brass wire. The oldest specimens of the clavichord still extant are from three to four feet in length, and about two feet in width. The lower keys are black, and the upper ones are white. There is only a single string for each tone and its upper semitone; thus, there is but one string for *C* and *C-sharp*, and likewise for *D* and *D-sharp*, and so on. The semitone is produced by a second tangent, which touches the string at a place a little distant from that at which it is touched by the tangent producing the whole-tone. On being pressed under the string, the tangent divides it into two vibrating parts, one of which is considerably longer than the other and gives the sound. The other part is too short to be distinctly audible, and therefore does not very perceptibly interfere with the clearness of the sound. Moreover, its vibration is checked by a strip of cloth interlaced with the strings. It will easily be understood that of the two tangents, the one which most shortens the sounding part of the string, must produce a tone of a higher pitch than the other.

Such was the construction of the *clavichord* until about the year 1700, when it was improved in so far as that each key was supplied with a separate string. The clavichord is pre-eminently a German instrument. Although now almost entirely supplanted by the pianoforte, it is still occasionally to be met with in the house of the German village schoolmaster and of the country parson. Though but weak in sound, it admits of much expression; and most of the German classical composers who lived before the invention of the pianoforte preferred the clavichord to the harpsichord. In England it has never become popular. Considering the simplicity of its construction, it might be surmised that the price of a clavichord was generally very moderate. In the latter half of the eighteenth century the prices charged for such instruments by some of the best

manufacturers were as follows:—Carl Lemme, in Brunswick, made clavichords of various qualities, which fetched from three to twelve Louis d'ors a-piece; he also made, for exportation to Batavia, clavichords with a compressed sounding-board, invented by his father in the year 1771; Krämer, in Göttingen, charged from four to fourteen Louis d'ors, according to size and finish; and Wilhelmi, in Cassel, charged from twenty to fifty thalers,—from about £3 to £7 10s.



**FIG. 66.—CLAVICEMBALO. Signed “Joanes Antonius Baffo, Venetus.”
Italian. Dated 1574. H. 9½ in., L. 83 in., W. 36 in. No. 6007-’59.
Victoria and Albert Museum.**

The *clavicembalo* (often designated merely *cembalo*) is called in German “Flügel,” on account of its shape somewhat resembling the wing of a bird. *Clavicembali* formerly in use generally had a compass of five octaves. The instrument was usually supplied with some stops by means of which the quality of sound could in some measure be modified. Furthermore, it was frequently made with two keyboards, one for the loud

and another for the soft tones. The harpsichord made in England was precisely of the same construction. In fact, the best harpsichord makers in England were emigrants from the continent, and the founders of some of the great pianoforte manufactories still flourishing in London. Burkhardt Tschudi, for instance, a harpsichord maker from Switzerland, was the founder of Broadwood's celebrated manufactory, which dates from the year 1732. Kirkman, a German (who, before he established himself in England, wrote his name Kirchmann) sold his harpsichords in London, according to the German Musical Almanac for the year 1782, at the price of from 60*l.* to 90*l.* apiece. In the beginning of the eighteenth century many of the harpsichords made in England had, according to Grassineau (Musical Dictionary, London, 1740), a compass of only four octaves.

However, already as early as in the sixteenth and seventeenth centuries, harpsichords or clavicembali, of a superior quality, manufactured by Hans Ruckers and his sons Jean and Andreas, were imported into England. The instruments of these celebrated Antwerp manufacturers were tastefully embellished, and the best Dutch painters not infrequently enriched them with devices. The consequence has been that after the invention of the pianoforte, many of these old harpsichords were taken to pieces in order to preserve the valuable panels. The price of a fine harpsichord by Ruckers about 1770, was £120.

The old *clavicembalo* by Antonio Baffo, of Venice ([Fig. 66](#)), has slips of prepared leather instead of the usual crowquills, which, if original, would show that the statement of some writers as to Pascal Taskin in Paris being the first to use leather is erroneous. Taskin, in constructing in the year 1768 the *Clavecin à peau de buffle*, may have revived an old invention, which, however, he seems to have much improved. He made a *clavecin* with three keyboards, two of which were connected with actions constructed of crowquills, and the third with an action of leather. The modification in quality of sound thereby obtained was greatly admired.

The illustration ([Fig. 67](#)) represents a clavecin made by Pascal Taskin in the year 1786. The case is highly ornamented with Japanese figures and gilding.

The invention of the *clavicembalo* as well as of the *clavicordo*, is by some old writers ascribed to Guido Aretinus (or Guido d'Arezzo), the famous monk who is recorded to have invented, in the year 1025, the Solmisation, and also to have first conceived the idea of employing lines and dots in the notation of musical sounds. Unauthentic though the tradition may be which assigns to Guido the invention of the stringed instruments with a keyboard, it appears very probable that some rude kind of clavichord was first constructed about his time, or soon after.

The *claviorganum*, or organ-harpsichord, consists of an organ and a harpsichord (or a spinet) combined. Either can be played separately or with the other together. The separation and the union are effected by means of a stop or a pedal. The claviorganum was, some centuries ago, not uncommon. It enables the performer to sustain the sound at pleasure, which on the harpsichord is as little possible as on the pianoforte. A *claviorganum* from Ightham Mote, near Sevenoaks, illustrated in [Fig. 68](#), affords evidence of a higher antiquity of instruments of this kind than might perhaps be expected. It bears the inscription, *Lodowicus Theewes me fecit*, 1579. There is scarcely more remaining of this interesting relic than the outer case; but this is so elaborately finished that, if the mechanism was constructed with equal care and success, it must have been a superior instrument. The maker is unknown in musical history. Perhaps he belonged to the family of Treu (also written Trew), musicians of repute in Anspach about the year 1600.



FIG. 67.—CLAVECIN. Made by Pascal Taskin of Paris. French. Dated 1786. H. 32¾ in., W. of keyboard, 30 in., L. 72 in. No. 1121-'69. Victoria and Albert Museum.



FIG. 68.—ORGAN-HARPSICHORD OR CLAVIORGANUM. Formerly in the chapel of Ightham Mote, near Sevenoaks, Kent. Probably English. Harpsichord, H. 9 in., L. 84 in., W. 35½ in. Organ case, H. 41 in., L. 91 in., W. 40 in. No. 125 125^a-’90. Victoria and Albert Museum.

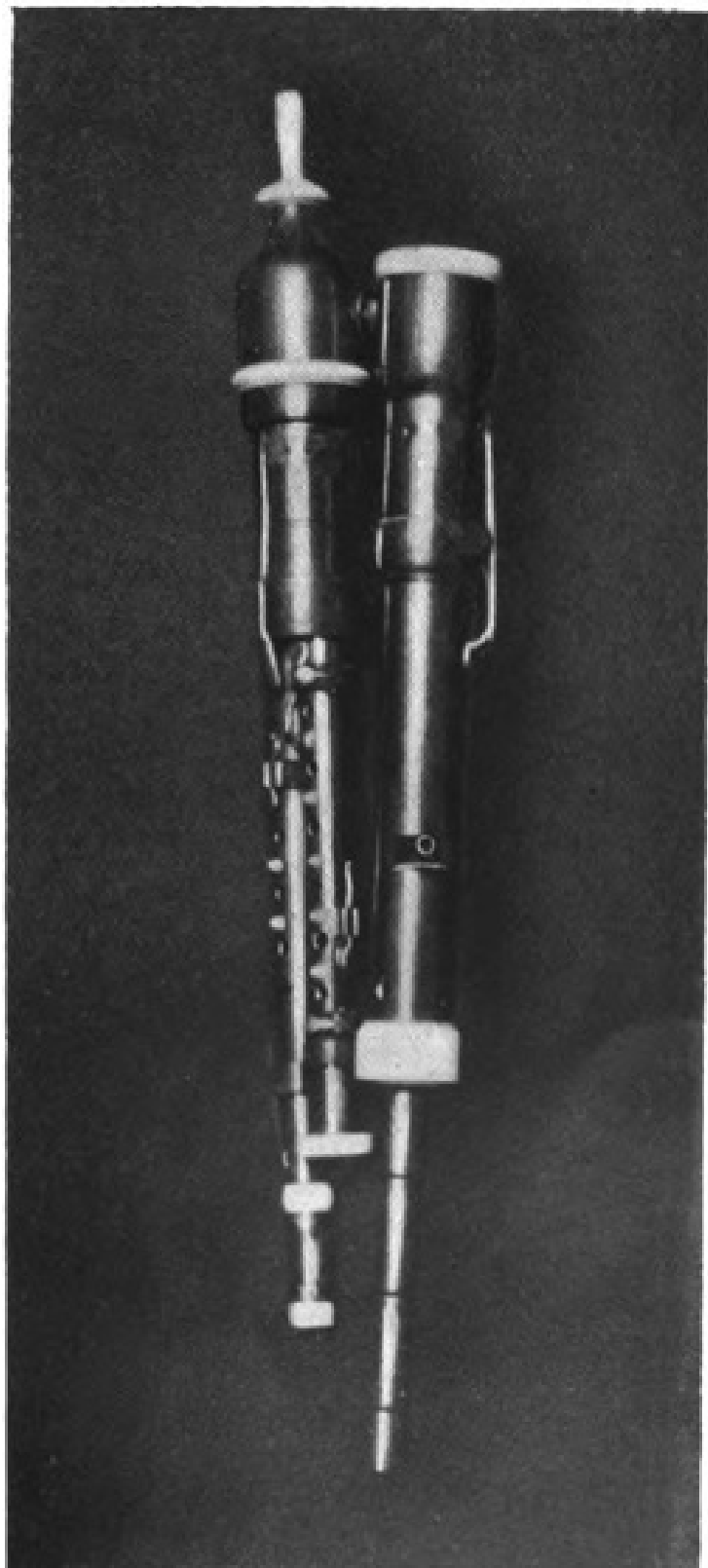


FIG. 69.—TRIPLE FLAGEOLET. Italian.

**About 1820. L. 20½ in. No. 295-'82.
Victoria and Albert Museum.**

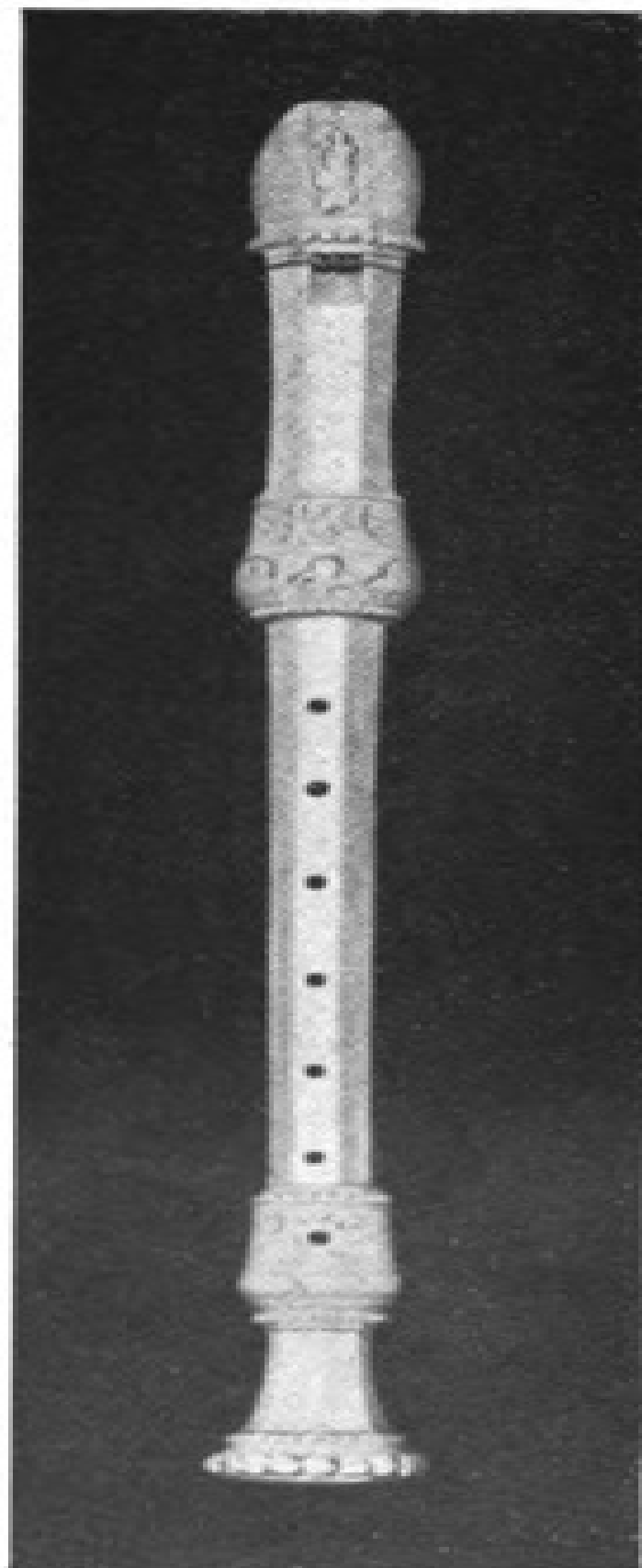


FIG. 70.—FLAUTO DOLCE, OR FLUTE.
Ivory. Inscribed “Anciuti a Milan, 1740.”
L. 18½ in. No. 7469-’61.
Victoria and Albert Museum.

The pianoforte, which now has entirely superseded the harpsichord, was first constructed at the beginning of the eighteenth century, in Italy and Germany. About the year 1767 it was from Germany introduced into England; but the English musicians for a considerable period objected to it, and preferred to retain the harpsichord.

That there was, in the time of Shakespeare, a species of flageolet, called *recorder*, is undoubtedly known to most readers from the stage direction in *Hamlet*: *Re-enter players with recorders*. The recorder is also mentioned by Milton, and described by Bacon, who states that “the figures of recorders, flutes and pipes are straight; but the recorder hath a less bore, and a greater above and below.” An illustration of this old instrument, which has now become very scarce, is given in “*The Genteel Companion; Being exact Directions for the Recorder: etc.*” London, 1683.

The *flauto dolce* (French, *flûte douce*, and *flûte à bec*), much in use some centuries ago, was made of various lengths (Fig. 70). The Germans called it *Pflockflöte*, i.e., a flute with a plug in the mouth-hole. The most common *flûte à bec* was made with six finger-holes, and its compass embraced somewhat more than two octaves. Several of the finger-holes required to be only partly covered in order to produce the desired tone. There was often a key on this instrument in addition to the finger-holes. This flute was much in favour in England; hence it was called in France “*Flûte d’Angleterre*.” It has gradually been supplanted by the “*Flûte traversière*,” or “German Flute.”

The *flageolet* (Fig. 71), the smallest *flûte à bec*, was formerly played in England even by ladies. Pepys, in his Diary (March 1st, 1666), records:—“Being returned home, I find Greeting, the flageolet-master, come, and teaching my wife; and I do think my wife will take pleasure in it, and it will be easy for her, and pleasant.”

The flageolet was made of various sizes. Pepys (Diary, January 20th, 1667) records:—“To Drumbleby’s, the pipemaker, there to advise about the

making of a flageolet to go low and soft; and he do show me a way which do, and also a fashion of having two pipes of the same note fastened together, so as I can play on one and then echo it upon the other, which is mighty pretty."

The *double flageolet* was invented by Bainbridge about the year 1800. The *triple flageolet* (Fig. 69) is less common but equally useless for musical performances of the present day. The "Harmonicon," London, 1830, records:—"Within these few years Mr. Bainbridge has added a bass joint to his double flageolet and the tone resembles the lower notes on a German flute. The effect produced by the combination of three notes is very good and mellifluous. The bass joint is fixed at the back of the double flageolet, and the breath is conveyed by means of a tube; and by the introduction of what are termed stop-keys, a solo, duet, or trio may be instantaneously performed. The bass notes are produced by keys pressed with the thumb of the left hand." The writer remarks that "this instrument being purely English, I consider it deserving of being recorded as a very ingenious invention."

The *hautboy* or *oboe* (Fig. 72) came into more general use about the year 1720.

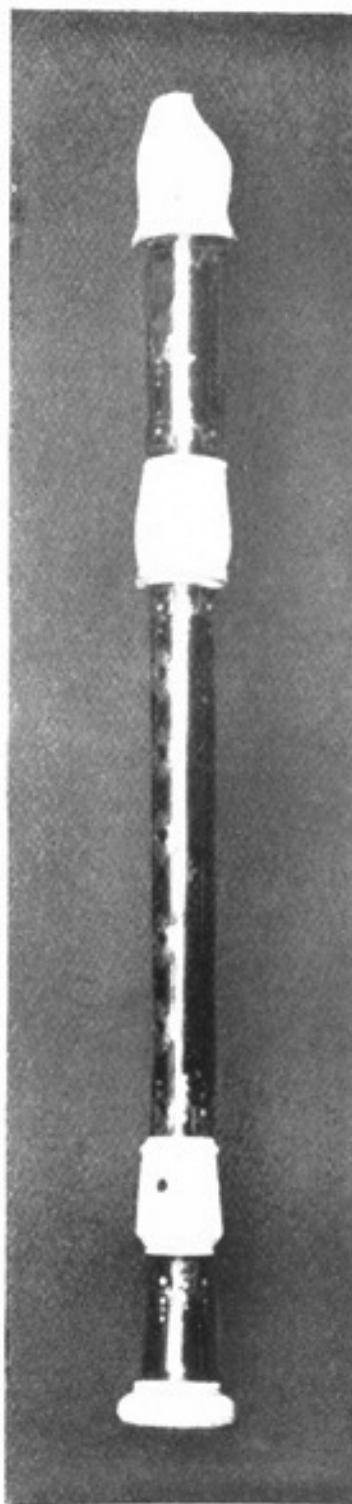


FIG. 71.—FLAGEOLET. Italian. Middle of 18th century. L. 20 in., Diam. of mouth, $1\frac{7}{8}$

in. No. 1124-'69.
Victoria and Albert Museum.

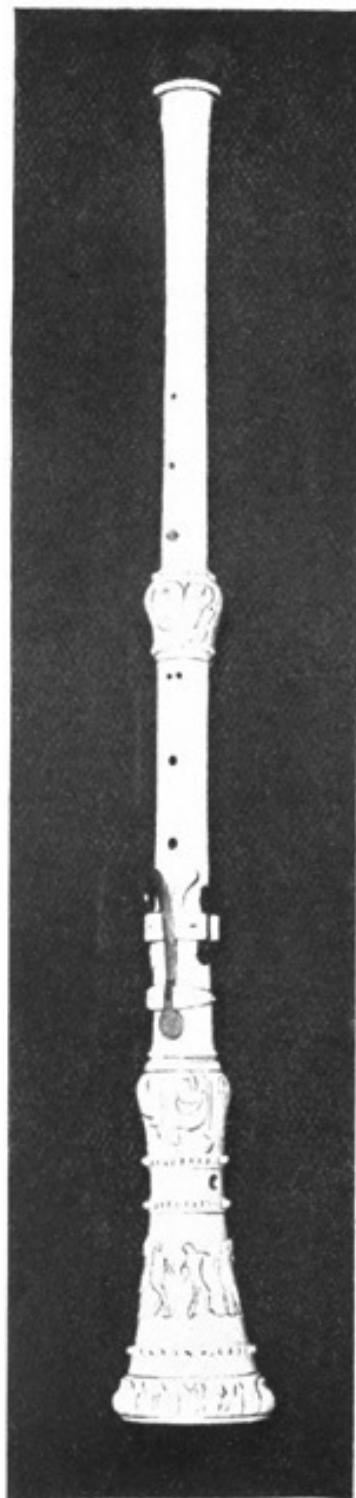


FIG. 72.—OBOE. Made by Anciuti of Milan; formerly in the possession of the composer Rossini. Latter half of 18th

century. L. 21½ in., Diam. of mouth, 2½ in.

No. 1127-69

Victoria and Albert Museum.

The most noteworthy kinds of the hautboy of the time of Handel and Sebastian Bach are,—the *oboe da caccia*, which is identical with the *corno inglese* (*English horn*, *cor anglais*), a large hautboy still occasionally employed in the orchestra, and the *oboe d'amore*, or *oboe lungo*, which has fallen into oblivion. The pitch of the *oboe d'amore* was a minor third lower than that of the common hautboy, or *oboe piccolo*; and its sound, owing to the narrowness of the bore at its further end, was rather weak, but particularly sweet.

The precursor of the hautboy was evidently the *bombardino*, or *chalumeau*. The *bombardino*, also called in Italian *bombardo piccolo*, was a small *bombardo*, an instrument of the hautboy kind, about three centuries ago much in use on the Continent.

The Germans called the *bombardo* “Pommer,” which appears to be a corruption of the Italian name. The *bombardo* was made of various sizes, and with a greater or smaller number of finger-holes and keys. That which produced the bass tones was sometimes of an enormous length, and was blown through a bent tube, like the bassoon, the invention of which it is said to have suggested.

The smallest instrument, called *chalumeau* (from *calamus*, “a reed”) is still occasionally to be found among the peasantry in the Tyrol and some other parts of the Continent. The Germans call it *Schalmei*, and the Italians *piffero pastorale*. In England it was formerly called *shawm* or *shalm*.

The *clarinet*, likewise an instrument of this class, is said to have been invented by Denner, in Nürnberg, about the year 1700. The clarinet has only a single vibrating reed in the mouth-piece; the hautboy has a double one.

The invention of the *bassoon* (Italian, *fagotto*; French *basson*; German, *Fagott*) is ascribed to Afranio, a canon of Ferrara, who constructed the first in the year 1539. The instrument was, however, an improved *bombardo* rather than a new invention. As early as the year 1550, the celebrated wind-instrument maker Schnitzer, in Nürnberg, manufactured bassoons which

were considered as very complete. [Fig. 73](#) illustrates a species of bassoon bound with brass with brass keys, and complete with mouth-piece and reed.

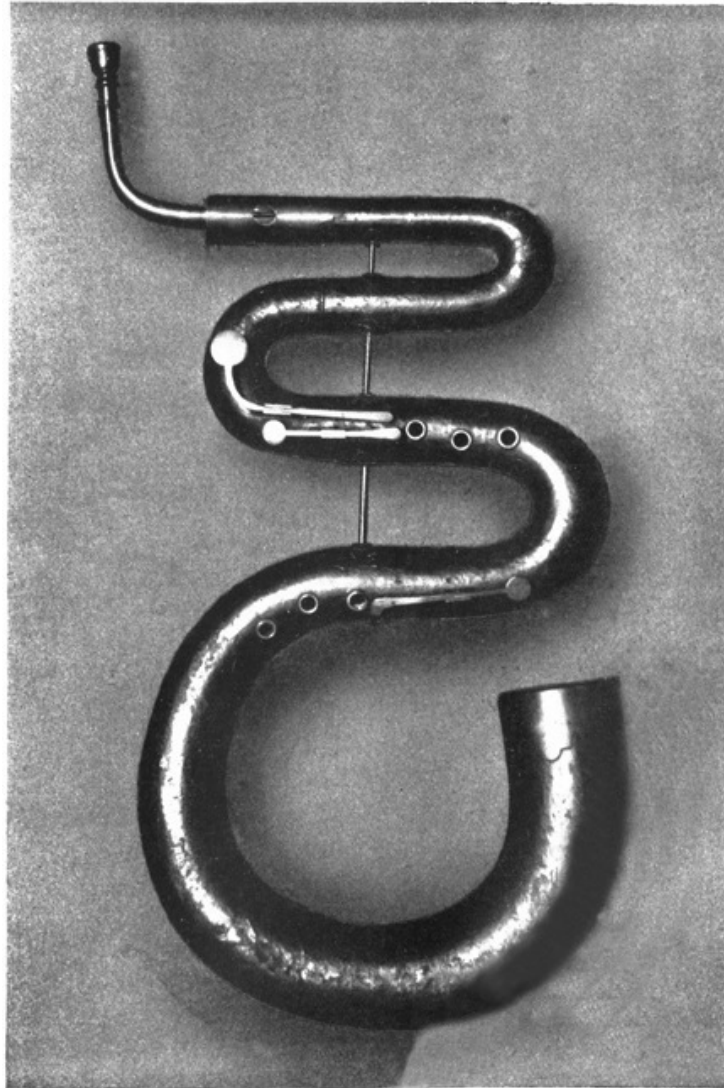
Various bassoons of small dimensions in use about two centuries ago, and earlier (the *dolciano*, *Quartfagott*, *Quintfagott*, *tenor-bassoon*, *corthol*, etc.), are now antiquated.



**FIG. 73.—BASSOON, species of. English.
Late 18th or early 19th century. L. 48¼ in.
No. 637-'72.
Victoria and Albert Museum.**

In the list of musical instruments of Sir Thomas Kytson, of Hengrave Hall, about the year 1600, recorded in the "History and Antiquities of Hengrave, Suffolk," by John Gage, London, 1822, is mentioned "A Curtall," which was probably the *corthol* or French *courtaut*, an early kind of bassoon, a specimen of which, dating from the fifteenth century, is preserved in the Conservatoire de Musique at Paris. According to Prætorius (anno 1619) the *fagotto piccolo*, a small species of bassoon, was called in England *single corthol*.

The invention of the *serpent* ([Fig. 74](#)) is attributed to Edme Guillaume, a canon of Auxerre in France, anno 1590. It was, however, no new invention, properly speaking, but merely an improvement upon the old *Basszinken*, the management of which was rendered more convenient by giving a serpentine winding to the tube. This instrument subsequently became rather popular. It was used in military bands and in processions until about the middle of the last century. The French made use of it also in church to support the voices. Towards the end of the eighteenth century it appears to have still been a common substitute for the organ in France. Dr. Burney, in his "Journal," London, 1773, states that he frequently met with it in the churches of that country, and he expresses a more favourable opinion of its suitableness for promoting edification than might have been expected from a refined musician:—"It gives the tone in chanting, and plays the bass when they sing in parts. It is often ill-played, but if judiciously used would have a good effect. It is, however, in general overblown, and too powerful for the voices it accompanies; otherwise, it mixes with them better than the organ, as it can augment or diminish a sound with more delicacy, and is less likely to overpower or destroy, by a bad temperament, that perfect one of which the voice only is capable."



**FIG. 74.—THE SERPENT. Made by
Gerock Wolf, in London. English. Early
19th century. L. 28 in. No. 286-'82.
Victoria and Albert Museum.**



FIG. 75.—SERINETTE OR BIRD ORGAN. French. Period of Louis XIV. H. 8½, L. 11½ in., W. 9 in. No. 629-'68. Victoria and Albert Museum.



**FIG. 76.—ORGAN (Positive). Bears the arms of John George I.,
Elector of Saxony (b. 1585, d. 1656). German. Dated 1627.**

H. 45½ in., W. 27½ in. No. 2-'67.

Victoria and Albert Museum.

The *serinette*, or bird organ ([Fig. 75](#)), was formerly used in France by ladies to teach airs to little singing birds, especially to a kind of siskin or canary, called in French *serin*; hence the name of the instrument.

The *organ positive* ([Fig. 76](#)) is distinguished from the *organ portative* in so far that the former was a larger instrument, generally placed on a table and blown by an attendant, while the latter was carried about by the performer in religious processions and on such-like occasions.

In England some rude species of organ is said to have been used in public worship as early as about the middle of the seventh century. It was, however, on the Continent, principally in Germany, that almost all the important improvements originated which gradually brought the organ to its

present high degree of perfection. Many old organs of fine workmanship are still extant in the churches of Germany. During the 18th century especially several large organs of deserved celebrity were built in that country; suffice it to instance those of the brothers Andreas and Gottfried Silbermann. In England the important inventions of the continental builders were not readily adopted. Recently, however, several huge organs of very fine workmanship have been constructed in England, chiefly for use in concert rooms, or public halls.

The *regal*, often mentioned in English literature of the time of Shakespeare, and earlier (*see also* [p. 96](#)), was a small *organ portative*. There was till about the end of the 18th century a “Tuner of the Regals,” in the Chapel Royal St. James’s, with a salary of 56*l*. The name *regal* is supposed to have been derived from *rigabello*, a musical instrument of which scarcely more is known than that it was played in the churches of Italy before the introduction of the organ.

The expression “a payre of regalls,” used by writers some centuries ago, evidently implies only a single instrument. Thus also the virginal is not unfrequently mentioned as “a payre of virginalls.” Moreover, it appears that the regal was occasionally made with two sets of pipes, so as to constitute a double organ of its kind.

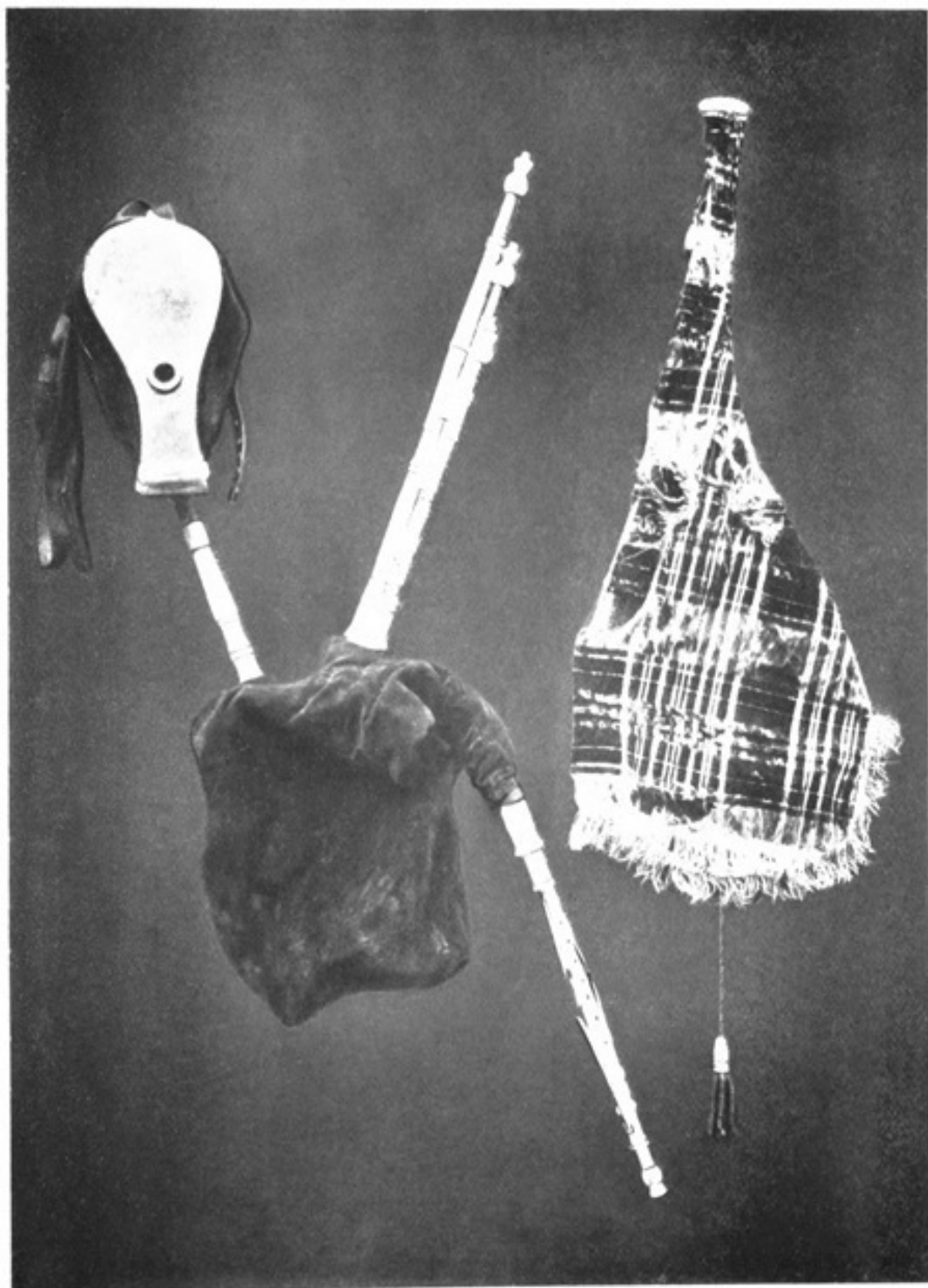
In the following lines from Sir W. Leighton’s “Teares or Lamentations of a Sorrowful Soule,” London, 1613, this little organ is mentioned in combination with other curious instruments now antiquated, most of which will be found in the present collection:—

"Praise him upon the claricoales,
The lute and simfonie:
With the dulsemers and the regalls,
Sweete sittrons melody."

The *bagpipe* ([Fig. 77](#)) appears to have been from time immemorial a special favourite instrument with the Celtic races; but it was perhaps quite as much admired by the Slavonic nations. In Poland, and in the Ukraine, it used to be made of the whole skin of the goat in which the shape of the animal, whenever the bagpipe was expanded with air, appeared fully

retained exhibiting even the head with the horns; hence the bagpipe was called *kosà*, which signifies a goat.

The bagpipe is of high antiquity in Ireland, and is alluded to in Irish poetry and prose said to date from the tenth century. A pig gravely engaged in playing the bagpipe is represented in an illuminated Irish manuscript, of the year 1300.



**FIG. 77.—BAGPIPES. English. 18th century. L. 30 in. No. 1197-'03.
Victoria and Albert Museum.**

The *bell* has always been so much in popular favour in England that some account of it must not be omitted. Paul Hentzner, a German, who visited England in the year 1598, records in his journal: "The people are vastly fond of great noises that fill the ear, such as firing of cannon, drums, and the ringing of bells; so that in London it is common for a number of them that have got a glass in their heads to go up into some belfry, and ring the bells for hours together for the sake of exercise." This may be exaggeration,—not unusual with travellers. It is, however, a fact that bell-ringing has been a favourite amusement with Englishmen for centuries.

The way in which church bells are suspended and fastened, so as to permit of their being made to vibrate in the most effective manner without damaging by their vibration the building in which they are placed, is in some countries very peculiar. The Italian *campanile*, or bell tower, is not unfrequently separated from the church itself. In Servia the church bells are often hung in a frame-work of timber built near the west end of the church. In Zante and other islands of Greece the belfry is usually separate from the church. The reason assigned by the Greeks for having adopted this plan is that in case of an earthquake the bells are likely to fall and, were they placed in a tower, would destroy the roof of the church and might cause the destruction of the whole building. Also in Russia a special edifice for the bells is generally separate from the church. In the Russian villages the bells are not unfrequently hung in the branches of an oak-tree near the church. In Iceland the bell is usually placed in the lych-gate leading to the graveyard.

The idea of forming of a number of bells a musical instrument such as the *carillon* is said by some to have suggested itself first to the English and Dutch; but what we have seen in Asiatic countries sufficiently refutes this. Moreover, not only the Romans employed variously arranged and attuned bells, but also among the Etruscan antiquities an instrument has been discovered which is constructed of a number of bronze vessels placed in a row on a metal rod. Numerous bells, varying in size and tone, have also been found in Etruscan tombs. Among the later contrivances of this kind in European countries the sets of bells suspended in a wooden frame, which we find in mediæval illuminations, deserve notice. In the British Museum is

a manuscript of the fourteenth century in which King David is depicted holding in each hand a hammer with which he strikes upon bells of different dimensions, suspended on a wooden stand.

It may be supposed that the device of playing tunes by means of bells merely swung by the hand is also of ancient date. In Lancashire each of the ringers manages two bells, holding one in either hand. Thus, an assemblage of seven ringers insures fourteen different tones; and as each ringer may change his two notes by substituting two other Dells if required, even compositions with various modulations, and of a somewhat intricate character, may be executed,—provided the ringers are good timeists; for each has, of course, to take care to fall in with his note, just as a member of the Russian horn band contributes his single note whenever it occurs.

Peal-ringing is another pastime of the kind which may be regarded as pre-eminently national to England. The bells constituting a peal are frequently of the number of eight, attuned to the diatonic scale. Also peals of ten bells, and even of twelve, are occasionally formed. A peculiar feature of peal-ringing is that the bells, which are provided with clappers, are generally swung so forcibly as to raise the mouth completely upwards. The largest peal, and one of the finest, is at Exeter Cathedral: another celebrated one is that of St. Margaret's, Leicester, which consists of ten bells. Peal-ringing is of an early date in England; Egelric, abbot of Croyland, is recorded to have cast about the year 960 a set of six bells.

The *carillon* is especially popular in the Netherlands and Belgium, but is also found in Germany, Italy, and some other European countries. It is generally placed in the church tower, and also sometimes in other public edifices. The statement repeated by several writers that the first carillon was invented in the year 1481 in the town of Alost is not to be trusted, for the town of Bruges claims to have possessed similar chimes in the year 1300. There are two kinds of carillons in use on the Continent, viz.: clock chimes, which are moved by machinery, like a self-acting barrel-organ; and such as are provided with a set of keys, by means of which the tunes are played by a musician. The carillon in the "Parochial-Kirche" at Berlin, which is one of the finest in Germany, contains thirty-seven bells; and is provided with a key-board for the hands and with a pedal, which together place at the disposal of the performer a compass of rather more than three octaves. The

keys of the manual are metal rods somewhat above a foot in length, and are pressed down with the palms of the hand. The keys of the pedal are of wood; the instrument requires not only great dexterity, but also a considerable physical power. It is astonishing how rapidly passages can be executed upon it by the player, who is generally the organist of the church in which he acts as *carillonneur*. When engaged in the last-named capacity he usually wears leathern gloves to protect his fingers, as they are otherwise apt to become ill fit for the more delicate treatment of the organ.

The want of a contrivance in the *carillon* for stopping the vibration has the effect of making rapid passages, if heard near, sound as a confused noise; only at some distance are they tolerable. It must be remembered that the *carillon* is intended especially to be heard from a distance. Successions of tones which form a consonant chord, and which have some duration, are evidently the most suitable for this instrument.

Indeed, every musical instrument possesses certain characteristics which render it especially suitable for the production of some particular effects. The invention of a new instrument of music has, therefore, not unfrequently led to the adoption of new effects in compositions. Take the pianoforte, which was invented in the beginning of the eighteenth century, and which has now obtained so great a popularity; its characteristics inspired our great composers to the invention of effects, or expressions, which cannot be properly rendered on any other instrument, however superior in some respects it may be to the pianoforte. Thus also the improvements which have been made during the present century in the construction of our brass instruments, and the invention of several new brass instruments, have evidently been not without influence upon the conceptions displayed in our modern orchestral works.

Imperfect though this essay may be it will probably have convinced the reader that a reference to the history of the music of different nations elucidates many facts illustrative of our own musical instruments, which to the unprepared observer must appear misty and impenetrable. In truth, it is with this study as with any other scientific pursuit. The unassisted eye sees only faint nebulae, where with the aid of the telescope bright stars are revealed.

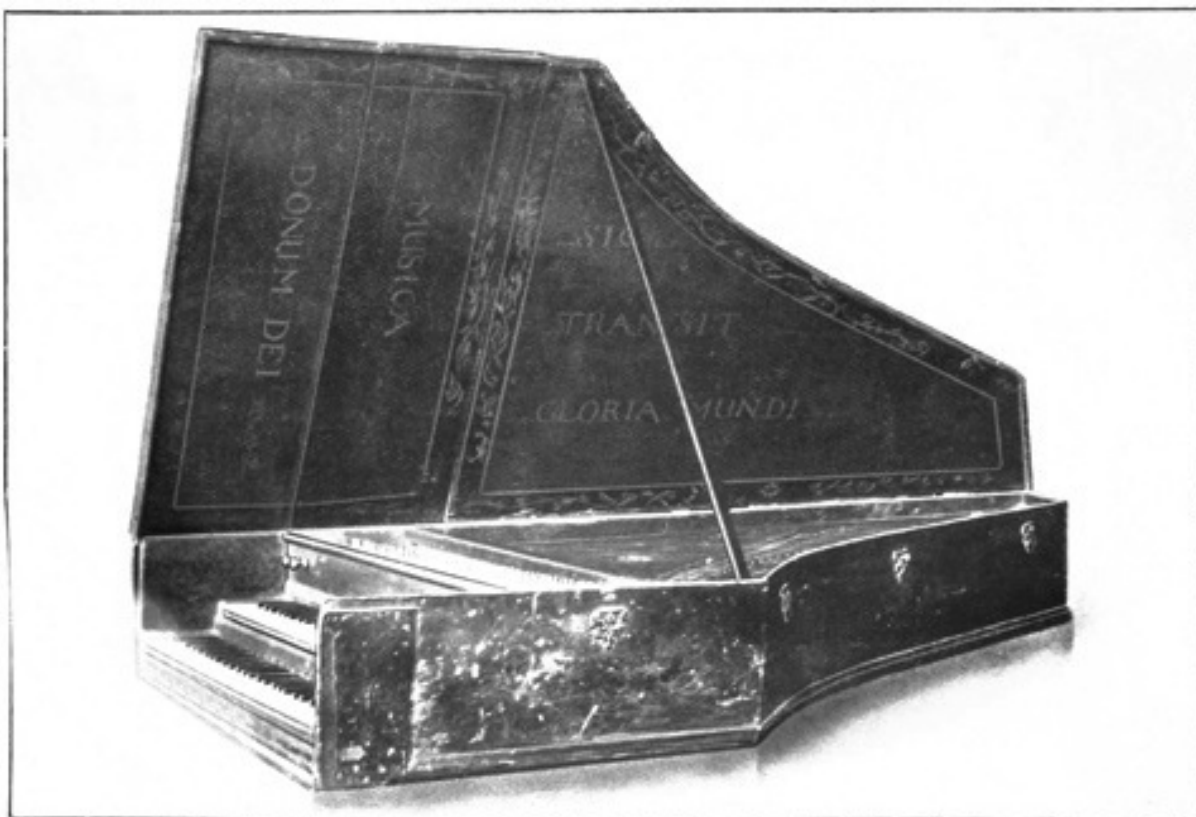


FIG. 78.—HANDEL'S HARPSICHORD. Made by Andreas Ruckers, of Antwerp, 1651. H. 36 in., L. of top 80 in., W. 36 in. No. 1079-'68. Victoria and Albert Museum.

APPENDIX.

HANDEL'S HARPSICHORD.

The following documentary evidence of this instrument's authenticity as Handel's harpsichord ([Fig. 78](#)) has been transmitted by Messrs. Broadwood:

33, Great Pulteney Street, London,
November 18th, 1868.

Handel's harpsichord was bought by us of Mr. Hooper, a pianoforte tuner at Winchester, in 1852. He had obtained it from Dr. Chard, the Cathedral organist of that city, who had taken pains to prove it to be the same instrument which Handel had left by will to his friend and amanuensis, Christopher Smith. In Handel's will, dated June, 1750, was the bequest:—'I give and bequeath to Christopher Smith my large harpsichord, my little house-organ, my music books, and 500*l.* sterling;' and in a codicil, dated 6th of August, 1756: 'I give to Christopher Smith 1,500*l.* additional to the legacy already given to him in my will.' Dr. Chard wrote to the Rev. George Coxe, of Twyford (Rector of St. Michael's, Winchester), to obtain his testimony to the identity of this harpsichord with the '*Large Harpsichord*' of the will. Mr. Coxe was nearly related to Smith, and had frequently heard him play upon it. On the 13th of May, 1842, and in the presence of witnesses, Mr. Coxe confirmed this. Dr. Chard states in the document signed by Mr. Coxe, that this harpsichord was left with a large collection of Handel's MSS. by Christopher Smith to his step-daughter, the Dowager Lady Rivers, who parted with it to Mr. Wickham, a surgeon, who, in his turn,

parted with it to the Rev. Mr. Hawtrey, Prebendary of Winchester, after whose death it came into the possession of Dr. Chard.

This interesting relic of Handel is also worthy of notice from having been one of the best-constructed instruments of the celebrated harpsichord makers, the Ruckers family of Antwerp. It is not remarkable for any beauty of decoration beyond the conventional ornamentation of the period; but the structure shows great skill in the manufacture, and that the harpsichord had become nearly perfected in the middle of the seventeenth century.

The two key-boards were used for variety of tone. The lower key-board, the *jacks* of which acted upon two sets of strings in unison, and one set an octave higher, was the louder in tone; the upper key-board, acting on one set of strings only, was the softer. But the lower key-board could be made to act upon one set of strings only, by means of stops drawn out by the hand of the performer. In touching the keys, a distinctive quality of tone may still be recognised, particularly in the higher notes, a reedy but soft and delicate *timbre* testifying to the former beauty of the instrument. It may be assumed as certain that the keys are not of Handel's time. We do not know when the present key-boards were put, or by whom, but the style of the white and black keys is undoubtedly modern. Neither can it be doubted that there were originally keys in keeping with the fashion of the harpsichord, which we may suppose to have been worn out, to account for the substitution of those existing. The case of deal, black japanned, the brass hinges, the ornamentation, and the mottoes are original. Inside the top is inscribed:—

Sic transit Gloria Mundi;

on the flap or folding of the top—

Musica Donum Dei;

and on the slip of wood above the upper keys—

Andreas Ruckers me fecit, Antwerpiae, 1651.

There is a date on the sounding-board "1651," and in the ornamental sound-hole are the initials "A. R." Among the flowers represented on the sounding-board may be seen a concert of monkeys, one beating time, another playing the viol da gamba, etc. A third motto existed until about fifteen years ago—*Acta Virum Probant*. This was rubbed off by a workman engaged in mending the lock-board (upon which this motto was), which had been split.

As a musical instrument, this harpsichord has lived its life. It is not now capable of being tuned, and any attempt to improve the accord of it might prove disastrous by the sounding-board giving way altogether. It is, therefore, of consequence to the preservation of the woodwork that tuning should not be attempted.

JOHN BROADWOOD & SONS.

Letter to the Rev. G. Coxe, Twyford, Rector of St. Michael's, Winchester:--

MY DEAR SIR,—Will you oblige me by certifying (if I am correct) the following:—

The celebrated Mr. Smith (or Schmidt) was Handel's private friend, and amanuensis. This said Mr. Smith was presented by Handel with his favourite fine double-keyed harpsichord, made by the best makers of the day, Andreas Ruckers of Antwerpia, 1651. This said instrument you have heard repeatedly Mr. Smith play on. Mr. Smith was father-in-law to you as well as your sister, the late Dowager Lady Rivers; and at his death, the said harpsichord, together with a large collection of Handel's oratorios, etc., etc., MSS., came into the hands of the Dowager Lady Rivers. This instrument was parted with to a Mr. Wickham, surgeon, who parted with it to the Rev. W. Hawtrey, Prebendary of Winchester Cathedral, upon the death of whom I purchased it at the sale of his effects; and in my possession it still remains. Is not this the identical instrument now spoken of? Your early answer to these queries, as the only living witness, will oblige.

Dear Sir,
Yours faithfully,
G. W. CHARD.

P.S.—Will you oblige me by certifying on this sheet of paper,
and returning it?

Answer.

I certify that the above statement is correct, as far as my
knowledge goes.

GEORGE COXE.

Twyford, *May 13th, 1842.*

Witness to the above signature,

SUSANNA GREGG.

JAMES HARRIS.

[1] Figured and described in Lartet & Christy's *Reliquiæ Aquitanicæ*, London, 1865-75, Pl. B. v., p. 48.

[2] The best instance is to be found in Lepsius' *Denkmäler*, III. 106a., where a music-school of the Akhenaten period (about 1400 B.C.) is depicted.

[3] For coloured plate after this painting see Wilkinson's *Ancient Egyptians*, Vol. I., Pl. xii. (facing page 480).

[4] See *Transactions of the Royal Society of Edinburgh*, Vol. xx., Part I (1850).

[5] See E. Aldis, *Carvings and Sculptures of Worcester Cathedral* (IV).

[6] See illustration in *Ann. Arch.*, IV., p. 37.

[7] See illustration in *Ann. Arch.*, iv., p. 98.

[8] For a more complete list of lute-makers see Von Lütgendorff, *Die Geigen- und Lautenmacher vom Mittelalter bis zur Gegenwart*, Frankfort, 1904.

INDEX

- Abyssinian instruments, [20](#).
- Acocotl, [67](#).
- Adair, quoted, [81](#).
- Adufe, [25](#).
- Æolian harp, [4](#).
- African instruments in America, [82](#).
- Ajacaxtli, [72](#).
- Al-Farabi, lutist, [55-57](#).
- American Indian instruments, [58 seq.](#)
- American Indians, metrical psalms of, [81](#).
- American Indians, musical performances of, [75](#).
- American Indians, North, musical talent of, [81](#).
- Anglo-Saxon instruments, [84](#), [86](#), [90](#), [94](#).
- Arab instruments, [3](#), [36](#), [48](#), [53 seq.](#), [108](#).
- Arabs in Spain, [36](#), [56](#).
- Archlute, [101](#), [105](#).
- Ashantee, trumpet from, [2](#).
- Ash-shakandi, [55](#).
- Asor, [20](#).
- Assyrian instruments, [16 seq.](#)
- Aulos, [31](#).
- Aztecs, instruments of the, [58](#), [59](#).

- Bach, [115](#).
- Bacon, quoted, [125](#).
- Bagpipe, Celtic, [130](#).
 - " Greek, [130](#).
 - " Hebrew, [23](#).
 - " Irish, [130](#).
 - " mediæval, [102](#).
 - " Persian, [53](#).

- ” Polish, [130](#).
- ” Roman, [35](#).

Bainbridge, inventor, [126](#).

Banduria, [110](#).

Bansi, [47](#).

Barbitos, [27](#), [30](#), [33](#).

Baryton, [116](#).

Bassoon, [127](#), [128](#).

Bass-viol, [114](#), [119](#).

Basszinken, [128](#).

Bells, Assyrian, [18](#).

- ” Buddhist, [80](#).
- ” Chinese, [39](#), [40](#).
- ” Egyptian, [14](#), [15](#).
- ” English, [131](#).
- ” Etruscan, [132](#).
- ” Hebrew, [25](#).
- ” Japanese, [46](#).
- ” Mexican, [73](#).
- ” Peruvian, [73](#), [80](#).
- ” Roman, [36](#), [132](#).
- ” hanging of, [131](#).
- ” ringing of, [131](#), [132](#).

Bene, [11](#).

Beni Hassan, painting at, [21](#).

Bernhard, inventor of the pedal, [96](#).

Beverley Minster, sculpture at, [97](#).

Bîn, [49](#).

Bird Organ, [129](#).

Biwa, [44](#).

Blasius, St., manuscript at, [89](#), [90](#).

Bombardino, [127](#).

Bombardo, [127](#).

Bombulom, [97](#).

Bone instruments, [58](#).

Boscherville, St. Georges de, sculpture from, [99](#).

Botuto, [68](#).

Bow, [50](#), [55](#), [88](#), [90](#), [113](#), [119](#).
Bridges, movable, [44](#).
Bruce, his discovery of harps on frescoes, [11](#).
Buccina, [35](#).
Bûche, [118](#), [119](#).
Budbudika, [47](#).
Buddhism, [39](#), [43](#), [52](#).
Buddhist Temples, bas-reliefs on, [43](#), [44](#).
Bunibulum, [97](#).
Bunting, quoted, [88](#).
Burmese instruments, [2](#), [3](#), [42](#).
Burney, Dr., quoted, [128](#).

Cachua, Peruvian dance, [79](#).
Calamus, [34](#).
Cambodia, temples in, [43](#).
Capistrum, [34](#).
Carians, pipes of the, [28](#).
Carillon, [131](#), [133](#).
Caroados, trumpet of the, [67](#).
Castanets, Egyptian, [14](#).
 " Greek, [32](#).
 " Roman, [36](#).
Cembalo, [122](#).
Ceylon, instruments of, [51](#).
Chalil, [23](#).
Chalumeau, [127](#).
Chang, [53](#).
Chanrares, [73](#).
Chatzozerah, [24](#).
Chayna, [62](#), [79](#).
Chelys, [28](#), [29](#), [33](#), [47](#).
Chên, [40](#).
Cheng, [6](#).
Chhilchiles, [72](#).
Ch'ih, [42](#).
Chimes, [133](#).

Ch'in, [43](#), [44](#).
Chinese "Board of Music," [78](#).
 " instruments, [2](#), [3](#), [4](#), [6](#), [37 seq.](#), [43](#).
Ch'ing, [37](#), [38](#), [39](#).
Chin-ku, [41](#).
Chiriqui Indians, pipe of, [60](#), [79](#).
Chiterna, [109](#).
Chitarrone, [106](#).
Ch'iu (wood), [41](#).
Ch'un-tu, [40](#).
Chorus, or choron, [93](#).
Chu, [41](#).
Chung, [39](#).
Cionar cruit, [89](#).
Cithara, [33](#), [84](#), [85](#), [94](#).
 " Anglica, [89](#).
 " Teutonica, [89](#).
Cither, [109](#), [119](#).
Cithern, or cittern, [102](#), [109](#).
Citole, [86](#).
Cittern, [102](#), [109](#).
Clarín, [67](#).
Clarinet, [127](#).
Clarion, mediæval, [102](#).
Clarseth, [110-112](#).
Clavecin, [124](#).
Clavicembalo, [6](#), [122](#), [124](#).
Clavichords, [121](#).
 " makers of, [122](#).
 " prices of, [122](#).
Clavicordo, [124](#).
Claviorganum, [124](#).
Conch trumpets, Hindu, [47](#).
 " " Mexican, [80](#).
Confucius, [37](#), [39](#), [40](#), [43](#).
Congo, instrument of the, [2](#).
Constantinople, obelisk at, [95](#).

Cor anglais, [127](#).
 Corno inglese, [127](#).
 Cornu, Etruscan, [32](#).
 " Roman, [33](#), [35](#).
 Corthol, [128](#).
 Courtaut, [128](#).
 "Chronicon picturatum Brunswicense," quoted, [91](#).
 Crotala, [36](#).
 Crowd, *see* Crwth.
 Crusaders, [36](#).
 Crusmata, [36](#).
 Crwth, [89](#), [90](#), [113](#).
 Cuddos nut, instrument made of, [52](#).
 "Curtail," A, [128](#).
 Cymbals, Assyrian, [18](#).
 " Egyptian, [14](#), [15](#).
 " Greek, [32](#).
 " Hebrew, [25](#).
 " mediæval, [103](#).
 " Roman, [36](#).
 Cymbalum, [36](#), [97](#).
 Cythera (cithara), [109](#).

Dalzell, Sir J. G., quoted, [116](#).
 Damaras, [6](#).
 Damaru, [47](#).
 Darabuka, [14](#), [24](#).
 Darius, [19](#).
 David, King, [19](#).
 Day, Major C. R., [49](#), [52](#).
 Diaulos, [31](#).
 Diff, [25](#).
 Doff, [25](#).
 Dōhachi, [45](#).
 Dolciano, [128](#).
 Dora, [45](#).
 Dordogne, [9](#).

Double-bass, [117](#).
" flageolet, [126](#).
" pipe, in Anglo-Saxon MS., [84](#).

Double-pipe, Egyptian, [13](#).

" " Greek, [31](#).
" " Phœnician, [36](#).
" " Roman [34](#), [35](#).

Dragonetti, Signor, [117](#), [118](#).

Drums, American Indian, [82](#).

" Assyrian, [17](#), [18](#).
" Chinese, [41](#).
" Egyptian, [14](#).
" Fiji, [80](#).
" Greek, [32](#).
" Hebrew, [24](#).
" Hindu, [47](#).
" Japanese, [45](#).
" mediæval, [56](#), [97](#).
" Mexican, [70](#).
" New Guinea, [2](#).
" Persian, [53](#).
" Peruvian, [72](#).
" of Tonga, [80](#).
" of Torres Strait Islands, [80](#).

Dublin Museum, harps in, [111](#).

Dulcimer, [6](#).

" Anglo-Saxon, [86](#).
" Assyrian, [17](#).
" Greek, [30](#).
" Hebrew, [19](#).
" Persian, [54](#), [55](#).

Egyptian instruments, [8](#), [10](#) *seq.*, [27](#), [98](#).

Elizabeth, Queen, [119](#), [120](#).

El-ood, [54](#), [56](#).

English instruments, [104](#).

Etruscan " , [32](#) *seq.*

Europe, introduction of instruments in, [36](#).
European instruments, [83 seq.](#)
Evelyn, quoted, [106](#), [116](#).
Exeter Cathedral, minstrel gallery in, [102](#).

Fagott, [127](#).
Fagotto piccolo, [128](#).
Fang-hsiang, [40](#).
Fiddle, Anglo-Saxon, [90](#).
 " Bengalese, [50](#).
 " Chinese, [51](#).
 " German, [90](#).
 " Hindu and Indian, [50](#), [88](#).
 " Moorish, [90](#).

Fidis or Fides, [33](#).
Fidla, [113](#).
Finnish instrument, [47](#), [88](#).
Fistula, [35](#).
Fithele (fiddle), [114](#).
Flageolet, English, [125](#), [126](#).
 " Japanese, [45](#).

Flauto dolce, [125](#).
Flutes, American Indian, [82](#).
 " Arab, [55](#).
 " Aztec, [60](#).
 " Chinese, [42](#).
 " Egyptian, [12](#).
Flutes, Etruscan, [32](#).
 " German, [126](#).
 " Greek, [31](#).
 " of Guiana Indians, [62](#).
 " Hebrew, [23](#), [26](#).
 " Hindu, [47](#).
 " Japanese, [45](#).
 " Mexican, [58 seq.](#)
 " Peruvian, [58 seq.](#)
 " Phrygian, [28](#).

” Roman, 34.
Flûte à bec, 125.
” d’Angleterre, 125, 126.
” traversière, 126.
Forkel, quoted, 23.
Fortunatus, quoted, 89, 90.
Franz, Karl, 115.
Free reed, 5.
French instruments, 112, 125, 126, 128, 129.
Frestele, Fretel or Fretiau, 94.
Fuye, 45.

Gage, John, quoted, 128.
Gaspard di Salo, 118.
Gerbert, Abbot, mentioned, 84, 89, 90.
Gittern, 56, 102, 108.
Gittith, 25, 26.
Gizeh, 13.
Gongs, Chinese, 45.
” Egyptian, 14.
” Japanese, 45.
” Mexican, 80.
” Tezcucan, 73.
Greek instruments, 27 *seq.*
Guatemala, instrument of, 82.
Guitar, instruction books for, 108, 109.
Guitar, Japanese, 44.
” mediæval, 102.
” post-mediæval, 108, 109.
” Spanish, 110.
Gut-komm, 43.
Gythorn, 108.

Handel’s harpsichord, 135.
Harmonica, 97.
Harmonicon, Chinese, 2, 37, 40.
Harmonicon, The, quoted, 126.

Harps, Anglo-Saxon, 87.
" Arabian, 53.
" Assyrian, 16, 28.
" Burmese, 16.
" Celtic, 87.
" Egyptian, 11.
" Finnish, 88.
" French, 112.
" German, 87.
" Greek, 28, 29.
" Hebrew, 19.
" Hindu, 50.
" Irish, 88, 110-112.
" mediæval, 89, 100-102.
" Persian, 53.
" Scandinavian, 87.
Harp-guitar, 110.
" lute, 110.
Harpsichord, 116, 121, 123.
" Handel's, authenticity of, 135 *seq.*
Harpsichord-makers, 123, 136.
Harp-theorbo, 110.
Harpu, 88.
Harp-ventura, 110.
Hautboy, 126.
Haydn, 116.
Hebrew instruments, 19 *seq.*
Hentzner, Paul, quoted, 131.
Hichiriki, 45.
Hindu instruments, 3, 46, 52, 88, 89, 93.
Hindus, musical scale of, 50.
Holmos or mouth-piece, 35.
Horn, English, 127.
" Greek, 32.
" Hebrew, 24.
Hsiao, 42.
Hsüan, 42.

Hsüan-chung, 39.

Huanca, 72.

Huayllaca, 62.

Huayra-puhura, 63, 79.

Huehuetl, 71, 80.

Hydraulis, 32.

Icelandic instrument, 114.

Ikuta-goto, 44, 45.

Instrument makers, 106, 111, 114-116, 118, 122-126, 128, 129, 136, 137.

Instruments, decoration of, 2, 8, 11, 16, 39, 41, 42, 109, 112, 113, 115, 116, 123, 136.

Intervals, diatonic, 112.

” in American Indian instruments, 79.

Intervals in Chinese instruments, 39.

Intervals in Persian instruments, 53.

Irish bards, meetings of, 111.

” instruments, 89.

Isis, worship of, 36.

Italian instruments, 106-109, 113, 120, 123, 130.

Japanese instruments, 3, 4, 44 *seq.*

Jars, musical, 69.

Javanese instruments, 2, 3.

Jerusalem, Temple of, 19, 23.

Jew's harp, 102.

Jinagovi, 52.

Jobel, 25.

Jones, Edward, quoted, 90.

Junk, 53.

Juruparis, 66.

Kach'-hapi, 47.

Kalmuks, trumpet of the, 80.

Kane, 46.

Kantele, 47, 88.

Kei, [45](#).
Kemángéh, [55](#).
Ken, [42](#).
Keras, [32](#).
Keren, [24](#).
Keyboards, instruments with, [120-125](#).
Khorsabad, [16](#).
Kinnor, [20](#).
Kiōto, bell at, [46](#).
Kithara, Asiatic, [27](#).
 ” Greek, [28](#), [29](#).
K’iu (wood), [41](#).
Ko-kiū, [44](#).
Kosà, [130](#).
Koto, [44](#).
Kouyunjik, [16](#).
Kratzenstein, [6](#).
Krotala, [32](#).
Ku, [41](#).
Kuan-tzŭ, [42](#).
K’uei, musician, [37](#).
Kuitra, [56](#), [108](#).
Kymbala, [32](#).

Langspiel, [114](#).
Laos, instruments of, [4](#), [42](#).
Launedda, [36](#).
Lay, T., quoted, [43](#).
Lei-ku, [41](#).
Leighton, Sir W., quoted, [130](#).
Lidl, Anton, [115](#).
Lionedda, [36](#).
Lira di braccio, [101](#).
Lituus, [35](#).
Lombrive, [10](#).
Lute, [104](#), [105](#), [116](#).
 ” Arab, [54](#).

- ” Hindu, [89](#).
- ” Japanese, [44](#).
- ” mediæval, [102](#).
- ” Moorish, [56](#).
- ” Tibetan, [43](#).

Lute-makers, principal, [106](#), [116](#).

Lutists, Arabian, [54](#), [55](#), [56](#).

Lydians, Kithara of, [28](#).

Lyra, German, [90](#).

- ” Greek, [28](#).

- ” Roman, [33](#).

Lyre, [84](#).

- ” Assyrian, [18](#).

- ” Greek, [27](#) *seq.*

- ” Hebrew, [20](#).

- ” Roman, [33](#).

Mace, Thomas, quoted, [104](#), [105](#), [119](#).

Machalath, [22](#), [25](#), [26](#).

Machol, [26](#).

Magadis, [27](#), [30](#), [52](#).

Magoudi, [52](#).

Magrepha, [23](#), [24](#).

Mam, [13](#).

Mandoline, [107](#), [108](#).

Mandora, [108](#).

Mandorina, [108](#).

Marimba, [82](#).

Martin, instrument-maker, [118](#).

Mattheson, quoted, [105](#).

Melozzo da Forlì, painting by, [97](#).

Melrose Abbey, sculpture at, [97](#).

Melville, Sir James, quoted, [120](#).

Menaaneim, [25](#).

Metzilloth, [25](#).

Metzilhaim, [25](#).

Mexican instruments, [59](#), [80](#) *seq.*

Miao-tsze, [43](#).
Middle Ages, instruments of the, [83](#).
Minnim, [22](#), [23](#).
Miriam, [25](#).
Mishrokitha, [23](#).
Monaulos, [31](#).
Monochord, [31](#), [92](#).
Moorish instruments, [56](#), [108](#).
Mosul, bas-relief from, [16](#).
Mozart, [107](#).
Munich Museum, vase in, [28](#).
Music, ancient books on, [48](#), [84](#).
 ” supposed origin of, [47](#).

Nabla, [30](#).
Nablas, [27](#).
Nablia, [34](#).
Nablum, [86](#), [100](#).
Naker, [56](#).
Naḳḳárah, [56](#).
Nakrys, [56](#).
Nara, bell near, [46](#).
Nebuchadnezzar, [18](#).
Nechiloth, [25](#), [26](#).
Nefer, [12](#).
Nekeb, [23](#).
Nevel, [19](#), [22](#), [30](#).
New Guinea, instruments of, [2](#).
New Zealand, instruments of, [2](#).
“Nibelungenlied,” The, [90](#).
Nimroud, [16](#), [18](#).
Nineveh, [16](#).
Nootka Sound, instrument of, [2](#).
Norwegian instruments, [113](#).
Nuy, [55](#).

Oboe da caccia, [127](#).

- ” d’amore, [127](#).
- ” Hindu, [79](#).
- ” lungo, [127](#).
- ” Persian, [55](#).
- ” piccolo, [127](#).
- Ocarina, Chinese, [42](#).
- Octave, Arabian, [54](#).
 - ” Chinese, [39](#).
- Octavina (Ottavino), [120](#).
- Oliphant, [94](#).
- Organ, Burmese, [42](#).
 - ” Chinese, [42](#).
 - ” English, [129](#).
 - ” French, [129](#).
 - ” Gamba stop in, [115](#).
 - ” German, [129](#).
 - ” Hebrew, [24](#).
 - ” hydraulic, [32](#).
 - ” pneumatic, [94](#).
 - ” portative, [129](#), [130](#).
 - ” positive, [129](#).
 - ” Siamese, [42](#).
- Organ-builders, German, [129](#).
- Organ-harpsichord, [124](#).
- Organistrum, [92](#), [99](#), [101](#).
- Orchestras, mediæval, [99](#).
- Orpheus, Chinese, [37](#).
- Ottavino, or Octavina, [120](#).
- Ovalle, Alonso de, quoted, [62](#).

- P’ai-hsiao, [42](#).
- Palenque, instruments from, [62](#).
- Pandean pipes, [23](#), [31](#), [35](#), [42](#), [53](#), [80](#).
- Pandoura, [30](#).
- Pandurina, [108](#).
- Pasquali, Signor, [117](#).
- Passerini, Signor, [117](#).

Pedal, invention of, 96.
 " in harpsichord, 124.
Pektis, 30.
Pepys, quoted, 120, 126.
Persian instruments, 3, 48, 52 *seq.*
Peruvian instruments, 58, 59.
Peruvians, songs of the, 80, 81.
Phaamon, 25.
Phœnicians, 36.
Phorbeia, 34.
Phorminx, 28, 29.
Pianoforte, 123, 125, 134.
Piao, 39.
Pien-ch'ing, 38, 39.
Pien-chung, 39.
Piffero pastorale, 127.
Pincullu, 62.
P'i-p'a, 43, 44.
Pipe of the Aztecs, 60.
 " Berecynthian, 27.
 " Carian, 28.
 " of Chiriqui Indians, 60, 79.
 " Egyptian, 12.
 " Greek, 31.
 " Hebrew, 23.
 " Japanese, 45.
 " Mexican, 58 *seq.*
 " Peruvian, 58 *seq.*
 " Phrygian, 27.
Pitch of Chinese instruments, 39.
 " the oboe, 127.
 " the ottavino, 120.
 " whistle sounds, 59.
Pito, 60.
Plectrum, 30, 40, 44, 45, 109, 110.
Plektron, *see* Plectrum.
Po-fu, 41.

Poitiers, [10](#).
 Post-mediaeval instruments, [104 seq.](#)
 Pottery, instruments of, [58 seq.](#)
 Prætorius, quoted, [111](#).
 Pre-historic relics, [9](#).
 Psalms, musical directions in, [26](#).
 Psalterion, [20](#).
 Psalterium, [33](#), [85](#), [86](#).
 Psaltery, [102](#), [116](#), [117](#).
 Psanterin, [20](#).
 Pungi, [52](#), [93](#).

Quanūn, [54](#), [55](#).
 Quartfagott, [128](#).
 Quills for twanging strings, [107](#), [109](#), [119](#).
 Quills in virginal, [120](#).
 Quinterna, [109](#).
 Quintfagott, [128](#).
 Quyvi, [62](#).

Rabôb, [55](#), [56](#).
 Ranking, J., quoted, [75](#).
 Rattles, [80](#).
 " American Indian, [72](#), [82](#).
 " Indian, [2](#).
 Ravanastra, [50](#).
 Rebec, [56](#), [102](#), [113](#).
 Rébek, [90](#).
 Recorder, [125](#).
 Regal, or regals, [96](#), [102](#), [129](#).
 Rigabello, [130](#).
 Rin, [46](#).
 Roman instruments, [32 seq.](#)
 Rote, [88](#).
 Rotta, [88](#), [89](#).

Sârangi, [50](#).

Sackbut, [94](#), [102](#).
Sainprae, Jaques, [115](#).
Salpinx, [32](#).
Salterio, [102](#).
Sambuca, [34](#), [94](#).
Sambyke, [27](#), [30](#).
Samisen, [44](#).
Sang, [43](#).
San-hsien, [44](#).
Sankha, [47](#).
Santiago de Compostella, sculpture at, [101](#).
Santir, [6](#), [20](#), [55](#).
Sardinia, [36](#).
Sârinda, [50](#).
Scabellum, [35](#).
Scale, Chinese, [37](#), [39](#).
 ” diatonic, [132](#).
 ” pentatonic, [42](#), [79](#).
Scandinavian harp, [87](#).
Schalmei, [127](#).
Scheitholz, [118](#), [119](#).
Schnitzer, instrument maker, [128](#).
Sê, [43](#).
Sebà, [12](#).
Serinette, [129](#).
Serpent, [128](#).
Seshesh, [15](#).
Shakespeare, quoted, [114](#).
Shakuhachi, [45](#).
Shalisbim, [25](#).
Shalm, or shawm, [102](#), [103](#), [127](#).
Shehna, [79](#).
Shêng, [42](#), [43](#), [45](#).
Shime-daiko, [45](#).
Shō, [45](#).
Shophar, [24](#).
Shwan-che, [43](#).

Siam, instruments used in, [3](#), [4](#), [42](#).
Simikon, [30](#).
Sistrum, Egyptian, [14](#), [98](#).
 ” Hebrew, [25](#).
 ” Roman, [36](#).
Sitar, [110](#).
Sitâra, [55](#).
Solomon, [19](#).
Sordino, [118](#).
Spain, Arabs in, [36](#), [56](#).
Spanish instruments, [36](#), [110](#).
Spinnet, [121](#).
Stones, sonorous, [39](#), [73](#).
Stops of the clavicembalo, [123](#).
Stop in organ-harpsichord, [124](#).
Strabo, quoted, [27](#).
Stradivarius, [118](#).
Strings, catgut, [1](#), [30](#), [108-110](#), [115](#).
Strings, silk, [1](#), [43](#), [44](#), [54](#), [109](#).
Strings, sympathetic, [115](#), [116](#).
 ” wire, [55](#), [108-110](#), [115-117](#), [120](#), [121](#).
Sultana, [116](#).
Sumphonia, [23](#).
Sung-ch’ing, [39](#).
Surnai, [55](#).
Suroda, [88](#), [89](#).
Syrinx, Greek, [31](#).
 ” Hebrew, [23](#).
 ” mediæval, [94](#), [99](#).
 ” Peruvian, [63](#).
 ” Roman, [35](#).

Tabret, [24](#).
Taiko, [45](#).
Talmud, The, [23](#).
Tamboura, Arabian, [54](#).
 ” Egyptian, [27](#).

- ” Hebrew, [22](#).
- Tambourine, Assyrian, [18](#).
- ” Egyptian, [14](#).
- ” Hebrew, [24](#).
- ” Peruvian, [72](#).
- ” Roman, [35](#).
- Tangents in the clavichord, [121](#).
- T’ê-ch’ing, [39](#).
- T’ê-chung, [39](#), [40](#).
- Tenor (violin), [113](#).
- Tenor-bassoon, [128](#).
- Tenor-viol, [119](#).
- Teponaztli, [70](#), [80](#).
- Testudo, [33](#).
- Tezcucans, instruments of the, [73](#).
- Thebes, [11](#), [12](#), [14](#).
- Theorbo, [101](#), [104](#), [105](#).
- Ti, [42](#).
- Tibetan instruments, [43](#), [80](#).
- Tibia, [34](#).
 - ” curva, [34](#).
 - ” dextra, [34](#).
 - ” gingrina, [34](#).
 - ” ligula, [34](#).
 - ” longa, [34](#).
 - ” obliqua, [34](#).
 - ” sinistra, [34](#).
 - ” utricularis, [34](#).
 - ” vasca, [35](#).
- Tibiæ impares, [34](#).
 - ” pares, [34](#).
- Timbrel, [24](#), [102](#).
- Timotheus, flutist, [57](#).
- Tintinnabula, [36](#).
- Tintinnabulum, [100](#).
- Tinya, [75](#).
- Titus, arch of, [24](#).

Tone of instruments, [112](#), [113](#).
 Toph, [24](#), [25](#).
 Toumrie, [52](#).
 Treble-viol, [119](#).
 Triangle, Hebrew, [25](#).
 Triangle, Roman, [36](#).
 Triangulum, [36](#).
 Trigonon, [17](#), [28](#), [30](#), [53](#).
 Trigonum, [34](#).
 Triple Flageolet, [126](#).
 Trombone, [94](#).
 Trumpets of South American Indians, [65](#).
 Trumpets, Anglo-Saxon, [94](#).
 " Ashantee, [2](#).
 " Assyrian, [18](#).
 " of the Caroados, [67](#).
 " Egyptian, [14](#).
 " Greek, [32](#).
 " Hebrew, [24](#) *seq.*
 " Hindu, [47](#), [79](#).
 " of the Kalmuks, [80](#).
 " Mexican, [80](#).
 " New Zealand, [2](#).
 " Persian, [53](#).
 " Thibetan, [80](#).
 Tschenk (Chang), [53](#).
 Tsu-ku, [41](#).
 Tsudzumi, [45](#).
 Tsurigane, [46](#).
 Tuba, [35](#).
 Tuckey, Captain, [2](#).
 Turé, [67](#), [79](#).
 "Tuner of the Regals," [130](#).
 Tuning of the spinet, [121](#).
 Tympanon, [32](#).
 Tympanum, [35](#).
 Tyrolean harp-makers, [112](#).

Tzeltzelim, 25.

Ugab, 23.

Ur-heen, 51, 52.

Ventura, Signor, 110.

Vielle, 101.

Vihuela, 102, 110.

Vina, 46, 47, 49.

” mahati, 49.

” rudra, 49.

Vinavah, 51.

Viol, mediæval, 99, 100.

” post-mediæval, 113, 119.

” Spanish, 102, 118.

Viola da gamba, 114, 115.

” d’amore, 116.

” di bardone, 115.

Violin, 91, 113, 114, 116.

” Japanese, 44.

” Persian, 55.

Violoncello, 114, 115.

Virginal, 119-121, 130.

Wait, the instrument, 103.

Walther, quoted, 121.

Welsh instruments, 89, 90.

Whistles, American Indian, 82.

” Mexican, 59, 60.

Wilkinson, Sir G., quoted, 21.

Ying-ku, 41.

Yotl, 73.

Yü, 40, 41.

” stone made into the ch’ing, 38.

Yüeh, 42.

Yüeh-ch’in, 43.

Zampogna, [23](#).

Zante, belfries in, [131](#).

Zither, or Zitter, [109](#).

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